

... complete document solutions

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All Notes Available

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1st order PDE!

(i) $yp - xq = xyz = x$

(ii) $xyzp - e^{x+y}q = xyz + 0x$

(iii) $x^2p + y^2q = xyz$

(iv) $p - q = z$

(v) $e^xp + yq = xyz$

Second Linear

(i) $e^xp - yq = z^2$

Quasi Linear: (i) $(x^2 + z^2)p - xyq = z^3 + yz$

(ii) $yz^2p + xy^2q = \sin z$

Note!

~~$pq = xy$~~ Non-linear

16/08/2014

Partial Differential Equⁿ : →

If $z = f(x, y)$

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}, \quad r = \frac{\partial^2 z}{\partial x^2}$$

$$s = \frac{\partial^2 z}{\partial x \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 1 \quad \text{is a partial Equⁿ .}$$

Order : → order of the pde is highest derivative occurring in the Equⁿ.

Degree : → The power of the highest derivative is known as degree :

Classification of first order partial P.D.E. : →

First order partial diff. Equ. is said to be linear if it is linear in p, q, z and of the form -

$$P(x, y) p + Q(x, y) q = R(x, y) z + S(x, y)$$

Semi Linear : → FOPDE is said to be ²semi linear if it is linear in p, q and of

the form -

$$P(x, y) p + Q(x, y) q = R(x, y, z)$$

Eg:- $xp + yq = x^2 z^2$

Quasi linear : → FOPDE is said to be quasi linear if it is p, q and z of the form -
linear in p, q .

$$P(x, y, z) p + Q(x, y, z) q = R(x, y, z)$$

$$\Rightarrow x^2 z^2 p + q y^2 = x^2 + z^2$$

- (1) S.L. $\Rightarrow$ Q $\Rightarrow$ L.
 (2) L. $\Rightarrow$ Q $\nRightarrow$ S.L.
 (3) L. $\Rightarrow$ S.L. $\Rightarrow$ Q $\checkmark$
 (4) Q $\Rightarrow$ S.L. $\Rightarrow$ L.

Formation of FODE : $\rightarrow$ There are three types -

(1) By eliminating arb. constants : $\rightarrow$

$$f(x, y, z, a, b) = 0 \quad \text{--- (1)}$$

P.D. w.r. to x & y

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial x} \quad \text{--- (2)}$$

($\because z$ is func of x & y)

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial y} \quad \text{--- (3)}$$

Eliminating a & b and using

$$F(x, y, z, p, q) = 0$$

Exp. ① $z = x + ax^2y^2 + b$

$$p = \frac{\partial z}{\partial x} = 1 + 2axy^2 = 0$$

$$\therefore x = -\frac{p-1}{2ay^2} \therefore a = -\frac{p-1}{2xy^2}$$

$$q = \frac{\partial z}{\partial y} = 2ax^2y = 0$$

$$\therefore a = q / 2x^2y$$

Comparing both value of 'a'

$$p = 1 + qy/x$$

$$\Rightarrow px - qy = x //$$

$$\underline{Q.} \quad z^2 (1+a^3) = 8(x+ay+b)^3 \quad \text{--- (A)}$$

$$p = \frac{\partial z}{\partial x} \quad (\text{P.D. w.r. to } x)$$

$$(1+a^3) 2z \frac{\partial z}{\partial x} = 8 \cdot 3(x+ay+b)^2 \cdot 1$$

$$(1+a^3) 2zp = 24(x+ay+b)^2 \quad \text{--- (1)}$$

P.D. w.r. to y.

$$(1+a^3) 2z \frac{\partial z}{\partial y} = 24(x+ay+b)^2 \cdot a$$

$$2z(1+a^3)q = 24a(x+ay+b)^2 \quad \text{--- (2)}$$

Dividing eq. (A) | (1)³

$$\Rightarrow \frac{z^4 (1+a^3)^2}{8z^3 p^3 (1+a^3)^3} = \frac{8^2 (x+ay+b)^6}{18^3 (x+ay+b)^6}$$

$$\Rightarrow \frac{z}{p^3 (1+a^3)} = \frac{1}{27}$$

$$\Rightarrow z = \frac{p^3 (1+a^3)}{27}$$

By eq. (2) | (3)

$$\frac{p}{q} = \frac{1}{a} \Rightarrow a = q/p$$

$$\therefore z = \frac{p^3}{27} \left(1 + \frac{q^3}{p^3} \right)$$

$$z = \frac{p^3 + q^3}{27}$$

$$\Rightarrow \boxed{27z = p^3 + q^3}$$

↓
Non linear

Q:- $lx + my + nz + 1 = 0$ ———

Find the p.d.e of all planes which are at constant distance from the origin.

$$l^2 + m^2 + n^2 = \frac{1}{a^2} \quad \text{--- (2)}$$

Sol.ⁿ

$$lx + my + nz + 1 = 0 \quad \text{--- (1)}$$

$$l + 0 + np + 0 = 0$$

$$p = -l/n \Rightarrow l = -np$$

P.D.W.R. to y of eqn. (1)

$$m + nq = 0$$

$$-nq = m \Rightarrow m = -nq$$

P.D.W.R. to z of eqn. (1)

$$n = 0$$

Now by eqn. (2)

$$l^2 + m^2 + n^2 = \frac{1}{a^2}$$

$$\Rightarrow n^2 p^2 + n^2 q^2 + n^2 = \frac{1}{a^2}$$

$$\Rightarrow n^2 (p^2 + q^2 + 1) = \frac{1}{a^2}$$

(2) By eliminating arbitrary func: $\rightarrow$

Q:-

$$Z = xy + F(x^2 + y^2) \quad \text{--- (1)}$$

p.d. w.r. to x

$$p = y + F'(x^2 + y^2) \cdot 2x \quad \text{--- (2)}$$

p.d. w.r. to y

$$q = x + F'(x^2 + y^2) \cdot 2y \quad \text{--- (3)}$$

by equ. (2) + (3)

equating value of $F'(x^2 + y^2)$

$$= \frac{p - y}{2x} = \frac{q - x}{2y}$$

$$\Rightarrow \boxed{py - qx = y^2 - x^2} \quad \text{Ans.}$$

Q:-

$$Z = F\left(\frac{xy}{z}\right)$$

$$p = F'\left(\frac{xy}{z}\right) \cdot \left[\frac{y}{z} - \frac{xy}{z^2} p\right] \times x$$

$$q = F'\left(\frac{xy}{z}\right) \cdot \left[\frac{x}{z} - \frac{xy}{z^2} q\right] \times y$$

$$px - qy = 0$$

$$\boxed{px - qy = 0} \quad \text{Ans.}$$

Q:-

$$\log(az - 1) = x + ay + b$$

$$\frac{1}{(az - 1)} \cdot ap = 1 \quad \text{--- (1)}$$

$$\frac{1}{(az - 1)} \cdot aq = a \quad \text{--- (2)}$$

$$q = az - 1$$

$$a = \frac{q+1}{z} \quad \text{--- (3)}$$

by eqn. (1) $ap = az - 1$

$$a(p-z) = -1$$

$$a = -\frac{1}{p-z} \quad \text{--- (4)}$$

by eqn. (3) & (4)

$$\frac{-1}{p-z} = \frac{q+1}{z}$$

$$\Rightarrow -z = pq + p - zq - z$$

$$\Rightarrow \boxed{pq + p - qz = 0} \quad \text{Ans.}$$

gate-07

Q.

Let $u(x, y) = f(xe^y) + g(y^2 \cos y)$

where f and g are infinitely diff. function

The pde, of minimum order satisfying by u .

(a) $u_{xy} + x u_{xx} = u_x$

(b) $u_{xx} + x u_{xx} = x \cdot u_x$

(c) $u_{xy} - x u_{xx} = u_x$

(d) $u_{xy} - x u_{xx} = x u_x$

= Solⁿ:- $u(x, y) = f(xe^y) + g(y^2 \cos y)$

$$u_x = f'(xe^y) \cdot e^y$$

$$u_{xx} = f''(xe^y) \cdot e^{2y}$$

$$u_{xy} = f'(xe^y) \cdot xe^{2y} + f'(xe^y) \cdot e^y$$

$$u_{xy} - x u_{xx}$$

$$f''(xe^y) \cdot xe^y + f'(xe^y) \cdot e^y - xf''(xe^y) \cdot e^y$$

$$= f'(xe^y) \cdot e^y = ux$$

$$\boxed{uxy - xuxx = ux}$$

Q:-

$$z = y^2 + 2f\left(-\frac{1}{x} + \log y\right)$$

$$p' = 2f'\left(-\frac{1}{x} + \log y\right) \cdot \left(\frac{1}{x^2}\right) \text{ --- (1)}$$

$$q = 2y + 2f'\left(-\frac{1}{x} + \log y\right) \cdot \left(\frac{1}{y}\right) \text{ --- (2)}$$

by equ. (1)

$$2f'\left(-\frac{1}{x} + \log y\right) = x^2 p$$

$$4 \cdot 2f'\left(-\frac{1}{x} + \log y\right) = yq - 2y^2$$

$$\boxed{x^2 p = yq - 2y^2}$$

Ans.

Q:-

$$z = y + x + F(xy)$$

$$p = 1 + F'(xy) \cdot y$$

$$F'(xy) = \frac{p-1}{y} \text{ --- (1)}$$

$$q = 1 + F'(xy) \cdot x$$

$$F'(xy) = \frac{q-1}{x} \text{ --- (2)}$$

by equ. (1) + (2)

$$\frac{p-1}{y} = \frac{q-1}{x}$$

$\Rightarrow$

$$\boxed{x(p-1) = y(q-1)}$$

Ans.

Q:-

$$z = y f(x) + x g(y)$$

$$p = y f'(x) + g(y)$$

$$q = f(x) + x g'(y)$$

Rules Regarding formation of PDE by removing arbitrary constant : $\rightarrow$

- 1) For a given relation involving variable and arbitrary constant, the relation is diff partially with respect to independent variables & arbitrary constant are eliminating to get the corresponding PDE.
- 2) If the No. of arbitrary constant are less than no. of independent variables then the elimination of arbitrary constant shall usually give rise to more than one diff. Equ. of order 1.
- 3) If the No. of arbitrary constant = Independent variable then give rise to unique partial diff. Equⁿ of order 1.
- 4) If the no. of arbitrary constant is greater than independent variable then the eliminating constant shall give rise to more than 1 P.D.E of order greater than 1.

Q:- $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ ——— ①

$$\Rightarrow \frac{2x}{a^2} + \frac{2z}{c^2} p = 0$$

$$\Rightarrow c^2 x + z a^2 p = 0 \text{ — ②} \Rightarrow p = -\frac{c^2 x}{a^2 z}$$

$$\therefore q = -\frac{c^2 y}{b^2 z} \Rightarrow c^2 y + b^2 z q = 0 \text{ — ③}$$

P. D. w. r. to x of ②

$$c^2 + a^2 p^2 + a^2 z r = 0$$

P.D.w.r. to x of eqn. (3)

$$b^2 (pq + rs) = 0 \Rightarrow b = 0$$

$$\boxed{(pq + rs) = 0} \rightarrow \text{P.d.E. of second order.}$$

TYPE - 3

If the Eqn. $F(u, v) = 0$ — (1)

where $u = u(x, y, z)$

$v = v(x, y, z)$

This is called Cauchy linear form.

P.D.E. (1) w.r. to x

$$\frac{\partial F}{\partial u} \left[\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial x} \right] + \frac{\partial F}{\partial v} \left[\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z} \cdot \frac{\partial z}{\partial x} \right] = 0$$

$$\Rightarrow \frac{\partial F}{\partial u} [u_x + p u_z] + \frac{\partial F}{\partial v} [v_x + p v_z] = 0 \quad \text{--- (2)}$$

P.D.E. (2) w.r. to y

$$\Rightarrow \frac{\partial F}{\partial u} [u_y + q u_z] + \frac{\partial F}{\partial v} [v_y + q v_z] = 0 \quad \text{--- (3)}$$

$\Rightarrow$ Divide eqn. (2) | (3)

$$\frac{(u_x + p u_z)}{(u_y + q u_z)} = \frac{(v_x + p v_z)}{(v_y + q v_z)}$$

$$\Rightarrow u_x v_y + q u_x v_z + p u_z v_y + p q u_z v_z = u_y v_x + p u_y v_z + q v_x u_z + p q u_z u_z$$

$$\Rightarrow p [u_z v_y - u_y v_z] + q [u_x v_z - v_x u_z] = (u_y v_x - u_x v_y) \quad \text{--- (6)}$$

$$\begin{bmatrix} p & q & -1 \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{bmatrix}$$

$$\Rightarrow p p + Q q = R$$

where $p = [u_3 v_4 - u_4 v_3]$

$$Q = [u_x v_3 - v_x u_3]$$

$$R = [u_y v_x - u_x v_y]$$

Q:- $\phi(x+y+z, x^2+y^2-z^2) = 0$

Find p.d.e. by eliminating y

$$u = x+y+z, \quad v = x^2+y^2-z^2$$

Solⁿ:-

$$\begin{vmatrix} p & q & -1 \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \begin{vmatrix} p & q & -1 \\ 1 & 1 & 1 \\ 2x & 2y & -2z \end{vmatrix}$$

$$= p(-2z - 2y) - q(-2z - 2x) - 1(2y - 2x) = 0$$

$$\Rightarrow \boxed{p(y+z) - q(z+x) = (x-y)}$$

Q:- $F(x+y, x-z^{1/2}) = 0$

$$\begin{vmatrix} p & q & -1 \\ 1 & 1 & 0 \\ 1 & 0 & -\frac{1}{2} z^{-1/2} \end{vmatrix} = 0$$

$$\Rightarrow p \left[\frac{-1}{2\sqrt{z}} \right] - q \left[\frac{-1}{2z^{1/2}} \right] - 1(-1) = 0$$

$$\Rightarrow \frac{p}{2\sqrt{z}} - \frac{q}{2\sqrt{z}} = 1$$

$$\Rightarrow \boxed{p - q = 2\sqrt{z}}$$

Q:- $z = xy + f(x^2 + y^2)$

$$z - xy = f(x^2 + y^2)$$

Here $u = z - xy$, $v = x^2 + y^2$

$$u = f(v)$$

$$u = f(v)$$

$$= \begin{vmatrix} p & q & -1 \\ -y & -x & 1 \\ 2x & 2y & 0 \end{vmatrix}$$

$$\Rightarrow p(-2y) - q(-2x) - 1(-2y^2 + 2x^2) = 0$$

$$\Rightarrow \boxed{py - qx = (x^2 - y^2)}$$

Types of Solution :->

(1) Complete Integral :-> For the p.d.e ,

$$f(x, y, z, p, q) = 0 \quad \text{--- (1)}$$

we can find the relation $F(x, y, z, a, b) = 0$ --- (2)
which contains as many arbitrary constant as
independent variable then equ. (2) is C.I. of (1)

(2) General Integral :-> The relation of the type
 $F(u, v) = 0$, where u, v are
arbitrary fun^c of x, y, z .

Method to find the solution of Linear Equⁿ of First order :->

$$\text{given the Equ. } Pp + Qq = R \quad \text{--- (1)}$$

The general Integral Equ. (I) is given by

$$F(u, v) = 0$$

$$u = u(x, y, z), \quad v = v(x, y, z)$$

Are the solⁿ of simultaneous Equⁿ -

$$\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}$$

u & v , should be I.I.

Q:- $x^2 p + y^2 q = (x+y)z$

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{(x+y)z}$$

$$\Rightarrow \frac{\frac{dx}{x} + \frac{dy}{y}}{(x+y)} = \frac{dz}{(x+y)z}$$

$$\Rightarrow \log x + \log y = \log z + \log C_1$$

$$\boxed{C_1 = xy/z} = v$$

$$\frac{dx}{x^2} = \frac{dy}{y^2}$$

$$\text{or } \boxed{C_2 = \frac{1}{x} - \frac{1}{y}} = u$$

$$F(u, v) = 0$$

$$F\left(\frac{1}{x} - \frac{1}{y}, \frac{xy}{z}\right) = 0$$

'OR' $\left(\frac{xy}{z}\right) = F\left(\frac{1}{x} - \frac{1}{y}\right)$

'OR' $\left(\frac{1}{x} - \frac{1}{y}\right) = F\left(\frac{xy}{z}\right)$

Ans.

Solve the determinate

$$\begin{vmatrix} x & y & z \\ \alpha & \beta & \gamma \\ \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} & -1 \end{vmatrix} = 0$$

$$\Rightarrow \beta [\gamma\gamma - \beta z] + \gamma [z\alpha - \gamma x] = (x\beta - \alpha\gamma)$$

$$\Rightarrow \frac{dx}{(y\gamma - \beta z)} = \frac{dy}{(z\alpha - \gamma x)} = \frac{dz}{(x\beta - \alpha\gamma)}$$

$$\Rightarrow \frac{x dx + y dy + z dz}{(y\gamma - \beta z)x + (z\alpha - \gamma x)y + (x\beta - \alpha\gamma)z} = \frac{dz}{(x\beta - \alpha\gamma)}$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$\Rightarrow \boxed{x^2 + y^2 + z^2 = c}$$

Now, $\frac{\alpha dx + \beta dy + \gamma dz}{\alpha y\gamma - \alpha\beta z + \beta z\alpha - \beta\gamma x + \gamma x\beta - \gamma\alpha y} = \frac{dz}{x\beta - \alpha\gamma}$

$$\Rightarrow \alpha dx + \beta dy + \gamma dz = 0$$

$$\Rightarrow \boxed{\alpha^2 + \beta^2 + \gamma^2 = c}$$

LET
Q.

The general Solution of -

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u$$

Passing through the centre $x=t, y=2t, u=1$
write in form of $u = f(v)$ or $v = f(u)$

Solⁿ

$$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{u}$$

$$x - y = c_1 \quad \text{--- (1)}$$

Now, $\frac{dx}{1} = \frac{du}{u}$

$$x = \log u + \log c_2$$

$$uc_2 = e^x \Rightarrow c_2 = u e^{-x}$$

Now $u e^{-x} = f(x - y) \quad \text{--- (2)}$

Put $x = t, y = 2t, u = 1$

$$e^{-t} = f(t - 2t)$$

$$e^{-t} = f(-t) \Rightarrow f(h) = e^h$$

By eqn. (2)

$$u e^{-x} = e^{x-y}$$

$$\boxed{u = e^{2x-y}}$$

NET
Q.

$$y \frac{\partial z}{\partial x} - x \frac{\partial z}{\partial y} = y z^2 e^{-(x^2+y^2)}$$

(1) $u = x^2 - y^2, v = x - \frac{1}{2} e^{-(x^2+y^2)}$

(2) $u = x^2 + y^2, v = x + \frac{1}{2} e^{(x^2+y^2)}$

(3) $u = x^2 + y^2, v = x - \frac{1}{2} e^{(x^2+y^2)}$

(4) $u = x^2 + y^2, v = x + \frac{1}{2} e^{(x^2+y^2)}$

Solⁿ:-

$$\frac{dx}{y} = \frac{dy}{-x} = \frac{dz}{\frac{2}{x y^2} e^{-(x^2+y^2)}}$$

$$\frac{dx}{y} = \frac{dy}{-x}$$

$$\Rightarrow x^2 + y^2 = c_1 \quad \text{--- (1)}$$

Now, $\frac{dx}{y} = \frac{dz}{\frac{2}{x y^2} e^{-(x^2+y^2)}} \Rightarrow x = \frac{-1}{z} e^{c_1} + c_2$

$$c_2 = x + \frac{1}{z} e^{(x^2+y^2)}$$

23 Aug 2014

- 1) By eliminating arbitrary fun^c we will always get Quasi linear P.D.E.
- 2) By eliminating arbitrary fun^c we will get both non linear & Quasi P.D.E.
- 3) No. of arbitrary fun^c to be eliminated = The order of PDE.

Solution of first order Quasi linear PDE : $\rightarrow$

$$pP + qQ = R$$

$$F(u, v) = 0, \quad u = \phi(v), \quad v = \phi(u)$$

TYPE-1 : By Taking two different pair of functions will get -

$$u(x, y, z) = c_1 \quad \& \quad v(x, y, z) = c_2$$

Eg:-

$$yzp + zxq = xy$$

$$\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy}$$

$$\Rightarrow \frac{x dx - y dy}{xyz - xyz} = \frac{dz}{xy}$$

$$\Rightarrow x dx - y dy = 0$$

$$\boxed{x^2 - y^2 = c_1}$$

$$\& \quad y dy - z dz = 0$$

$$\boxed{y^2 - z^2 = c_2}$$

TYPE-2 :- By Taking two fraction we get $u(x, y, z) = c_1$. Then other variables $v(x, y, z) = c_2$ is obtained by making use of $u(x, y, z) = c_1$.

Exp:- $5p - 6q = 5x^4 \cos(6x+5y)$

Sol:- $\frac{dx}{5} = \frac{dy}{-6} = \frac{dz}{5x^4 \cos(6x+5y)}$

$$\boxed{6x + 5y = c_1}$$

Now, $\frac{dx}{5} = \frac{dy}{-6} = \frac{dz}{5x^4 \cos c_1}$

$$\frac{dx}{5} = \frac{dz}{5x^4 \cos c_1}$$

$\Rightarrow \frac{x^5}{5} = \frac{z}{\cos c_1} + c_2$

$$\therefore \boxed{c_2 = \frac{x^5}{5} - \frac{z}{\cos(6x+5y)}}$$

Exp:- $p - q = \log(x+y)$

$$\frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{\log(x+y)}$$

$\Rightarrow \boxed{x + y = c_1}$

Now $dx = \frac{dz}{\log c_1}$

$$x = z \log c_1 + c_2$$

$$\boxed{c_2 = x - z \log(x+y)}$$

Type-3 : $\rightarrow$ Let $Pp + Qq = R$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

Choose multipliers as S, T, U s.t.

$$SP + QT + UR = 0$$

Exp. $(z-y)p + (x-z)q = y-x$

$$\Rightarrow \frac{dx}{z-y} = \frac{dy}{x-z} = \frac{dz}{y-x}$$

$$\Rightarrow dx + dy + dz = 0$$

$$\Rightarrow \boxed{x + y + z = c_1}$$

Now, $x dx + y dy + z dz = 0$

$$\boxed{x^2 + y^2 + z^2 = c_2}$$

Type-IV $\rightarrow$ choose multiply S, T, U s.t. $Sdx + Tdy + Udz$ is exact diff. equ. of $Sp + qR + UR$

Exp. $(y+z)p + (z+x)q = (x+y)$

$$\frac{dx}{(y+z)} = \frac{dy}{(z+x)} = \frac{dz}{(x+y)}$$

$$\Rightarrow \frac{dx - dy}{-(x-y)} = \frac{dx + dy + dz}{2(x+y+z)}$$

$$\Rightarrow -\log(x-y) = \frac{1}{2} \log(x+y+z) + \log c_1$$

$$\Rightarrow \boxed{c_1 = (x-y) \cdot (x+y+z)^2}$$

Now, $\frac{dx - dy}{-(x-y)} = \frac{dy - dz}{-(y-z)}$

$$\Rightarrow \log(x-y) = \log(y-z) + \log c_2$$

$$\boxed{c_2 = \left(\frac{x-y}{y-z} \right)}$$

Surface & Normal : $\rightarrow$ Let Ω be a domain in $\mathbb{R}^3$
 & let $f(x, y, z)$ be a funcⁿ in class $C^1(\Omega)$ if -
 ($f', f'' \dots f^k$ are conti)

$$\text{grad}(F) = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) \text{ --- (1)}$$

Assume $\text{grad}(F)(x, y, z)$ is non zero at every point
 in Ω : $[f(x, y, z) \neq (0, 0, 0)] \text{ --- (2)}$

i.e. P.d of f do not vanish simultaneously at any
 point in Ω .

Under the assumption (2) The set of point (x, y, z)
 in Ω which satisfied the Eqnⁿ.

$$F(x, y, z) = c \text{ --- (3) is surface in } \Omega.$$

Eqn. (3) is called "level Surface".

Level Surface : $\rightarrow$ For different value of c in eqn. (3)
 we get different level surface on one
 level surface F will have same value at every point.

Exp:- $F(x, y, z) = x^2 + y^2 + z^2 = c$

$$\text{grad } F = (2x, 2y, 2z)$$

In $\mathbb{R}^3$ except origin is surface.

Here, $\$$ level surfaces are concentric sphere centres
 at origin.

Exp:- $F(x, y, z) = x^2 + z^2$

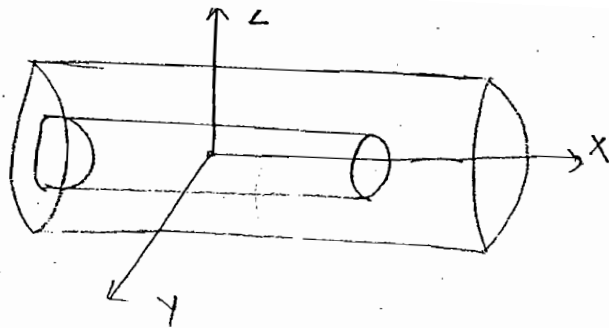
$$\text{grad } F = (2x, 0, 2z)$$

(whole y -axis is do-
 main)

It is a surface in $\mathbb{R}^3$ except origin & y -axis.

The level surfaces are concentric cylinder with
 make axis as y .

Equ. (3) represents one parameter of family of surfaces.



Each point in domain lies exactly one level surface.

And any two points in domain (x_0, y_0, z_0) & (x_1, y_1, z_1) lie on same level surface iff -

$$F(x_0, y_0, z_0) = F(x_1, y_1, z_1)$$

Exp-

$$F(x, y, z) = z$$

$$\text{grad } F = (0, 0, 1)$$

$$\text{Domain} = \mathbb{R}^3$$

level surface $z=1, z=2, z=3 \dots$

planes parallel to xy .

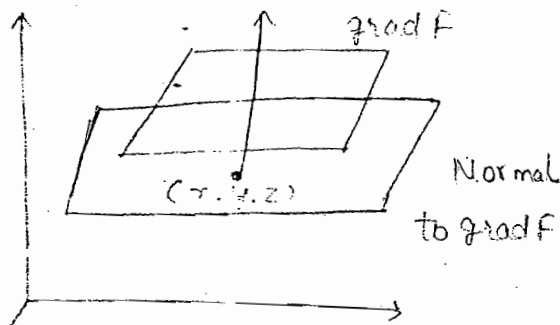
Tangent planes to a point on a surface: →

Consider a level surface

$$S_c = F(x, y, z) = c$$

We can have tangent plane at every point of the circles.

Now $\text{grad } F$ is a vector normal to the tangent plane at every point of tangency.



Let p is a plane passing through a fixed point (x_0, y_0, z_0) .

$$r_0 = (x_0, y_0, z_0)$$

r_0 is the distance (position vector) of fixed pts.

Let r be any arbitrary point on a plane p .

Let n be a normal vector at Point $r-r_0$.

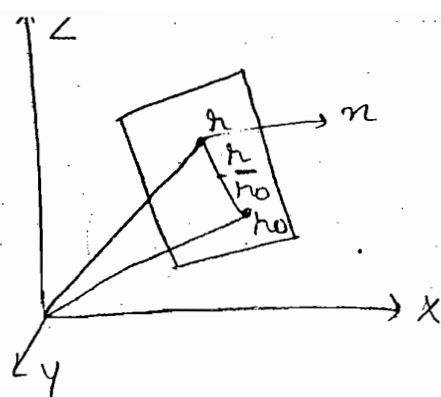
Vector $r-r_0$ will lie on plane and it will be orthogonal to n .

$$\text{s.t. } (r-r_0) \cdot n = 0$$

$$(x-x_0) \left[\frac{\partial f}{\partial x} (x_0, y_0, z_0) \right] +$$

$$(y-y_0) \left[\frac{\partial f}{\partial y} (x_0, y_0, z_0) \right] + (z-z_0) \left[\frac{\partial f}{\partial z} (x_0, y_0, z_0) \right] = 0$$

This is the eqⁿ of tangent plane to the surface F where x_0, y_0, z_0 is the point of tangency.



Exp:- $x^2 + y^2 + z^2 = 6$

Find the tangent plane at $1, 2, -1$

Solⁿ:- $\Delta f = \frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial z} = 2z$

$$\text{Grad } f = (2, 4, -2)$$

$$\Rightarrow (x-1) \cdot 2 + (y-2) \cdot 4 + (z+1) \cdot (-2) = 0$$

$$\Rightarrow 2x + 4y - 2z = 12$$

$$x + 2y - z = 6$$

Parametric Eqⁿ of Surfaces $\rightarrow$ Now consider

a surface $S_c = f(x, y, z) = c$ is a possible to solve $z = f(x, y)$ — (1)

Consider, $x = f_1(u, v), \quad y = f_2(u, v)$

$$z = f_3(u, v) \quad \text{--- (2)}$$

For every pair u & v will have a point (x, y, z) in a space.

However, every point in a space need not correspond to pair u & v .

By Implicit Function Thm ,

$$\text{if } \frac{\partial(f_1, f_2)}{\partial(u, v)} \neq 0$$

Then by using first two equ. of (2) . you find

$$u = \lambda(x, y), \quad v = \mu(x, y)$$

$$z = f_3[\lambda(x, y), \mu(x, y)]$$

Hence, any point x, y, z obtain from (2) will lie on a fix surface.

$$F(x, y, z) = f(x, y) - z = 0$$

For this type of surface, the normal vector $\text{grad } F$.

$$\text{w.r. to } x \Rightarrow \frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} \cdot \frac{\partial z}{\partial x} = 0 \quad \text{--- (3)}$$

$$\text{w.r. to } y = \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} \cdot \frac{\partial z}{\partial y} = 0 \quad \text{--- (4)}$$

$$\frac{\partial z}{\partial x} = \frac{-\partial F / \partial x}{\partial F / \partial z}$$

$$\partial z / \partial x = -\partial F / \partial x$$

$$\Rightarrow \partial F / \partial x = p.$$

$$\text{Similarly } \partial F / \partial y = q$$

Vectors to normal to a surface.

$$F(x, y, z) = F(x, y) - z = 0 \text{ is } (p, q, -1)$$

Equ. (2) is a parametric equ. of surface.

" parametric equ. of a surface need not to be unique."

exp:-

$$x^2 + y^2 + z^2 = r^2$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$x = \frac{r(1-\phi^2)}{(1+\phi^2)} \cos \theta, \quad y = \frac{r(1-\phi^2)}{(1+\phi^2)} \sin \theta$$

$$z = \frac{2r\phi}{1+\phi^2}$$

24/08/2014

Curves and Tangent :→ The most common way of describing a curve in $\mathbb{R}^3$ by means of parametric representation.

Let r denotes the position vector of a point on curve C .

Then $r = f(t)$, $t \in I$

f $F(t) = \{f_1(t), f_2(t), f_3(t)\}$ is a vector valued function of a parameter t .

$$x = f_1(t), \quad y = f_2(t) = z = f_3(t)$$

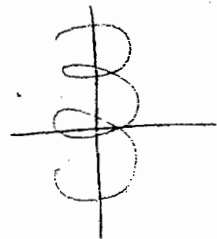
Now assuming $f_1, f_2, f_3 \in C^1(I)$ & Their derivative donot vanish simultaneously at any point of I .

$$\frac{dr}{dt} = \left(\frac{df_1(t)}{dt}, \frac{df_2(t)}{dt}, \frac{df_3(t)}{dt} \right) \neq (0, 0, 0)$$

Then for each $t \in I$, the non vanishing vector dr/dt is a tangent to the curve C at point $(f_1(t), f_2(t), f_3(t))$.

$$x = \cos t, \quad y = \sin t, \quad z = t$$

It is a parametric Equ. of helix.



$$\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} = (-\sin t, \cos t, 1)$$

at $\pi/2$ $(-1, 1, 0)$ is a tangent at point $(0, 1, \pi/2)$

Intersection of two surfaces be a curve provided both surfaces are independent.

$$\text{Let } F_1(x, y, z) = c_1, \quad F_2(x, y, z) = c_2 \quad \text{--- (1)}$$

are two surfaces.

The set up point satisfying both equation must lie on the intersection of two surfaces.

The intersection if non empty, is generally a curve. But if $\text{grad } F_1$ & $\text{grad } F_2$ are not collinear then the intersection is always a curve.

$$\text{i.e. } \text{grad } F_1(x_0, y_0, z_0) \times \text{grad } F_2(x_0, y_0, z_0) \neq 0$$

$$\forall x_0, y_0, z_0 \in \mathcal{N} \quad \text{--- (2)}$$

Proof :- $F_1(x, y, z) = c_1$ &

$$F_2(x, y, z) = c_2$$

Using assumption Eau. (2)

$$\begin{aligned} \text{grad } F_1 \times \text{grad } F_2 &= \left(\frac{\partial F_1}{\partial x}, \frac{\partial F_1}{\partial y}, \frac{\partial F_1}{\partial z} \right) \times \left(\frac{\partial F_2}{\partial x}, \frac{\partial F_2}{\partial y}, \frac{\partial F_2}{\partial z} \right) \\ &= \begin{bmatrix} \frac{\partial (F_1, F_2)}{\partial (y, z)} & \frac{\partial (F_1, F_2)}{\partial (z, x)} & \frac{\partial (F_1, F_2)}{\partial (x, y)} \end{bmatrix} \neq 0 \end{aligned}$$

$$\text{Let } \frac{\partial (F_1, F_2)}{\partial (y, z)} \neq 0 \quad \text{at } (x_0, y_0, z_0)$$

(By Implicit thm)

So $y = g(x), \quad z = h(x)$

$$\text{Let } x = t, \quad \boxed{y = g(t), \quad z = h(t)}$$

This show that near x_0, y_0, z_0 the set of point (x, y, z) which satisfying eau. (1) forms a curve whose parametric representation given by eau (3)

Eg:- $F_1 = x^2 + y^2 - z$

$F_2 = z$

$\text{grad } F_1 = (2x, 2y, -1)$

$\text{grad } F_2 = (0, 0, 1)$

if $\mathcal{A}$ in $\mathbb{R}^3$ with z -axis removed.

$\text{grad } F_1 \times \text{grad } F_2 \neq 0$

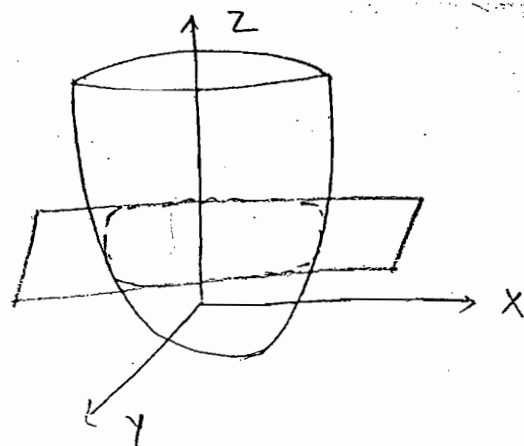
on z -axis $x, y = 0$

$(0, 0, -1) \times (0, 0, 1) = 0$

$z = 1, \quad x^2 + y^2 - 1 = 0$
 $y = \pm \sqrt{1-x^2}$

$\therefore (t, \sqrt{1-t^2}, 1)$

$(0, 1, 1)$ satisfy both eqn.



Formation of two surfaces from a curve $\rightarrow$

Suppose C is given parametrically.

$x = f_1(t), \quad y = f_2(t), \quad z = f_3(t)$

Satisfying $\left(\frac{df_1(t)}{dt}, \frac{df_2(t)}{dt}, \frac{df_3(t)}{dt} \right) \neq (0, 0, 0)$

Then near any point on C , This curve is represent by the intersection of two surfaces.

$F_1(x, y, z) = C_1$

$F_2(x, y, z) = C_2$

Solution of PDE of first order $\rightarrow$

Let $F(x, y, z, a, b) = 0$ — (1)

$F(x, y, z, p, q) = 0$ — (2)

Let $Pp + Qq = R$ — (3)

Sol. of (3) is a surface lying in (x, y, z) space and called an integral surface.

Let $z = f(x, y)$ — (A) is an integral surface of eqn. (3)

Normal to this (A) surface is given by $(P, Q, -1)$

$$P(P) + Q(Q) - (1)R = 0$$

$$[P, Q, R] \cdot (P, Q, -1) = 0$$

Any Integral surface must be tangential to a vector with component (P, Q, R) .

Integral surface passes through a given curve.

'OR' Cauchy problem for FODE of quasi linear.

To find Integral surface for Q.L. P.D.E.

$$Pp + Qq = R \text{ — (1)}$$

Passing through $(x_0(t), y_0(t), z_0(t))$ — (2)

a curve with t as a parameter.

$$\text{I)} \quad \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\text{we find } u(x, y, z) = C_1$$

$$v(x, y, z) = C_2$$

Put initial condition in eqn. (3)

$$\text{II)} \quad u[x_0(t), y_0(t), z_0(t)] = C_1 \text{ — (4)}$$

$$v[x_0(t), y_0(t), z_0(t)] = C_2 \text{ — (5)}$$

III) Eliminating ' t ' from (4) & (5)

$$\psi(C_1, C_2) = 0$$

$$\boxed{\psi(u, v) = 0} \rightarrow \text{Integral surface}$$

Sometimes solution can also be obtained by assuming $V = \ln(u)$ + determining $\ln$.

Q. $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u$

Passing through $x = t$, $y = 2t$, $u = 1$

Solⁿ

$\frac{dx}{dt} = \text{solve. ?}$

Q: $\rightarrow$ Find the Integral solⁿ of -

$$y p + x q = x^2 + y^2$$

passing through the curve

$$x = 1-t, \quad y = 1+t, \quad z = 1+t^2$$

(a) $z = xy + \frac{1}{2} (x^2 - y^2)^2$

(b) $z = xy + \frac{1}{4} (x^2 - y^2)^2$

(c) $\checkmark z = xy + \frac{1}{8} (x^2 - y^2)^2$

(d) $z = xy + \frac{1}{16} (x^2 - y^2)^2$

Solⁿ:-

$$\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{x^2 + y^2}$$

$$x dx - y dy = 0$$

$$u = \boxed{x^2 - y^2 = C_1}$$

$$\frac{y dx + x dy}{y^2 + x^2} = \frac{dz}{x^2 + y^2}$$

$$d(xy) = dz$$

$$\Rightarrow \boxed{v = xy - z = C_2}$$

$$\text{put } x = 1-t, \quad y = 1+t, \quad z = 1+t^2$$

$$u = (1-t)^2 - (1+t)^2$$

$$= -4t$$

$$c_1 = -4t \quad \text{--- (1)}$$

$$v = (1-t)(1+t) - (1+t^2)$$

$$= -2t^2$$

$$c_2 = -2t^2 \quad \text{--- (2)}$$

$$t = -c_1/4, \quad t = \left(-\frac{c_2}{2}\right)^{1/2}$$

$$\therefore \frac{c_1^2}{16} = \frac{c_2}{2} \Rightarrow \frac{c_1^2}{8} + c_2 = 0$$

$$\Rightarrow (xy - z) + \frac{1}{8} (x^2 - y^2)^2 = 0$$

$$\boxed{z = xy + \frac{1}{8} (x^2 - y^2)^2}$$

md
method.

$$xy - z = g(x^2 - y^2)$$

$$\Rightarrow (1-t)(1+t) - (1+t^2) = g[(1-t)^2 - (1+t)^2]$$

$$\Rightarrow -2t^2 = g(-4t)$$

$$\Rightarrow -2t^2 = g(-4t)$$

$$\left(\begin{array}{l} \because 4t = h \\ t = -h/4 \end{array} \right)$$

$$g(h) = -2\left(-\frac{h}{4}\right)^2$$

$$g(h) = -h^2/8$$

$$(xy - z) = -\frac{1}{8} (x^2 - y^2)^2$$

$$\boxed{z = xy + \frac{1}{8} (x^2 - y^2)^2}$$

OR

$$z = xy + A(x^2 + y^2)^2$$

$$1 + t^2 = 1 - t^2 + A(-4t)^2$$

$$\boxed{A = 1/8}$$

Exp.

$$x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z$$

Passing through —

$$x + y = 0, \quad z = 1$$

Solⁿ:

$$x = t, \quad y = -t, \quad z = 1$$

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(z^2 + x^2)} = \frac{dz}{(x^2 - y^2)z}$$

$$\Rightarrow x dx + y dy + z dz = 0$$

$$\Rightarrow u = \boxed{x^2 + y^2 - 2z = c_1} \quad \text{--- (1)}$$

Now $\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0$

$$\Rightarrow \log x + \log y + \log z = \log c_2$$

$$\therefore v = \boxed{xyz = c_2} \quad \text{--- (2)}$$

$$u = t^2 + (-t)^2 - 2$$

$$c_1 = 2(t^2 - 1) \quad \text{--- (3)}$$

$$v = t(-t) \cdot 1$$

$$c_2 = -t^2 \quad \text{--- (4)}$$

$$t^2 = \frac{c_1}{2} + 1, \quad t^2 = -c_2$$

$$\frac{c_1}{2} + 1 = -c_2$$

$$\Rightarrow \boxed{(x^2 + y^2 - 2z) + 2xyz + 2 = 0}$$

Q → Let $u(x, y)$ be a solⁿ of $xu_x + uy = 1$
 with $u(x, 0) = 2 \log x$, $x > 1$
 Then value at $u(e, 1) = ?$

- 1) 1
- 2) -1
- 3) 0
- 4) None of these.

Solⁿ: - $\frac{dx}{x} = \frac{dy}{1} = \frac{du}{1}$

$$\log x = y + c$$

$$x = C_1 e^y \Rightarrow \boxed{C_1 = x e^{-y}}$$

Now, $\frac{dy}{1} = \frac{du}{1}$

$$\Rightarrow y = u + c_2 \Rightarrow \boxed{C_2 = y - u}$$

Passing through $(t, 0, 2 \log t)$ $\frac{y-u}{x} = \phi(x e^{-y})$

$$u_1 = \frac{t}{e^0}$$

$$C_1 = t$$

$$(y-u) = \phi(x e^{-y})$$

$$(0 - 2 \log t) = \phi(t e^0)$$

$$\phi(t) = -2 \log t$$

$$\therefore (y-u) = -2 \log(x e^{-y})$$

$$y-u = -2 \log(x e^{-y})$$

$$u = y + 2 \log(x e^{-y})$$

at $(e, 1)$

$$u = 1 + 2 \log(e \cdot e^{-1})$$

$$\boxed{u = 1}$$

Q.

$$(2xy-1)p + (z-2x^2)q = 2(x-yz)$$

passing through $(1, 0, 1)$ Solⁿ

$$\frac{dx}{2xy-1} = \frac{dy}{z-2x^2} = \frac{dz}{2(x-yz)}$$

$$\Rightarrow x dx + y dy = \frac{dz}{-2}$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = \frac{-z}{2} + C_1$$

$$\therefore \boxed{C_1 = x^2 + y^2 + z}$$

$$z dx + dy + x dz = 0$$

$$zx + y + xz = C_2$$

$$\boxed{C_2 = 2xz + y}$$

$$u = (1)^2 + 0 + 1$$

$$C_1 = 1 + 1 \Rightarrow 1 = C_1 - 1$$

$$v = 2 \times 1 \times 1 + 0$$

$$C_2 = 2 \Rightarrow 2 = C_2 / 2 \Rightarrow 1 = C_2 / 2$$

$$\Rightarrow C_1 - 1 = \frac{C_2}{2}$$

$$\Rightarrow \boxed{2(x^2 + y^2 + z - 1) = 2xz + y}$$

Q:-

$$x^3 p + y(3x^2 + y)q = z(2x^2 + y)$$

passing through $(1, 1, 1+1)$ Solⁿ:-

$$\frac{dx}{x^3} = \frac{dy}{y(3x^2 + y)} = \frac{dz}{z(2x^2 + y)}$$

$$\Rightarrow 3 dy = y dz$$

$$\Rightarrow 3y = yz + C_1 \Rightarrow \boxed{C_1 = 0}$$

30/ Aug / 2014

Existence And uniqueness of Integral surface
passes through given curve: $\rightarrow$

Let P, Q, R are function of class C^1 in domain Ω of $\mathbb{R}^3$.

$$\text{And } Pp + Qq = R \quad \text{--- (1)}$$

If at any point $(x_0, y_0, z_0) \in \Omega$

$$\text{if } \left[P(x_0, y_0, z_0) \frac{dy_0(t)}{dt} - Q(x_0, y_0, z_0) \frac{dx_0(t)}{dt} \right] \neq 0 \quad \text{--- (2)}$$

then $\exists$ a unique solⁿ ^① in the nbd of x_0, y_0, z_0
it will satisfied the initial curve,

$$x = x_0(t), \quad y = y_0(t), \quad z = z_0(t)$$

Geometrically $\text{Eau. (2)} \Rightarrow$ The projection of vector $v(P, Q, R)$ at (x_0, y_0, z_0) on x - y plane is
not tangent to a curve in x - y plane at (x_0, y_0)

Q:-

Find $\rightarrow$

$$ux - uy = 2 \text{ passes through } (2, 1, 2)$$

Then no. of solⁿ of this Eauⁿ -

Sol.

$$\begin{vmatrix} P & Q \\ \frac{dx_0}{dt} & \frac{dy_0}{dt} \end{vmatrix} \neq 0$$

$$P = 1, \quad Q = -1, \quad R = 2$$

$$\frac{dx_0}{ds} = 2, \quad \frac{dy_0}{ds} = 1, \quad \frac{dz_0}{ds} = 2$$

$$= \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 3 \neq 0$$

This curve has unique solⁿ.

$$\frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{-2}$$

$$\Rightarrow x + y = c_1 = u$$

$$\Rightarrow 2y + z = c_2 = v$$

Passing through $(2\Delta, \Delta, 2\Delta)$

$$u = 2\Delta + \Delta = 3\Delta = c_1$$

$$v = 2\Delta + 2\Delta = 4\Delta = c_2$$

$$\Delta = \frac{c_1}{3}, \quad \Delta = \frac{c_2}{4}$$

$$\Rightarrow \frac{c_1}{3} = \frac{c_2}{4}$$

$$\Rightarrow \boxed{4(x+y) = 3(2y+z)} \rightarrow \text{unique sol}^n$$

'OR'

$$(2y+z) = f(x+y)$$

$$2\Delta + 2\Delta = f(3\Delta)$$

$$f(3\Delta) = 4\Delta$$

$$3\Delta = h \Rightarrow \Delta = h/3$$

$$\boxed{f(h) = 4h/3}$$

#

Non Existence & Non uniqueness : $\rightarrow$

$$\frac{\frac{dx_0(t)}{dt}}{P(x_0, y_0, z_0)} = \frac{\frac{dy_0(t)}{dt}}{Q(x_0, y_0, z_0)} = u$$

$$\Rightarrow \frac{\frac{dz_0(t)}{dt}}{R(x_0, y_0, z_0)} \neq u$$

Then there is no solⁿ in nbd of (x_0, y_0, z_0)

Q:

Q.

$$u_x + u_y = u$$

$$u(s, s) = \sin s$$

$$0 \leq s \leq 1$$

Solⁿ

$$\begin{vmatrix} P & Q \\ \frac{dx_0}{ds} & \frac{dy_0}{ds} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

we can't say anything about uniqueness.

$$\frac{\frac{dx_0}{ds}}{P} = \frac{1}{1} = \frac{\frac{dy_0}{ds}}{Q}$$

$$\Rightarrow \frac{\frac{dz_0}{ds}}{R} = \frac{\cos s}{\sin s} = \cot s \neq 1$$

$$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{u}$$

$$\Rightarrow \boxed{x - y = c_1}$$

$$\Rightarrow x = \log u + \log c_2$$

$$\boxed{c_2 = u e^{-x}}$$

$$\therefore c_1 = s - s = 0 \quad \therefore c_1 = 0$$

$$c_2 = u e^{-s} \Rightarrow c_2 = \sin s \cdot e^{-s}$$

$$\sin s \cdot e^{-s} = 0$$

$$\sin s \cdot e^{-s} = k$$

$$\boxed{\sin s = k e^s}$$

They are L.D.

which is no solution.

#

Infinite Solution : $\rightarrow$

$$\text{If } \frac{\frac{dx_0}{ds}}{P} = \frac{\frac{dy_0}{ds}}{Q} = \frac{\frac{dz_0}{ds}}{R} = u$$

Then given pde have infinite solⁿ.
i.e. There will be infinite integral surface passes through

Q:

$$4x - 4y = 2$$

$$\Gamma: (1, -1, 2)$$

$$\begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} = -1 + 1 = 0$$

$$\frac{1}{1} = \frac{-1}{-1} = \frac{2}{2} = 1$$

$$\frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{2}$$

$$\Rightarrow \boxed{x + y = c_1}$$

$$\boxed{2y + z = c_2}$$

$$c_1 = 1 - 1 = 0$$

$$\therefore \boxed{c_1 = 0}$$

$$c_2 = -2 + 2 = 0$$

$$\therefore \boxed{c_2 = 0}$$

$$0 = 0$$

So Infinite solⁿ.

#

Surface orthogonal to given surface $\rightarrow$

Let $f(x, y, z) = c$ be a given family of surfaces:

$$\text{grad } f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

Let $z = z(x, y)$ is a surface — (2)
which is orthogonal to surface (1).

Normal to equ. (2) is $(p, q, -1)$ — (3)

$$p \frac{\partial f}{\partial x} + q \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z}$$

$$\text{i.e. } p^2 + q^2 = R$$

$$\boxed{\frac{dx}{\partial f / \partial x} = \frac{dy}{\partial f / \partial y} = \frac{dz}{\partial f / \partial z}}$$

Q. find the surface which is orthogonal to
 $(x+y)z = c(3z+1)$

passing through $x^2+y^2=1$, $z=1$

Solⁿ:-

$$\frac{(x+y)z}{(3z+1)} = c$$

$$F = \frac{(x+y)z}{(3z+1)}$$

$$\frac{dx}{f_x} = \frac{dy}{f_y} = \frac{dz}{f_z}$$

$$\Rightarrow f_x = \frac{z}{3z+1}, \quad f_y = \frac{z}{3z+1}$$

$$f_z = \frac{(3z+1) \cdot (x+y) - (x+y)z \cdot 3}{(3z+1)^2} = \frac{(x+y)}{(3z+1)^2}$$

$$\Rightarrow \frac{\frac{dx}{z}}{\frac{z}{3z+1}} = \frac{\frac{dy}{z}}{\frac{z}{3z+1}} = \frac{\frac{dz}{x+y}}{\frac{x+y}{(3z+1)^2}}$$

$$\Rightarrow \boxed{x-y=c_1}$$

$$\begin{aligned} x dx + y dy &= z(3z+1) dz \\ &= \frac{x^2}{2} + \frac{y^2}{2} = \frac{3z^3}{3} + \frac{z^2}{2} + c_2 \end{aligned}$$

$$\therefore \boxed{c_2 = x^2 + y^2 - 2z^3 - z^2}$$

$$c_1 = x^2 + y^2 = 1$$

$$z = 1, \quad x = t, \quad y = \sqrt{1-t^2}$$

$$(x^2 + y^2 - 2z^3 - z^2) = \phi(x-y)$$

$$1 - 2 - 1 = \phi(x-y)$$

$$\phi(x-y) = -2$$

$$\boxed{x^2 + y^2 - 2z^3 - z^2 = -2}$$

Q:- $z = cxy (x^2 + y^2)$

Passes through $x^2 - y^2 = a^2, z = 0$

$$f = \frac{z}{xy(x^2 + y^2)}$$

$$f_x = \frac{xy(x^2 + y^2) \cdot 0 - [y(x^2 + y^2) + xy \cdot 2x]}{xy(x^2 + y^2)^2}$$

$$= \frac{-y - x(x^2 + y^2)(xy - y) - 2x^2y}{xy(x^2 + y^2)^2}$$

$$3x^3y - x^2y + xy^3 - y^3 - 2x^2y$$

$$= -3x^2y + xy^3 - y^3 + x^3y$$

$$c' = f = \frac{xy(x^2 + y^2)}{z} \quad f_x = \frac{y(x^2 + y^2) + xy(2x)}{z^2}$$

$$f_y = \frac{x(x^2 + 3y^2)}{z}, \quad f_z = -\frac{(x^2 + y^2)(xy)}{z^2}$$

$$dx = \frac{dy}{\frac{y(3x^2 + y^2)}{z} - \frac{x(x^2 + 3y^2)}{z} - \frac{(x^2 + y^2)(xy)}{z^2}}$$

$$3x^3y + y^3x^2 - x^3y^2 - 3xy^4$$

$$xy(x^2 - y^2)$$

$$= xy(x^2 - y^2)(x^2 + y^2)$$

Compatible System of Equation : $\rightarrow$
 Let two p.d.e's $\quad \swarrow$ (They have one parameter)

$$f(x, y, z, p, q) = 0 \quad \text{--- (1)}$$

$$g(x, y, z, p, q) = 0 \quad \text{--- (2)}$$

Equ. (1) & (2) are compatible in domain D .

$$\text{if (i) } \frac{\partial(f, g)}{\partial(p, q)} \neq 0 \quad \text{on } D \quad \text{--- (3)}$$

$$(ii) \quad dz = p dx + q dy \text{ is Integrable. --- (4)}$$

where p, q are given by solving (3) & (4)

$$p = \phi(x, y, z)$$

$$q = \psi(x, y, z)$$

The Necessary & Sufficient condition for $dz = p dx + q dy$ to be integral if -

$$[f, g] = \frac{\partial(f, g)}{\partial(x, p)} + p \frac{\partial(f, g)}{\partial(z, p)} + \frac{\partial(f, g)}{\partial(y, q)} + q \frac{\partial(f, g)}{\partial(z, q)} = 0$$

If equ. (1) & (2) are compatible they have common solution of one parameter family of surfaces.

Every solⁿ of equ. (1) is not a solⁿ of equ. (2) & vice versa.

Common solⁿ comes from equ. (4)

Q. Show that $p^2 + q^2 = 1$
 $+ (p^2 + q^2)x = pz$

two p.d.e's are compatible.

Solⁿ:- $f = p^2 + q^2 - 1 = 0$
 $g = (p^2 + q^2)x - pz = 0$

$$= \begin{vmatrix} f_p & f_q \\ g_p & g_q \end{vmatrix} = \begin{vmatrix} 2p & 2q \\ 2px-z & 2qx \end{vmatrix} = 2qz \neq 0$$

$$[f, g] = 0$$

$$\frac{\partial(f, g)}{\partial(x, p)} = \begin{vmatrix} 0 & 2p \\ p^2+q^2 & 2px-z \end{vmatrix} = -(2p^3+2pq^2)$$

$$\frac{\partial(f, g)}{\partial(z, p)} = \begin{vmatrix} 0 & 2p \\ -p & 2px-z \end{vmatrix} = 2p^2$$

$$\frac{\partial(f, g)}{\partial(y, q)} = \begin{vmatrix} 0 & 2q \\ 0 & 2qx \end{vmatrix} = 0$$

$$\frac{\partial(f, g)}{\partial(z, q)} = \begin{vmatrix} 0 & 2q \\ -p & 2qx \end{vmatrix} = 2pq$$

$$[f, g] = -2p^3 - 2pq^2 + 2p^3 + 2pq^2 = 0$$

$$dz = p dx + q dy$$

$$p^2 + q^2 = 1$$

$$(p^2 + q^2)x = pz$$

$$x = pz \quad \therefore p = x/z$$

$$q = \sqrt{1-p^2} \Rightarrow \pm \frac{\sqrt{z^2-x^2}}{z}$$

$$\therefore dz = \frac{x}{z} dx \pm \frac{\sqrt{z^2-x^2}}{z} dy$$

$$\Rightarrow z dz = x dx \pm \sqrt{z^2-x^2} dy \quad \left[\int f^n(x) \cdot f'(x) dx = \frac{f^{n+1}(x)}{n+1} \right]$$

$$\Rightarrow \frac{z dz - x dx}{\sqrt{z^2-x^2}} = dy$$

$$\frac{d(z^2-x^2)}{2\sqrt{z^2-x^2}} = dy \quad \Rightarrow \pm \sqrt{z^2-x^2} = y + c$$

$$\boxed{z^2-x^2 = (y+c)^2} \quad \text{Ans.}$$

31 | Aug | 2014

Q:- Show that following are compatibility

$$P = 5x - 7y$$

$$Q = -7x + 8y$$

Sol:-

$$f = p - 5x + 7y$$

$$g = q + 7x - 8y$$

$$\Rightarrow \begin{vmatrix} f_p & f_q \\ g_p & g_q \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0$$

$$dz = Pdx + Qdy$$

$$dz = \underbrace{(5x - 7y)}_M dx + \underbrace{(-7x + 8y)}_N dy$$

$$M_y = N_x = -7$$

This is a exact Eqn.

(P & Q already
solved
[f, g] don't
show)

$$Z = \int M dx + \int N dy$$

$$= \int (5x - 7y) dx + \int 8y dy$$

$$Z = \left[\frac{5x^2}{2} - 7xy + 4y^2 \right] + C$$

NOTE:- If $P = P(x, y)$ & $Q = Q(x, y)$ Then given pde's
are compatibility iff -

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

$$\# \quad xp - yq = x$$

$$+ \quad x^2p + q = xz$$

Ans:- $z - x = c(1 + xy)$

$$\# \quad p = (x+y)^2$$

$$q = x^2 + 2xy - y^2$$

Ans:- $z = \frac{x^3}{3} + xy^2 + x^2y - \frac{y^3}{3} + c$

Characteristics Curve of Semi Linear Equⁿ :-

$$P(x, y) p + Q(x, y) q = R(x, y, z)$$

Solⁿ is -

$$\frac{dx}{P(x, y)} = \frac{dy}{Q(x, y)}$$

'OR'

$$\boxed{\frac{dy}{dx} = \frac{Q(x, y)}{P(x, y)}}$$

Is called characteristics Curve.

Gate

Q:-

The characteristics curve of given Equⁿ is -

$$x \frac{du}{dy} - y \frac{du}{dx} = u$$

(a) Straight line with slope 1.

(b) " " " " = -1

(c) Circle with centre at origin.

(d) Circle touching on y axis & centred at x axis.

Sol:-

$$\frac{dx}{-y} = \frac{dy}{x} \Rightarrow x^2 + y^2 = c_1$$

IInd :- if $u(x, y)$ is solⁿ of above Equⁿ with $u(x, 0) = \sin \frac{\pi}{4} x$, Then $u\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = ?$

$$(1) \quad \frac{1}{2} e^{\pi/4}$$

$$(2) \quad \frac{1}{\sqrt{2}} e^{\pi/4}$$

$$(3) \quad -\frac{1}{\sqrt{2}} e^{\pi/4}$$

$$(4) \quad \frac{1}{\sqrt{2}} e^{-\pi/4}$$

Solⁿ:

$$\frac{dx}{-y} = \frac{dy}{x} = \frac{du}{u}$$

$$\Rightarrow x^2 + y^2 = c_1$$

$$4 \quad \frac{y}{x} = \log u + c_2$$

$$\therefore c_2 = u e^{-y/x}$$

$$u(x, 0) = \sin \frac{\pi}{4} x$$

$$\frac{dx}{-y} = \frac{du}{u}$$

$$\Rightarrow \frac{du}{u} = \frac{-dx}{\sqrt{a^2 - x^2}} \Rightarrow \log u = -\sin^{-1} \left(\frac{x}{c_1} \right) + \log c_2$$

$$\frac{u}{c_2} = e^{\sin^{-1} \left(\frac{x}{c_1} \right)}$$

$$c_2 = u e^{\sin^{-1} \left(\frac{x}{\sqrt{x^2 + y^2}} \right)}$$

$$\Rightarrow u e^{\sin^{-1} \left(\frac{x}{\sqrt{x^2 + y^2}} \right)} = f(x^2 + y^2)$$

Given $u(x, 0) = \sin \frac{\pi}{4} x$

$$\sin \frac{\pi}{4} x \cdot e^{\sin^{-1} (1)} = f(x^2)$$

$$= e^{\pi/4} \cdot \sin \frac{\pi}{4} x = f(x^2)$$

$$\Rightarrow e^{\pi/4} \cdot \sin \frac{\pi}{4} \sqrt{t} = f(t)$$

$$x^2 = t$$

$$\text{and } x = \sqrt{t}$$

$$\Rightarrow u e^{\sin^{-1}\left(\frac{x}{\sqrt{x^2+y^2}}\right)} = e^{\pi/2} \cdot \sin \frac{\pi}{4} \cdot \sqrt{x^2+y^2}$$

$$\text{find } u\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = ?$$

$$u e^{\sin^{-1}(1/\sqrt{2})} = e^{\pi/2} \cdot \sin \frac{\pi}{4}$$

$$\left(\because \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4} \right)$$

$$u = e^{\pi/2} \cdot \sin \frac{\pi}{4} \cdot e^{-\pi/4}$$

$$\boxed{u = \frac{1}{\sqrt{2}} e^{\pi/2}}$$

Ans.

NET

Q :-

The c.e. of p.d.e. —

$$u_{xx} + u_y = u$$

(i) Exponential curve along which u remain constant

(ii) " " " " " u varies along with linearly with x .

(iii) A straight line along with u remain const.

(iv) " " " " " u varies exponentially with x ✓

Solⁿ:-

$$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{u}$$

$$\boxed{x - y = c_1} \rightarrow \text{Straight line.}$$

$$\frac{dx}{1} = \frac{du}{u}$$

$$x = \log u + \log c_2$$

$$u c_2 = e^x \quad \therefore \boxed{u = \frac{1}{c_2} e^x}$$

NET.

Q.

The c.e. curve of p.d.e

$$2y u_x + (2x + y^2) u_y = 0 \text{ passing through}$$

(0, 0) —

~~(a)~~ $y^2 = 2(e^x - x - 1)$

(b) $y^2 = 2(e^x - x + 1)$

(c) $y^2 = 2(e^x + x + 1)$

(d) $y^2 = 2(e^x + x + 1)$

Solⁿ:- put (0, 0) in options

(b) $y^2 = 2(e^x - x + 1)$

$0 = 2(e^0 - 0 + 1) = 0 \neq 2$

(d) $y^2 = 2(e^x + x + 1)$

$0 = 2(1 + 0 + 1) = 0 \neq 4$

Let $y^2 = 2(e^x + ax - 1)$ is Solⁿ

$$\frac{dx}{2y} = \frac{dy}{(2x + y^2)}$$

$$2y dy = 2(e^x + a) dx$$

$\Rightarrow (2x + y^2) dx = 2y dy$

$= [2x + 2(e^x + ax - 1)] \frac{dx}{2} = 2(e^x + ax - 1)$

$2x + 2e^x + 2ax - 2 = 2e^x + 2a$

$2(x - 1) = 2a(1 - x)$

$a = -1$

'OR'

$$\frac{dx}{2y} = \frac{dy}{2x + y^2}$$

$$2y \frac{dy}{dx} - y^2 = 2x$$

Solve it by Bernoulli's $\rightarrow$

Types Of Integral : $\rightarrow$

(1) Complete Integral (Solⁿ) : $\rightarrow$ Let $F(x, y, z, p, q) = 0$ (1)
be a given first order PDE Then -

$$F(x, y, z, a, b) = 0 \quad \text{--- (2)}$$

Is a complete solⁿ of eqn. (1)

(i) If eqn. (2) satisfy eqn. (1)

(ii) $M = \begin{bmatrix} F_a & F_{xa} & F_{ya} \\ F_b & F_{xb} & F_{yb} \end{bmatrix}$ Rank is 2

(2) General Integral (Solⁿ) : $\rightarrow$

i) Linear : - General sol of the Equation

$$F(u, v) = 0$$

where u, v are functionally indep. solⁿ of

$$\boxed{\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}}$$

ii) Non Linear : $\rightarrow$ general solⁿ of first order non linear
Eqn -

$$f(x, y, z, p, q) = 0 \quad \text{--- (1)}$$

Let $z = f(x, y, a, b)$ is a complete solⁿ of (1)

Choose $b = \phi(a)$ or $a = \psi(b)$ --- *

where ϕ & ψ are arb. func.

$$z = F[x, y, a, \phi(a)]$$

or $G(x, y, a, \phi(a)) = 0$ (one parameter family of surfaces)
(3)

The Envelope eqn. (3) is general solⁿ of eqn. (1)

P.D. Equ. (3) with r. to 'a'.

$$\frac{\partial u}{\partial a} = 0 \quad \text{--- (4)}$$

Eliminating 'a' using equ. (3) & (4).

That will give the general solⁿ of Eqⁿ (1)

Note:-

General Integral of Non linear first order P.D.E. is envelope of one parameter family of surfaces.

Particular Integral : $\rightarrow$ If we are given that with particular relⁿ b/w a & b then the envelope obtain using equ. (3) & (4) will be a p.I.

Singular Integral : $\rightarrow$ Singular solⁿ is a envelope of two parameter family of surfaces.

& let $u(x, y, z, a, b) = 0$ be a complete solⁿ.
--- (5)

P.D. w.r. to a equ. (5)

$$\frac{\partial u}{\partial a} = 0 \quad \text{--- (6)}$$

P.D. w.r. to b

$$\frac{\partial u}{\partial b} = 0 \quad \text{--- (7)}$$

Using equ. (5), (6), (7) eliminate a & b

we will get a singular solⁿ of F.O. P.D.E.
(Non-linear).

Q:- Let $z = ax + by + a^2 + b^2$ — (1) is a solⁿ of

$$z = px + qy + p^2 + q^2 \text{ — (2)}$$

Prove equ. (1) is complete solⁿ of equ. (2)

+ find P.I. + Singular Integral where $a^2 + b^2 = 1$

Sol:-

$$F = z - ax - by + a^2 - b^2$$

$$\Rightarrow \begin{bmatrix} x+2a & 0 & 1 & 0 \\ y+2b & 0 & 0 & 1 \end{bmatrix} \quad \text{Rank} = 2$$

So it is a complete Integral.

Particular Int. : $\rightarrow$

$$z = ax + \sqrt{1-a^2}y + a^2 + (1-a^2)$$

$$\boxed{z = ax + \sqrt{1-a^2}y + 1} \text{ — (3)}$$

P.D. w.r. to (a)

$$0 = x - \frac{a}{\sqrt{1-a^2}} y$$

$$x = \frac{ay}{\sqrt{1-a^2}} \Rightarrow \sqrt{1-a^2} = \frac{ay}{x} \text{ — (5)}$$

$$z = ax + \frac{ay}{x} \cdot y + 1$$

$$a(x^2 + y^2) = (z-1)x$$

$$a = \frac{(z-1)x}{(x^2 + y^2)}$$

by equ. (4)

$$x = \frac{(z-1)x y}{(x^2 + y^2)(\sqrt{1-a^2})}$$

$$\therefore \boxed{(z-1)^2 = x^2 + y^2}$$

$$f = z - ax - by - a^2 - b^2 = 0$$

$$\frac{\partial f}{\partial a} = -x - 2a = 0 \quad \therefore \boxed{a = -x/2}$$

$$\frac{\partial f}{\partial b} = -y - 2b = 0 \quad \therefore \boxed{b = -y/2}$$

$$\therefore z = \left(-\frac{x^2}{2} - \frac{y^2}{2} + \frac{x^2}{4} + \frac{y^2}{4} \right) = -\left(\frac{x^2}{4} + \frac{y^2}{4} \right)$$

Solution of 1st order Non linear p.d.e $\rightarrow$

(1) Charpit Method $\rightarrow$ Let $f(x, y, z, p, q) = 0$ $\vdash$ (1)
be a given. p.d.e.

Charpit method consist in choosing one parameter family of p.d.e of order 1. s.t. every member of that family is compatible to given p.d.e.

Let $g(x, y, z, p, q, a) = 0$ $\vdash$ (2) be a one parameter of family.

s.t. equ. (2) is compatible with equ. (1) $\forall a$.

Since (1) & (2) are compatible then $\vdash$

$$[f, g] = 0 \quad \forall a$$

$$\frac{\partial(f, g)}{\partial(x, p)} + p \frac{\partial(f, g)}{\partial(z, p)} + \frac{\partial(f, g)}{\partial(y, q)} + q \frac{\partial(f, g)}{\partial(z, q)} = 0$$

Solve this and collect the terms of g_x, g_y, g_z, g_p, g_q .

$$\Rightarrow \frac{\partial f}{\partial x} - f_p g_x - f_q g_y + [-p f_p - q f_q] g_z + (f_x + p f_z) g_p + (f_y + q f_z) g_q = 0$$

This is linear in 'g' with Inde. var. as x, y, z, p .

By Lagrangian Equⁿ -

$$\frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dz}{-pf_p - qf_q} = \frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z}$$

This is called Charpit Equⁿ.

Solve Equ. (4) in a simplex way containing p & q or both and one arbitrary constant.

$$g(x, y, z, p, q, a) = 0$$

Since equ. (1) & (2) are compatible.

$$p = \phi(x, y, z, a)$$

$$q = \psi(x, y, z, a)$$

$$dz = p dx + q dy$$

$$dz = \phi(x, y, z, a) dx + \psi(x, y, z, a) dy$$

$$z = f(x, y, a, b)$$

Is called complete solⁿ.

14 Sep 2014

Q. $(p^2 + q^2)y - qz = 0$ — (1)

Sol. $\frac{dx}{2py} = \frac{dy}{2qy - z} = \frac{dz}{[2p^2y + 2q^2y - qz]} = \frac{dp}{[0 + p(-q)]}$

$$= \frac{dq}{-[(p^2 + q^2) + q(-q)]}$$

$$\Rightarrow \frac{dp}{pq} = \frac{dq}{-p^2}$$

$$\Rightarrow -p dp - q dq = 0$$

$$\Rightarrow \boxed{p^2 + q^2 = a^2} \text{ — (3)}$$

by (1) + (3)

$$a^2 y - qz = 0$$

$$\Rightarrow q = \frac{a^2 y}{z} \quad \text{--- (4)}$$

$$\begin{aligned} \therefore p^2 &= a^2 - q^2 \\ &= a^2 \left[1 - \frac{a^2 y^2}{z^2} \right] \end{aligned}$$

$$p^2 = \frac{a^2}{z^2} [z^2 - a^2 y^2]$$

$$\therefore p = \pm \frac{a}{z} \sqrt{z^2 - a^2 y^2} \quad \text{--- (5)}$$

$$dz = p dx + q dy$$

$$dz = \pm \frac{a}{z} \sqrt{z^2 - a^2 y^2} + \frac{a^2 y}{z} dy$$

$$\frac{z dz - a^2 y dy}{\sqrt{z^2 - a^2 y^2}} = \pm a dx$$

Put $z^2 - a^2 y^2 = t$
 $2z dz - 2a^2 y dy = dt$

$$= \frac{1}{2\sqrt{t}} dt = \pm a dx$$

$$\Rightarrow \frac{1}{2} \frac{t^{1/2}}{1/2} = \pm ax + b$$

$$\Rightarrow \sqrt{z^2 - a^2 y^2} = \pm ax + b$$

Squaring on both sides

$$\boxed{z^2 = a^2 y^2 + (ax + b)^2}$$

Q.

$$2(xp + yq + z) = yp^2$$

Sol.

$$\begin{aligned} \frac{dx}{2x - 2yp} &= \frac{dy}{2y} = \frac{dz}{- [2px - 2yp^2 + 2yq]} = \frac{dp}{- [2p + 2p]} \\ &= \frac{dq}{- [2q + 2q] - p^2} \end{aligned}$$

$$\Rightarrow \frac{dy}{2y} = \frac{dp}{-4p}$$

$$= 2 \log y + \log p = \log a$$

$$\boxed{p = a/y^2}$$

by ①

$$\begin{aligned} \frac{2xa}{y^2} + 2yq + 2z &= \frac{ya^2}{y^4} \\ &= \frac{2xa}{y^2} + 2z - \frac{a^2}{y^3} = -2yq \\ &= - \left[\frac{xa}{y^3} + \frac{z}{y} - \frac{a^2}{2y^4} \right] = q \end{aligned}$$

$$dz = p dx + q dy$$

$$dz = \frac{a}{y^2} dx + \left[\frac{a^2}{2y^4} - \frac{z}{y} - \frac{xa}{y^3} \right] dy$$

$$dz = a dx + \left[\frac{a^2}{2y^2} - zy - \frac{ax}{y} \right] dy$$

$$= ax - \frac{a^2}{4y^3} - \frac{zy^2}{2} + \frac{ax}{y^2} + b$$

$$z = ax - \frac{a^2}{2y} - \frac{zy^2}{2} - ax \log y + b$$

$$y dz = \frac{a}{y} dx + \left(\frac{a^2}{2y^3} - z - \frac{ax}{y^2} \right) dy$$

$$y dz + z dy = a \left[\frac{y dx - x dy}{y^2} \right] + \frac{a^2}{2y^3} dy$$

$$= d(yz) - a d\left(\frac{x}{y}\right) - \frac{a^2}{2y^3} dy = 0$$

$$\Rightarrow \boxed{yz - \frac{ax}{y} + \frac{a^2}{4y^2} + b = 0} \quad \text{Ans.}$$

Special Cases : $\rightarrow$

1) $f(p, q) = 0$ or it reducible in this form.

$$f_x = f_y = f_z = 0$$

$$\frac{dp}{0} = \frac{dq}{0}$$

$$p = a \text{ or } q = a$$

If $p = a$ — (3)

$f(a, q) = 0$, $q = Q(a) \rightarrow$ Some func.

$$dz = p dx + q dy$$

$$= a dx + Q(a) dy$$

$$\boxed{z = ax + Q(a)y + b} \rightarrow \text{Complete Integral}$$

$$z = ax + by + c$$

where $f(a, b) = 0$

$\Rightarrow$ In this form Singular solⁿ does not exist

$$\left(\because \frac{\partial z}{\partial b} \neq 0 = 1 \right)$$

Q.

$$f(x, y, z, p, q) = p + q - pq = 0$$

$$z = ax + by + c$$

where $a + b - ab = 0$

$$a + b(1-a) = 0$$

$$b = a / (a-1)$$

$$\boxed{z = ax + \left(\frac{a}{a-1} \right) y + c}$$

or

$$a = b / (b-1)$$

$$\boxed{z = \left(\frac{b}{b-1} \right) x + by + c}$$

Imp.
2)

Reducable to $f(p, q)$ form:-

Exp.

$$p^2 x + q^2 y = z$$

$$\Rightarrow x \left(\frac{\partial z}{\partial x} \right)^2 + y \left(\frac{\partial z}{\partial y} \right)^2 = z$$

$$\Rightarrow \text{Divide} \left(\frac{\sqrt{x}}{\sqrt{z}} \frac{\partial z}{\partial x} \right)^2 + \left(\frac{\sqrt{y}}{\sqrt{z}} \frac{\partial z}{\partial y} \right)^2 = 1$$

$$= \left(\frac{\partial z / \sqrt{z}}{\partial x / \sqrt{x}} \right)^2 + \left(\frac{\partial z / \sqrt{z}}{\partial y / \sqrt{y}} \right)^2 = 1$$

$$\frac{\partial x}{\partial \sqrt{x}} = \partial x, \quad \frac{\partial y}{\partial \sqrt{y}} = \partial y, \quad \frac{\partial z}{\partial \sqrt{z}} = \partial z$$

$$= \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = 1 \quad x =$$

$$\Rightarrow p^2 + q^2 = 1 \quad \Rightarrow a^2 + b^2 = 1$$

$$z = ax + by + c$$

$$z = ax + \sqrt{1-a^2}y + c$$

(Integrate x, y, z then find x, y, z)

$$x = 2\sqrt{x}, \quad y = 2\sqrt{y}, \quad z = 2\sqrt{z}$$

$$2\sqrt{z} = 2a\sqrt{x} + \sqrt{1-a^2} \cdot 2\sqrt{y} + c$$

$$\Rightarrow \boxed{\sqrt{z} = a\sqrt{x} + \sqrt{1-a^2} \cdot \sqrt{y} + c}$$

Q:-

$$p^2 x^2 + q^2 y^2 = z^2$$

Sol:-

$$x^2 \left(\frac{\partial z}{\partial x} \right)^2 + y^2 \left(\frac{\partial z}{\partial y} \right)^2 = z^2$$

$$\Rightarrow \left(\frac{x}{z} \frac{\partial z}{\partial x} \right)^2 + \left(\frac{y}{z} \frac{\partial z}{\partial y} \right)^2 = 1$$

$$\Rightarrow \frac{\partial x}{\partial x} = \partial x, \quad \frac{\partial y}{\partial y} = \partial y, \quad \frac{\partial z}{\partial z} = \partial z$$

$$= \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = 1$$

$$= p^2 + q^2 = 1$$

$$z = ax + \sqrt{1-a^2}y + c$$

$$x = \log x, \quad y = \log y$$

$$\sqrt{z} = \log z = a \log x + \sqrt{1-a^2} \log y + c$$

$$\boxed{z = x^a \cdot y^{\sqrt{1-a^2}} \cdot c}$$

Ans.

Q.

$$x^2 p^2 + q^2 = 1$$

$$\left(x \frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = 1$$

$$\frac{\partial x}{\partial x} = \partial x, \quad \frac{\partial y}{\partial y} = \partial y, \quad \partial z = \partial z$$

Case-II: - $f(p, q, z) = 0$

$$\frac{dp}{-pf_z} = \frac{dq}{-qf_z}$$

$$\Rightarrow \frac{dp}{p} = \frac{dq}{q} \Rightarrow p = aq \text{ or } q = ap$$

$$q = ap$$

$$dz = p dx + q dy$$

$$dz = p dx + ap dy$$

$$dz = p [d(x + ay)]$$

$$u = x + ay$$

$$dz = p du \Rightarrow p = \frac{dz}{du}$$

$$q = a \frac{dz}{du}$$

$$f\left(\frac{dz}{du}, a \frac{dz}{du}, z\right) = 0$$

$$f(p, q, z) = 0$$

↓

$$f\left(\frac{\partial z}{\partial u}, a \frac{\partial z}{\partial u}, z\right) = 0 \quad \text{where } u = x + ay \text{ or } ax + y$$

'OR'

$$f(p, q, z) = 0$$

$$p = aq$$

$$f(aq, q, z) = 0$$

$$q = q(a, z)$$

$$dz = p dx + q dy$$

$$= aq dx + q dy$$

$$= Q(a, z) [a dx + dy]$$

$$\Rightarrow \int \frac{dz}{Q(a, z)} = ax + y + b$$

Q.

$$z = b^2 + q^2, \quad f = z - b^2 - q^2$$

$$\frac{dz}{du} = \frac{dz}{du}, \quad q = a \frac{dz}{du}$$

$$u = x + ay$$

$$z - \left(\frac{dz}{du} \right)^2 - \left(a \frac{dz}{du} \right)^2 = 0$$

$$\Rightarrow z - \left(\frac{dz}{du} \right)^2 [1 + a^2] = 0$$

$$\Rightarrow \left(\frac{dz}{du} \right)^2 = \frac{z}{1 + a^2}$$

$$\Rightarrow \frac{dz}{du} = \pm \sqrt{\frac{z}{1 + a^2}}$$

$$\Rightarrow \int \frac{dz}{\sqrt{z}} = \pm \int \frac{du}{\sqrt{1 + a^2}}$$

$$= 2\sqrt{z} = \pm u / \sqrt{1 + a^2} + b$$

$$\Rightarrow 4z(1 + a^2) = (x + ay + c)^2 \quad \text{--- (1) Find singular sol}^n.$$

D.w.r. to a

$$\Rightarrow 2a(4z) - 2(x + ay + c) \cdot y$$

$$\Rightarrow 8az = 2(x + ay + c) \cdot y \quad \text{--- (2)}$$

D.w.r. to c

$$0 = 2(x + ay + c) \quad \text{--- (3)}$$

by eqn. (2) + (3)

$$\boxed{z = 0}$$

So it is singular solⁿ.

$$\left[\begin{array}{l} \frac{z^2}{z^2} + \left(z \frac{\partial z}{\partial x} \right)^2 + \left(z \frac{\partial z}{\partial y} \right)^2 = 0 \\ \swarrow \text{it is not reducible.} \\ \text{Not } \frac{\partial z}{\partial x} \\ \text{--- (4)} \end{array} \right.$$

Case-3:- when the given equⁿ does not contain z and can be written in this form -

$$f_1(p, x) = f_2(q, y)$$

$$\frac{dx}{\partial f_1 / \partial p} = \frac{dy}{\partial f_2 / \partial q} = \frac{dz}{p \partial f_1 / \partial p - q \partial f_2 / \partial q} = \frac{-dp}{-\partial f_1 / \partial x}$$

$$= \frac{-dq}{-\partial f_2 / \partial y}$$

$$\Rightarrow \frac{dx}{[\partial f_1 / \partial p]} = \frac{-dp}{[\partial f_1 / \partial x]}$$

$$\Rightarrow \frac{\partial f_1}{\partial x} dx + \frac{\partial f_1}{\partial p} dp = 0$$

$$\Rightarrow df_1 = 0 \Rightarrow f_1 = \text{Const.} = k$$

$$f_1(p, x) = f_2(q, y) = k$$

$$p = \phi(k, x)$$

$$q = \psi(k, y)$$

$$dz = \phi(k, x) dx + \psi(k, y) dy$$

$$\boxed{z = h(k, x, y) + b}$$

(Singular solⁿ
does not exist)

1st Q.

$$y(1+x^2)p^2 - qx^2 = 0$$

$$\Rightarrow \left(\frac{1+x^2}{x^2} \right) p^2 = \frac{q}{y} = a^2$$

$$\frac{dx}{2p \left[1 + \frac{1}{x^2} \right]} = \frac{-dp}{-2p}$$

$$\Rightarrow p = \pm \frac{ax}{\sqrt{1+x^2}}, \quad q = a^2 y$$

$$dz = p dx + q dy$$

$$dz = \pm \frac{ax}{\sqrt{1+x^2}} dx + a^2 y dy$$

$$0t = 2(at^2 + b) \times 2at$$

$$2 = a(at^2 + b)$$

$$at^2 = \frac{2}{a} - b \Rightarrow \boxed{t^2 = \frac{2}{a^2} - \frac{b}{a}} \quad \text{--- (2)}$$

Using Equ. (2) in (1)

$$4\left(\frac{2}{a^2} - \frac{b}{a}\right) = \left[a\left(\frac{2}{a^2} - \frac{b}{a}\right)^2 + b\right]^2$$

$$4\left(\frac{2}{a^2} - \frac{b}{a}\right) = \frac{4}{a^2}$$

$$\frac{b}{a} = \frac{1}{a^2} \Rightarrow \boxed{b = \frac{1}{a}}$$

$$z^2 = a^2x^2 + \left(ay + \frac{1}{a}\right)^2$$

$$a^2z^2 = a^4x^2 + (a^2y + 1)^2 \quad \text{--- (3)}$$

$$a^2z^2 = a^4x^2 + a^4y^2 + 2a^2y + 1$$

$$a^4(x^2 + y^2) + a^2(2y - z^2) + 1 = 0$$

Eliminating 'a'. D.w.r. to a.

$$4a^3(x^2 + y^2) + 2a(2y - z^2) = 0$$

$$\Rightarrow \frac{4a^3}{2a} = -\left(\frac{2y - z^2}{x^2 + y^2}\right)$$

$$\Rightarrow 2a^2 = \left(\frac{z^2 - 2y}{x^2 + y^2}\right) \Rightarrow a^2 = \frac{1}{2} \left(\frac{z^2 - 2y}{x^2 + y^2}\right)$$

Put the value of a^2 in equ. (3)

$$\frac{1}{2} \left(\frac{z^2 - 2y}{x^2 + y^2}\right) \cdot z^2 = x^2 \cdot \frac{1}{4} \left(\frac{z^2 - 2y}{x^2 + y^2}\right)^2 + \left[\frac{1}{2} \left(\frac{z^2 - 2y}{x^2 + y^2}\right) y + 1\right]^2$$

$$\frac{z^2}{2} = \frac{x^2}{4} \left(\frac{z^2 - 2y}{x^2 + y^2}\right) + \frac{1}{4} \left[y + \left(\frac{x^2 + y^2}{z^2 - 2y}\right)\right]^2 \cdot \left(\frac{z^2 - 2y}{x^2 + y^2}\right)$$

$$\boxed{z^2 = 2\left(y \pm \sqrt{x^2 + y^2}\right)}$$

NOTE:- we can derive any other c.I. from the given c.I. for first order p.d.e.

Eg:- Let $F(x, y, z, a, b) = 0$ — (1) is the complete integral of equ^n $f(x, y, z, p, q) = 0$ — (2)

Let $u(x, y, z, h, k) = 0$ — (3) is another complete integral of equ^n (2)

we will show that it is possible to derive $\text{equ}(3)$ as an envelope of one parameter subfamily of $\text{equ}(1)$

I) we first choose a curve c which lies on $\text{equ}^n(3)$ (or surface (3)). This curve will contain two constants h and k .

II) Using this curve we will find a subfamily of the given c.I. (equ^n i) and then we will find the envelope of the subfamily.

Eg:- Show that $2xz + q^2 = x(xp + yq)$ — (1)
This has a complete integral $z + a^2x = axy + bx^2$
and deduce $x(y + hx)^2 = 4(3 - kx^2)$ is also a complete integral.

Solⁿ:- By checking that $\text{rank} = 2$

$$\begin{bmatrix} f_a & f_{xa} & f_{ya} \\ f_b & f_{xb} & f_{yb} \end{bmatrix}$$

$$F = z + a^2x - axy - bx^2$$

$$\begin{bmatrix} 2ax - xy & 2a - y & -x \\ -x^2 & -2x & 0 \end{bmatrix}$$

$$\text{Rank} = 2 \quad \text{if } x \neq 0$$

$$Z = \pm a \sqrt{1+x^2} + \frac{a^2 y^2}{2} + c$$

$$⑥ \quad x^4 + y^3$$

$$Q. \quad Z(p^2 - q^2) = x + y$$

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(IV) Clairaut's form: $\rightarrow$ If the given Equ^n is of the form $Z = px + qy + f(p, q)$ Then its complete integral is —

$$Z = ax + by + f(a, b)$$

Eq:- $6Z \left(\frac{\partial Z}{\partial x} \right) \left(\frac{\partial Z}{\partial y} \right) = Z \left(\frac{\partial Z}{\partial x} \right)^2 \left[3 \left(\frac{\partial Z}{\partial x} \right)^3 + 2x \left(\frac{\partial Z}{\partial y} \right) \right] + 2 \left(\frac{\partial Z}{\partial y} \right)^2 \left[2 \left(\frac{\partial Z}{\partial y} \right)^3 + 3y \left(\frac{\partial Z}{\partial x} \right) \right]$

Solⁿ:- $6Zpq = 3p^2(3p^3 + 2qx) + 2q^2(2q^3 + 3py)$

$$Z = \frac{1}{2} \frac{p}{q} (3p^3 + 2qx) + \frac{2}{6} \frac{q}{p} (2q^3 + 3py)$$

$$= \frac{3}{2} \frac{p^4}{q} + xp + \frac{2q^4}{3p} + yq$$

$$= px + qy + \frac{3}{2} \frac{p^4}{q} + \frac{2}{3} \frac{q^4}{p}$$

$$Z = ax + by + \frac{3}{2} \frac{a^4}{b} + \frac{2}{3} \frac{b^4}{a}$$

Cauchy problem for Non-linear Equ^n : $\rightarrow$

Let $f(x, y, z, p, q) = 0$ be a Non linear PDE — (1)
 Let $(x = x_0(t), y = y_0(t), z = z_0(t))$ — (2) be a given curve.

We have to find an integral surface of equ^n (1) which passes through Equ (2)

Step-I :- find complete Sol^n
 $f(x, y, z, a, b) = 0$ — (3)

Step-II :- Using (2) in Equ. (3) we get a fun^c of a & b i.e. $F(t, a, b) = 0$ — (4)

Step-III → Eliminate 't' from equⁿ (4) by $\frac{\partial F}{\partial t} = 0$ — (5)

sl using Equⁿ (4) & (5) we get $\eta(a, b) = 0$. — (6)

Step-IV → Solve Equⁿ (6) we get —

$$b = \phi_1(a) \quad \text{or} \quad b = \phi_2(a) \quad \text{— (7)}$$

Each one of equⁿ (7) defines a subfamily of equⁿ (3) i.e. complete sol^n & the envelope of these subfamily is exist it is also a solution of Cauchy problem. for Non-linear Equⁿ. It may be unique.

Using Equ. (3) & (7)

$$\psi(x, y, z, a) = 0 \quad \text{— (8)}$$

Envelope of Equⁿ (8) is the sol^n of our Cauchy problem.

$$\text{Envelope} \left\{ \begin{array}{l} \frac{\partial \psi}{\partial a} = 0 \quad \text{— (9)} \\ \text{Eliminate 'a' using Equⁿ (8) and (9)} \end{array} \right\}$$

Eg :- $(p^2 + q^2)x = pz$ find the integral sol^n passing through the curve $x_0 = 0, y_0 = t^2, z_0 = 2t$.

Solⁿ - Its complete sol^n (by Charpit method)

$$z^2 = a^2 x^2 + (ay + b)^2$$

$$4t^2 = 0 + (at^2 + b)^2 \quad \text{— (1)}$$

D.w. equ. (1) h. to t

It defines on domain $(x \neq 0)$

Let $x = t$, $y = -ht$ and $z = kt^2$ be a curve ($t \neq 0$)
lies on Eqn. (3)

$$kt^2 + a^2t = at(-ht) + bt^2$$

$$kt^2 + a^2t + aht^2 - bt^2 = 0$$

$$kt + a^2 + aht - bt = 0$$

$$(kt + a^2) + (ah - b)t = 0$$

$$\therefore (k + ah - b)t + a^2 = 0 \quad \text{--- (4)}$$

D. w. r. to t

$$k + ah - b = 0 \quad \Rightarrow \quad \boxed{k = b - ah} \quad \text{--- (5)}$$

put Eqn. (5) in (2)

$$z + a^2x = axy + (k + ah)x^2 \quad \text{--- (6)}$$

Now we find the envelope of Eqn. (6) by eliminating 'a' by diff. Eqn. (6) w.r. to a .

$$2ax = xy + hx^2$$

$$a = \frac{y}{2} + \frac{hx}{2}$$

put in eqn. (6)

$$z = \left(\frac{y+hx}{2} \right) xy + \left(k + \frac{(y+hx)h}{2} \right) x^2 - \left(\frac{y+hx}{2} \right)^2 x$$

$$\Rightarrow (z - kx)^2 = \frac{y+hx}{2} \left[xy + hx^2 - \frac{(y+hx)x}{2} \right]$$

$$\Rightarrow (z - kx)^2 = \frac{y+hx}{2 \times 2} (2xy + 2hx^2 - xy - hx^2)$$

$$\Rightarrow 4(z - kx)^2 = (y + hx)(xy + hx^2) \\ = (y + hx)(y + hx) \cdot x$$

$$\boxed{4(z - kx)^2 = (y + hx)^2 \cdot x}$$

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PDE of order n with constant coeff. :- $\rightarrow$

Homogeneous

Non-Homo.

Reduced into linear factor

Not reducible

$$\left[K_0 \frac{\partial^n z}{\partial x^n} + K_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + K_n \frac{\partial^n z}{\partial y^n} \right] +$$

$$\left[L_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + L_1 \frac{\partial^{n-1} z}{\partial x^{n-2} \partial y} + \dots + L_{n-1} \frac{\partial^{n-1} z}{\partial y^{n-1}} \right] + \dots$$

$$+ \left[M_0 \frac{\partial z}{\partial x} + M_1 \frac{\partial z}{\partial y} \right] + N_0 z = f(x, y)$$

where, K_i 's, L_i 's, M_i 's, N_i 's are const func^s of x & y
 this is called general PDE of order 2.

(i) If all the coeff. are constant then Eqnⁿ will be linear PDE of order n.

(ii) If $K_i \neq 0$, $L_i = 0$, $M_i = 0$, $N_i = 0$

$$\left[K_0 \frac{\partial^n z}{\partial x^n} + K_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots \right] = f(x, y)$$

It is called homogeneous linear PDE.

$$u_{xx} + u_{yy} + u_{zz} = x$$

¶

Homogeneous Linear PDE of order n $\rightarrow$

$$[K_0 D_x^n + K_1 D_x^{n-1} D_y + \dots + K_n D_y^n] z = f(x, y) \quad \text{--- (1)}$$

$$\boxed{\phi(D_x, D_y) z = f(x, y)}$$

The general solⁿ of Eqn. (1) is given by —

$$z = \underline{z_c} + \underline{z_p} \Rightarrow C.F. + P.I. \quad z_c \rightarrow C.F., \quad z_p \rightarrow P.I.$$

Complementary fun^c $\rightarrow$ let

$$[K_0 D_x^n + K_1 D_x^{n-1} D_y + \dots + K_n D_y^n] z = 0$$

$$\Rightarrow [(D_x - \lambda_1 D_y)(D_x - \lambda_2 D_y) \dots (D_x - \lambda_n D_y)] z = 0$$

$$\begin{aligned} \frac{\partial z}{\partial x} - \lambda_1 \frac{\partial z}{\partial y} &= 0 \Rightarrow \frac{dz}{dx} = \frac{dy}{-d_1} = \frac{dz}{0} \\ z &= \phi(y + \lambda_1 x) \end{aligned}$$

$$\boxed{\begin{array}{l} \text{Symbolic form} \\ \frac{\partial z}{\partial x} = D_x \quad \phi(D_x, D_y) z = f(x, y) \\ \frac{\partial z}{\partial y} = D_y \quad \frac{\partial z}{\partial x} \end{array}}$$

$$z = \phi_1(y + \lambda_1 x) + \phi_2(y + \lambda_2 x) + \dots + \phi_n(y + \lambda_n x)$$

if all λ_i 's are distinct.

$\phi_i(y + \lambda_i x)$ --- are arbitrary functions

Auxillary Eqnⁿ $\rightarrow D_x = \lambda, D_y = 1$

$$[K_0 \lambda^n + K_1 \lambda^{n-1} + \dots + K_n] = 0$$

(if highest order of D_x is non zero then put $D_x = \lambda$)

(ii) If λ_i 's are repeated —

$$\lambda_1 = \lambda_2 = \lambda'$$

$$\Rightarrow (D_x - \lambda' D_y)^2 z = 0$$

$$\Rightarrow (D_x - \lambda' D_y) \underbrace{(D_x - \lambda' D_y) z}_u = 0$$

$$(D_x - \lambda' D_y) u = 0$$

$$(D_x - \lambda' D_y) u = u$$

$$u = \phi_1(y + \lambda' x)$$

$$\Rightarrow (D_x - \lambda' D_y) z = \phi_1(y + \lambda' x)$$

$$\frac{dx}{1} = \frac{dy}{-d} = \frac{dz}{\phi_1(y+dx)}$$

$$C_1 = y + dx$$

$$\frac{dx}{1} = \frac{dy}{-d} = \frac{dz}{\phi_1(C_1)}$$

$$x = \frac{z}{\phi_1(C_1)} + C_2$$

$$x\phi_1(C_1) = z + C_2\phi_1(C_1)$$

$$C_2 = x\phi_1(C_1) - z$$

$$x\phi_1(C_1) = \phi_2(y+dx) + z$$

$$z = \phi_2(y+dx) + x\phi_1(y+dx)$$

Simultaneously $(D_x - dD_y)^n z = 0$ $z = \phi_1(y+dx) + x\phi_2(y+dx) + \dots + x^{n-1}\phi_n(y+dx)$

$$\# (D_x^3 D_y^2 - D_x^2 D_y^3) z = 0$$

$$D_y^2 [D_x^3 - D_x^2 D_y] z = 0$$

$$D_y^2 \cdot D_x^2 [D_x - D_y] z = 0$$

$$D_x^2 = 0, \\ d^2 = 0$$

$$D_y^2 = 0, \\ d^2 = 0$$

$$\phi_1(y) + x\phi_2(y) \quad \phi_3(x) + y\phi_4(x)$$

(highest order of $D_x = 3$)
(firstly take common)

$$D_x - D_y = 0 \\ \frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{0} \\ \text{or } d-1=0 \\ \phi_5(y+x)$$

$$C.F. = \phi_1(y) + x\phi_2(y) + \phi_3(x) + y\phi_4(x) + \phi_5(y+x)$$

$$Q. \frac{\partial^4 z}{\partial y^4 \partial x} + \frac{\partial^4 z}{\partial y^4} = 0$$

$$D_x D_y^3 + D_y^4 = 0$$

$$\Rightarrow D_y^3 [D_x + D_y] = 0$$

$$D_y^3 = 0 \\ d^3 = 0$$

$$\phi_1(x) + y\phi_2(x) + y^2\phi_3(x) + \phi_4(y-x) \\ = \phi_1(x) + y\phi_2(x) + y^2\phi_3(x) + \phi_4(x-y)$$

$$D_x + D_y = 0 \\ d+1=0 \\ \phi_4(y-x)$$

Q:- A Surface passing through two lines

$$Z = x = 0, \quad z-1 = x-y = 0$$

+ Satisfying $\lambda - 4\delta + 4t = 0$

$$(i) \quad z = \frac{3y}{y+2x} \quad \checkmark (ii) \quad z = \frac{3x}{y+2x}$$

$$(iii) \quad z = \frac{2}{y+3x} \quad (iv) \quad z = \frac{3}{x+3y}$$

Solⁿ $\rightarrow$
$$Z_{xx} - 4Z_{yy} + 4Z_{xy} = 0$$

$$D_x^2 + 4D_x D_y + 4D_y^2 = 0$$

$$\lambda = 2, 2$$

Put $D_x = \lambda, D_y = 1$

c.f. $Z = \phi_1(y+2x) + x\phi_2(y+2x)$

Put $z = x = 0$

$$0 = \phi_1(y) \Rightarrow \phi_1(y) = 0 \Rightarrow z = x\phi_2(y+2x)$$

Put $z-1 = x-y$

$$1 = x\phi_2(3x)$$

$$\phi_2(3x) = \frac{1}{x} \Rightarrow 3x = t$$

$$\phi_2(t) = \frac{3}{t} \Rightarrow z = x \times \frac{3}{(y+2x)}$$

Particular Integral $\rightarrow$

$$[K_0 D_x^n + K_1 D_x^m D_y + \dots + K_n D_y^n] z = f(x, y)$$

$$\phi(D_x, D_y) z = f(x, y)$$

$$z_p = \frac{1}{\phi(D_x, D_y)} f(x, y)$$

$$(i) \text{ If } f(x, y) = \psi(\alpha x + \beta y) \quad \left(\begin{array}{l} \sin(3x+2y), (x+2y)^2 \\ \log(3x+y) \end{array} \right)$$

$$\text{then P.I.} = \frac{1}{\phi(\alpha, \beta)} \int \int \dots \int \psi(u) \underbrace{du \dots du}_{n \text{ times}}$$

$$\text{Provided } \phi(\alpha, \beta) \neq 0 \quad u = \alpha x + \beta y$$

$$(ii) \quad \phi(\alpha, \beta) = 0$$

$$\phi(Dx, Dy) = (\beta Dx - \alpha Dy)^n \cdot g(Dx, Dy)$$

$$\text{where } g(\alpha, \beta) \neq 0$$

$$\Rightarrow \frac{1}{(\beta Dx - \alpha Dy)^n \cdot g(Dx, Dy)} \cdot \psi(\alpha x + \beta y)$$

$$\Rightarrow \frac{x^n}{n! \cdot \beta^n} \left[\frac{1}{g(Dx, Dy)} \cdot \psi(\alpha x + \beta y) \right]$$

$$Q. \quad [Dx^2 + Dx Dy - 2Dy^2] z = \sqrt{x+2y} \quad \text{--- (1)}$$

$$\text{C.F.} \rightarrow (d^2 + d - 2) = 0 \quad (Dx = d, Dy = 1)$$

$$\Rightarrow (d+2)(d-1) = 0 \Rightarrow d = -2, 1$$

$$z = \phi_1(y+x) + \phi_2(y-2x)$$

$$\text{P.I.} \Rightarrow f(x, y) = \sqrt{x+2y}$$

$$\alpha = 1, \beta = 2 \quad \text{in (1)}$$

$$\phi(\alpha, \beta) = 1 + 2 - 4 = -1 \neq 0$$

$$= \frac{1}{\phi(\alpha, \beta)} \iint \sqrt{u} \, du \cdot du, \quad u = x+2y$$

$$= -\frac{1}{5} \iint \sqrt{x+2y} \sqrt{u} \, du \cdot du$$

$$= -\frac{4}{75} (x+2y)^{5/2}$$

$$z = \phi_1(y+x) + \phi_2(y-2x) - \frac{4}{75} (x+2y)^{5/2}$$

$$\# \quad [Dx^3 - 4Dx^2 Dy + 4Dx Dy^2]z = 8 \cos(2x+y)$$

$$C.F. \rightarrow d^3 - 4d^2 + 4d = 0$$

$$d[d^2 - 4d + 4] = 0$$

$$d = 0, \quad d = 2, 2$$

$$\alpha = 2$$

$$\beta = 1$$

$$\phi_1(x) + \phi_2(xy + 2x) + x\phi_3(y + 2x)$$

$$P.I. = \frac{1}{Dx(Dx - 2Dy)^2} \cdot 8 \cos(2x+y)$$

$$= \frac{x^2}{2!(1)^2} \left[\frac{1}{Dx} \cdot 8 \cos(2x+y) \right]$$

$$= \frac{x^2}{2} \cdot \frac{8 \sin(2x+y)}{2} = 2x^2 \sin(2x+y)$$

$$\# \quad f(x, y) = \psi(\alpha x + \beta y)$$

$$(i) \quad \text{If } f(x, y) = e^{\alpha x + \beta y}$$

$$P.I. = \frac{1}{\phi(Dx, Dy)} \cdot e^{\alpha x + \beta y}$$

$$= \frac{1}{\phi(\alpha, \beta)} \cdot e^{\alpha x + \beta y}, \quad \phi(\alpha, \beta) \neq 0$$

$$(ii) \quad f(x, y) = \sin(\alpha x + \beta y) \quad \text{or} \quad \cos(\alpha x + \beta y)$$

$$D_x^2 \rightarrow -\alpha^2$$

$$D_y^2 \rightarrow -\beta^2$$

$$D_x D_y \rightarrow -\alpha\beta$$

Q.

$$(3D_x^2 - 10D_x D_y + 3D_y^2)z = 10e^{2x-y} + \sin(2x+y)$$

Case - $f(x, y) = x^m y^n$

If $m = n$ then, expressed either in x & y .
 If $m > n$ " " " power of Dy
 If $m < n$ " " " " " Dx

Q. $(Dx^2 - Dy^2)z = x^2y$
 $d^2 - 1 = 0, d = \pm 1$

$z_c = \phi_1(y+x) + \phi_2(y-x)$

$z_p = \frac{1}{(Dx^2 - Dy^2)} \cdot x^2y$

$= \frac{1}{Dx^2 \left[1 - \frac{Dy^2}{Dx^2} \right]} \cdot x^2y$

$= \frac{1}{Dx^2} \left[1 - \frac{Dy^2}{Dx^2} \right]^{-1} \cdot x^2y$

$= \frac{1}{Dx^2} \left[1 + \frac{Dy^2}{Dx^2} \right] \cdot x^2y$

$= \frac{1}{Dx^2} [x^2y + 0]$

$z_p = \frac{x^4y}{12}$

(i) $m > n$ Dy/Dx
 (ii) $m < n$ Dx/Dy
 (iii) $m = n$ $x \& y$
 $m > n$

General Method to find P.I. $\rightarrow$

$z_p = \frac{1}{\phi(Dx, Dy)} \cdot f(x, y)$

$= \frac{1}{(Dx - d_1 Dy) (Dx - d_2 Dy) \dots (Dx - d_n Dy)} \cdot f(x, y)$ (if homo.)

$\Rightarrow \frac{1}{(Dx - d_1 Dy)} \cdot f(x, y)$

$= \int f(x, c - d_1 x) dx$ where $y = c - d_1 x$

$$Q. \quad [Dx^2 + Dx Dy - 6 Dy^2] z = y \sin x$$

$$z_p = \frac{1}{(Dx+3Dy)(Dx-2Dy)} \cdot y \sin x$$

$$= \frac{1}{(Dx+3Dy)} \cdot \left[\frac{1}{(Dx-2Dy)} \cdot y \sin x \right]$$

$$= \int (c-2x) \sin x \, dx \quad y = c-x$$

$$= \left[\int (c-2x) \cos x - 2 \sin x \right] \cdot \frac{1}{(Dx+3Dy)}$$

$$= \frac{1}{(Dx+3Dy)} \left[-y \cos x - 2 \sin x \right]$$

$$y = c+3x$$

$$= \int -[(c+3x) \cos x - 2 \sin x] \, dx$$

$$z_p = -y \sin x + \cos x$$

$$\# \quad (D_x^2 - Dx Dy - 2 Dy^2) = (y-1)e^x$$

$$z_c = \frac{1}{(Dx-2Dy)(Dx+Dy)} (y-1)e^x$$

$$= \frac{1}{(Dx-2Dy)} \left[\int (c+x-1) e^x \, dx \right]$$

$$y = c+x$$

$$c = y-x$$

$$= \frac{1}{(Dx-2Dy)} \left[(y-2) e^x \right]$$

$$\Rightarrow y = c-2x$$

$$\Rightarrow \int (c-2x-2) e^x \, dx$$

$$\Rightarrow c = y+2x$$

$$\Rightarrow \boxed{z_p = y e^x}$$

#

$$x+y-6t = x^2 \cos(x+y)$$

#

$$D_x^2 + 2Dx Dy + D_y^2 = x e^{x+y}$$

Linear PDE $\rightarrow$ A PDE said to be linear if it is linear in dependent variable and all its derivative and also coff. of higher derivative should not contain either the dependent variable or its derivative.
 Ex: $u_{xx} + u_{yy} = 0$ free from dependent variable & its derivative

Semi linear PDE $\rightarrow$ A PDE said to be semi linear if it is linear in its higher derivative and also coff. of higher deri. should not contain either the dependent variable or its derivative.
 Ex: $u_{xx} + u_{yy} = 1$

Quasi linear $\rightarrow$ A PDE is said to be Quasi linear if it is linear in its higher derivative.

Eg:- $(u_x)^4 u_{xx} + u_{yy} + (u_y)^2 = 0$ is Quasi linear.

Non homogeneous PDE Linear with constant coefficient $\rightarrow$

If the total order is not same throughout the Eqn^n is called Non-homo.

$$u_{xx} + u_y = 0$$

$$\phi(D_x, D_y)z = f(x, y)$$

Case-I $\rightarrow$ If $\phi(D_x, D_y)$ can be factorised linearly and factor are not repeated that of eqn -

$$\phi(D_x, D_y) = (\alpha_1 D_x + \beta_1 D_y + \gamma_1) (\alpha_2 D_x + \beta_2 D_y + \gamma_2) \dots (\alpha_n D_x + \beta_n D_y + \gamma_n)$$

$$\Rightarrow (\alpha_1 D_x + \beta_1 D_y + \gamma) z = 0$$

$$\frac{dx}{\alpha_1} = \frac{dy}{\beta_1} = \frac{dz}{-\gamma_1 z}$$

$$\beta_1 x - \alpha_1 y = c_1$$

$$-\frac{\gamma_1}{\alpha_1} x = \log z + \log c_2$$

$$c_2 = z^{\frac{\gamma_1}{\alpha_1}} e^{\frac{\gamma_1}{\alpha_1} x}$$

$$z e^{\frac{\gamma_1}{\alpha_1} x} = \phi(\beta_1 x - \alpha_1 y)$$

$$z = e^{-\frac{\gamma_1}{\alpha_1} x} \phi(\beta_1 x - \alpha_1 y)$$

or

$$z = e^{-\frac{\gamma_1}{\beta_1} y} \phi(\beta_1 x - \alpha_1 y)$$

G.S.

$$z = e^{-\frac{\gamma_1}{\alpha_1} x} \phi_1(\beta_1 x - \alpha_1 y) + e^{-\frac{\gamma_2}{\alpha_2} x} \phi_2(\beta_2 x - \alpha_2 y) \dots$$

If factors are repeated $\rightarrow$

$$z = e^{-\frac{\gamma_1}{\alpha_1} x} [\phi_1(\beta_1 x - \alpha_1 y) + x \phi_2(\beta_1 x - \alpha_1 y)]$$

$$\# \quad [(Dx + Dy - 2)(Dx - Dy)]z = 0$$

$$(Dx + Dy - 2)z = 0$$

$$\alpha = 1, \beta = 1, \gamma = -2$$

$$(Dx - Dy)z = 0$$

$$\phi_2(x+y)$$

$$z = e^{2x} \phi_1(x-y)$$

$$z = e^{2x} \phi_1(x-y) + \phi_2(x+y)$$

$$\text{OR} \quad e^{2y} \phi_1(x-y) + \phi_2(x+y)$$

#

$$S + P = Q + Z$$

$$Dx Dy + Dx = Dy + Z = 0$$

$$\Rightarrow (Dx Dy + Dx - Dy - 1)z = 0$$

$$\Rightarrow [Dx(Dy+1) - (Dy+1)]z = 0$$

$$[(Dx-1)(Dy+1)]z = 0$$

$$\alpha=1, \beta=0, \gamma=-1$$

$$\alpha=0, \beta=1, \gamma=1$$

$$z = e^{(-1)y} \phi_1(-y)$$

$$z = e^y \phi_2(x)$$

$$z = e^x \phi_1(-y) + e^y \phi_2(x)$$

Case-II $\rightarrow$ If $\phi(Dx, Dy) = 0$ can not be factorised linearly $\rightarrow$

if $z = A e^{hx+ky}$ is the solution of (1)

$$\& \eta(h, k) e^{hx+ky} = 0$$

$$\Rightarrow \eta(h, k) = 0$$

$$z = \sum_{i=1}^{\infty} A_i \cdot e^{h_i x + k_i y}$$

$$\text{where } \eta(h_i, k_i) = 0$$

#

$$(3Dx + 4Dy - 6)(Dx^2 + Dy)z = 0$$

$$\alpha=3, \beta=4, \gamma=-6$$

$$e^{2x} \phi_1(4x-3y)$$

$$\text{Now } (Dx^2 + Dy)z = 0$$

$$z = \sum A_i e^{h_i x + k_i y}$$

$$h_i^2 + k_i = 0 \Rightarrow k_i = -h_i^2$$

$$z = \sum_i A e^{h_i x - h_i^2 y}$$

#

$$[Dx^2 - Dx Dy - 2Dy^2 + 2Dy + 2Dx]z = 0$$

$$[Dx^2 + 2Dx - Dx Dy - 2Dy^2 + 2Dy]z = 0$$

$$(Dx + Dy)(Dx - 2Dy + 2) = 0$$

$$\alpha=1, \beta=-2, \gamma=2$$

$$\phi_1(-y+x), \quad e^{-2x}\phi_2(-2x-2y)$$

$$z = \phi_1(x-y) + e^{-2x}\phi_2(-2x-y)$$

Particular Integral in Non-homo. Case $\rightarrow$

Case (i) $\phi(Dx, Dy)z = f(x, y)$

if $f(x, y) = e^{\alpha x + \beta y}$

P.I. = $\frac{1}{\phi(\alpha, \beta)} e^{\alpha x + \beta y}$, $\phi(\alpha, \beta) \neq 0$

(ii) $f(x, y) = \sin(\alpha x + \beta y)$ or $\cos(\alpha x + \beta y)$

P.I. = $D_x^2 \rightarrow -\alpha^2$, $D_y^2 \rightarrow -\beta^2$
 $D_x D_y \rightarrow -\alpha\beta$

(iii) $f(x, y) = e^{\alpha x + \beta y} \cdot v(x, y)$

P.I. = $\frac{1}{\phi(Dx, Dy)} \cdot e^{\alpha x + \beta y} \cdot v(x, y)$

= $e^{\alpha x + \beta y} \cdot \frac{1}{\phi(Dx+\alpha, Dy+\beta)} \cdot v(x, y)$

Equation Reducible into constant coefficient $\rightarrow$

$$[(\alpha_0 x^n D_x^n + \alpha_1 y x^{n-1} D_x^{n-1} D_y + \dots + \alpha_n y^n D_y^n) + \dots + (\beta_0 x D_x + \beta_1 y D_y + \gamma_0)] z = f(x, y)$$

$$\boxed{\phi(x D_x, y D_y) z = f(x, y)}$$

$$x = e^X, \quad y = e^Y$$

$$X = \log x, \quad Y = \log y$$

$$x D_x = D_X$$

$$x^2 D_x = D_X(D_X - 1)$$

$$y D_y = D_Y$$

$$y^2 D_y = D_Y(D_Y - 1)$$

$$\# \left[x^2 D_x^2 - 4xy D_x D_y + 4y^2 D_y^2 + 6y D_y \right] = x^7 y^9$$

$$\left[D_x(D_x-1) - 4 D_x D_y + 4 D_y(D_y-1) + 6 D_y \right] = e^{7x+9y}$$

Put $D_x = \lambda, D_y = 1$

$$\lambda^2 - \lambda - 4\lambda + 4$$

$$D_x^2 - D_x - 4 D_x D_y + 4 D_y^2 - 4 D_y + 6 D_y$$

$$D_x^2 - D_x - 4 D_x D_y + 4 D_y^2 + 2 D_y$$

$$= (D_x - 2 D_y)^2 - (D_x - 2 D_y)$$

$$\left[(D_x - 2 D_y)(D_x - 2 D_y - 1) \right] Z = e^{7x+9y}$$

$$A=1, B=-2, C=-1$$

$$\frac{D_x}{1} = \frac{D_y}{-2} = \frac{DZ}{0}$$

$$\phi(-y - 2x)$$

$$-2x = y + c \Rightarrow$$

Ans $\rightarrow \phi_1 \left[\log x^2 y \right] + x \phi_2 \left[-\log x^2 y \right] + \frac{1}{132} x^7 y^9$

Second Order partial differential Equⁿ →

A PDE is said to be Second order Semi linear

Equⁿ →

$$R(x, y) u_{xx} + S(x, y) u_{xy} + T(x, y) u_{yy} +$$

$$g(x, y, u, u_x, u_y) = 0$$

where

$$R^2 + S^2 + T^2 \neq 0$$

and R, S, T are conti fun^c on $x, y, \mathbb{R}$.

" A fun^c $u(x, y)$ is said to be regular solⁿ of equⁿ in domain $D \subseteq \mathbb{R} \times \mathbb{R}$ " if $u \in C^2(D)$

and satisfied equ. (i)

Classification of Second order PDE →

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0$$

- (i) If $S^2 - 4RT > 0$ in $D \rightarrow$ Hyperbola
- (ii) If $S^2 - 4RT = 0$ in $D \rightarrow$ Parabola
- (iii) If $S^2 - 4RT < 0$ in $D \rightarrow$ Elliptic
- $\begin{cases} H > \\ P < \\ E = \end{cases}$

Laplace Equⁿ $u_{xx} + u_{yy} = 0$

$$R=1, S=0, T=1 \rightarrow \text{Elliptic}$$

Heat Equⁿ $u_{xx} = \frac{1}{c} u_t$

$$R=1, S=0, T=0$$

Parabola

Wave Equⁿ, $u_{xx} = \frac{1}{c^2} u_{tt}$

$$R=1, S=0, T=1 \rightarrow \text{Hyperbola}$$

$xy_{xt} - (x^2 - y^2)_t - xy_t + py - qx - 2(x^2 - y^2) = 0$
 where $x^2 + y^2 \neq 0$

$R = xy, \quad S = -(x^2 - y^2), \quad T = -xy$

$S^2 - 4RT = (x^2 + y^2)^2 > 0 \rightarrow \text{Hyperbolic}$

Q $\rightarrow x u_{xx} + 2xy u_{xy} + y u_{yy} + x u_y + y u_x = 0$

$R = x, \quad S = 2xy, \quad T = y$

$S^2 - 4RT = 4x^2y^2 - 4xy \Rightarrow 4xy(4xy - 1) = 0$

(i) E, $x < 0, y < 0, xy > 1$

(ii) P, $x < 0, y < 0, xy = 1$

(iii) E, $x > 0, y > 0, xy < 1$

(iv) H, $x < 0, y < 0, xy > 1$

Q $\rightarrow (x-1)^2 u_{xx} - (y-2)^2 u_{yy} + 2x u_x + 2y u_y + 2xy u = 0$

is parabolic in $S \subseteq \mathbb{R}^2$ but not in $\mathbb{R}^2 \setminus S$ then

S is -

(i) $\{ (x, y) \in \mathbb{R}^2 : x = 1 \text{ or } y = 2 \}$

(ii) $\{ (x, y) \in \mathbb{R}^2 : x = 1 \text{ and } y = 2 \}$

(iii) $\{ (x, y) \in \mathbb{R}^2 : x = 1 \}$

4 $(x-1)^2 (y-2)^2$

(iv) $\{ (x, y) \in \mathbb{R}^2 : y = 2 \}$

Sol $\rightarrow$

$S = \{ (1, y), (x, 2), (1, 2) \}$

$S = \{ (1, 2) \} \rightarrow (1, 2) \text{ is a point}$

$S = \{ (1, y) \}$

$S = \{ (x, 2) \}$

04/Oct/2017

Classification of 2nd order PDE $\rightarrow$

(a) Characteristic Roots $\rightarrow$ Let $R\lambda + S\lambda + T\lambda + f(x, y, z, p, q) = 0$ — (1)
be a Second order P.D.E.

The root of the Equation

$Rd^2 + Sd + T = 0$ — (2) are called characteristic roots of eqn (1).
and Equation given by this $\rightarrow$

If d is root of eqn (2)

$$\boxed{\frac{dy}{dx} + d = 0}$$

In Hyperbolic ($S^2 - 4RT > 0$) there exist 2 real c. roots

In parabolic ($S^2 = 4RT$), family of real ch. curve = 1

In Elliptic ($S^2 - 4RT < 0$), family of real ch. curve = 0

$y^2\lambda - x^2\lambda = 0$: $x \neq 0, y \neq 0$

Find characteristic curve?

Sol. $R = y^2, S = 0, T = -x^2$

$$\Rightarrow S^2 - 4RT = 0 + 4x^2y^2 > 0$$

Hyperbolic $\therefore$ 2 real characteristic roots

$$y^2(d^2) + 0 \cdot d + (-x^2) = 0$$

$$y^2d^2 - x^2 = 0$$

$$d = \pm \frac{x}{y}$$

$$d_1 = \frac{x}{y}, \quad d_2 = -\frac{x}{y}$$

$$\frac{dy}{dx} + \frac{x}{y} = 0, \quad \frac{dy}{dx} - \frac{x}{y} = 0$$

$$x^2 + y^2 = c_1$$

$$y^2 - x^2 = c_2$$

Circle & Hyperbolic curves

Canonical | Normal form $\rightarrow$ "we change the independent variable to reduce Second order PDE into some simpler form that form is called Canonical | Normal form."

(I) Reduce Hyperbolic Equⁿ into Canonical form $\rightarrow$

Ex 1 $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0$

$R = 1, S = 0, T = -1$

$Rd^2 + Sd + T = 0$

$d^2 - 1 = 0 \Rightarrow d = \pm 1$

(1) Find char. roots.

(2) Find char. curve

$\frac{dy}{dx} + 1 = 0$

$\frac{dy}{dx} - 1 = 0$

Say $\eta_1(x,y) = c_1$

$\eta_2(x,y) = c_2$

$y+x = c_1$

$y-x = c_2$

Let $u = \eta_1(x,y), v = \eta_2(x,y)$

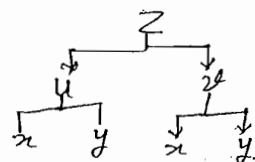
Find p, q, r, s, t (if char. required)

$u = y+x$

$v = y-x$

$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}$

$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot 1 + \frac{\partial z}{\partial v} \cdot 1$



$\frac{\partial^2 z}{\partial u \partial v} = \phi(u, v, z, z_u, z_v)$

$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial x} \right) \frac{\partial u}{\partial x} + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial x} \right) \frac{\partial v}{\partial x}$

$= \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \cdot 1 + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \cdot (-1)$

$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) (-1)$

$= \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} \right) + \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} \right) - \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial v} \right)$

$= \frac{\partial}{\partial u} \left[\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right] + \frac{\partial}{\partial v} \left[\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right] (-1)$

$= \frac{\partial^2 z}{\partial u^2} - \frac{\partial^2 z}{\partial u \partial v} - \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$

$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$

$\frac{\partial u}{\partial x} = 1$
 $\frac{\partial v}{\partial x} = -1$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

Now, $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0$

$$4 \frac{\partial^2 z}{\partial u \partial v} = 0 \Rightarrow$$

$$\boxed{\frac{\partial^2 z}{\partial u \partial v} = 0}$$

$$DuDv = 0, \quad Du = 0 \quad \text{or} \quad Dv = 0$$

$$\Rightarrow z' = \phi(u) + \phi(v)$$

$$\text{or } D_x^2 - D_y^2 = 0$$

$$\lambda^2 - 1 = 0, \quad \lambda = \pm 1$$

$$\phi_1(y+x) + \phi_2(y-x)$$

Result $\rightarrow$ If the given Eau^n is hyperbolic then the canonical form will be —

$$\boxed{\frac{\partial^2 z}{\partial u \partial v} = \phi(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v})}$$

2. Reduce parabolic Eau^n into Canonical form $\rightarrow$

Working Rule $\rightarrow R\lambda^2 + S\lambda + T = 0 \quad \text{--- (1)}$
 $\lambda_1 = \lambda_2 = \lambda$

This Eau^n have only one chara. root.

Corresponding this Eau^n , characteristic curve is

$$\frac{dy}{dx} + \lambda = 0$$

$$f(x, y) = c_1$$

choose let $u = f(x, y), \quad v = g(x, y) \quad \text{s.t.} \quad \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} \neq 0$

choose $v = g(x, y) \quad \text{s.t.} \quad f + g$ are I.F.

find p, q, r, s, t in terms of u, v , the resulting canonical form will be one of type.

$$\frac{\partial^2 z}{\partial u^2} = \phi\left(z, u, v, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}\right)$$

Or $\frac{\partial^2 z}{\partial v^2} = \phi\left(z, u, v, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}\right)$

Q. Reduce $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$ into canonical. — ①

$$R = 1, S = 2, T = 1$$

$$S^2 = 4RT$$

$$4 = 4 \quad \text{It is parabolic.}$$

$$Rd^2 + Sd + T = 0$$

$$d^2 + 2d + 1 = 0$$

$$(d+1)^2 = 0, \quad d = -1, -1$$

$$\frac{dy}{dx} + d = 0$$

$$\frac{dy}{dx} + 1 = 0 \Rightarrow y - x = c_1$$

choose.

$$u = y - x, \quad v = y + x$$

$$\frac{\partial z}{\partial x} = -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}$$

$$r = z_{uu} - z_{uv} + z_{vv}$$

Now

$$s = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) \frac{\partial u}{\partial y} + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial x} \right) \frac{\partial v}{\partial y}$$

$$= \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial x} \right) + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial x} \right)$$

$$= + \frac{\partial}{\partial u} \left[-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right] (1) + \frac{\partial}{\partial v} \left[-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right] (1)$$

$$= -\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial u \partial v}$$

$$s = \frac{\partial^2 z}{\partial v^2} - \frac{\partial^2 z}{\partial u^2}$$

from ① we have

$$\frac{\partial^2 z}{\partial u^2} = 0$$

$$= \frac{\partial}{\partial u} \left[\frac{\partial z}{\partial x} \right] \frac{\partial y}{\partial y} + \frac{\partial}{\partial v} \left[\frac{\partial z}{\partial x} \right] \frac{\partial v}{\partial v}$$

$$= \frac{\partial}{\partial u} \left[-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right] + \frac{\partial}{\partial v} \left[-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right]$$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

by Eqn. (1)

$$2 \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial v^2} - 2 \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial v^2}$$

$$\Rightarrow 4 \frac{\partial^2 z}{\partial v^2} = 0$$

$$\Rightarrow \boxed{\frac{\partial^2 z}{\partial v^2} = 0}$$

$$D_v^2 = 0$$

$$D_v = 0$$

$$\phi_1(v) + v\phi_2(v)$$

$$\# \quad u_{xx} + 2u_{xy} + u_{yy} = 0$$

(i) parabolic & has characteristic

$$\xi(x, y) = x + 2y + \eta(x, y) = x - 2y$$

(ii) Reducible to Canonical form

$$\frac{\partial^2 u}{\partial \xi^2} = 0, \quad \eta(x, y) = x + y$$

~~Ans~~ parabolic & has General solⁿ, $u = (x-y)f_1(x+y) + f_2(x+y)$

(iii) Reducible to Canonical form -

$$\frac{\partial^2 u}{\partial \eta^2} = 0, \quad \eta(x, y) = x + 2y$$

Solⁿ →

$$D_x^2 + 2D_x D_y + D_y^2 = 0$$

$$d^2 + 2d + 1 = 0$$

$$d = -1, -1$$

$$\left\{ \begin{array}{l} \phi(y+x) + x\phi_2(y+x) \end{array} \right.$$

Reduce into Elliptic form $\rightarrow$

$$R\lambda + S\mu + T\nu + f(x, y, z, p, q) = 0$$

$$S^2 - 4RT < 0$$

$$Rd^2 + Sd + T = 0$$

d_1, d_2 (complex roots)

$$\frac{dy}{dx} + d_1 = 0, \quad \frac{dy}{dx} + d_2 = 0$$

$$f_1(x, y) = c_1, \quad f_2(x, y) = c_2$$

$$u = \frac{f_1 + f_2}{2}, \quad v = \frac{f_1 - f_2}{2i}$$

$$\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} = \phi(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v})$$

Q $\rightarrow$

$$Zxx + x^2 zyy = 0 \quad \text{--- (1)}$$

$$R=1, \quad S=0, \quad T=x^2$$

$$S^2 = 4RT$$

$$0 - 4x^2 < 0$$

$$Rd^2 + Sd + T = 0$$

$$d^2 + x^2 = 0$$

$$d = \pm ix$$

$$\frac{dy}{dx} + ix = 0, \quad \frac{dy}{dx} - ix = 0$$

$$y + \frac{ix^2}{2} = c_1, \quad y - \frac{ix^2}{2} = c_2$$

$$u = y, \quad v = x^2/2$$

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial y} \right) \frac{\partial y}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} \\ &= \frac{\partial z}{\partial u} \cdot 0 + x \frac{\partial z}{\partial v} \end{aligned}$$

$$p = \frac{\partial z}{\partial x} = x \frac{\partial z}{\partial v}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial u} \left[x \frac{\partial z}{\partial v} \right] = x \frac{\partial}{\partial x} \left[\frac{\partial z}{\partial v} \right] + \frac{\partial z}{\partial v}$$

$$Z_{xx} = \frac{\partial}{\partial u} \left(\frac{\partial Z}{\partial x} \right) \frac{\partial u}{\partial x} + \frac{\partial}{\partial v} \left(\frac{\partial Z}{\partial x} \right) \frac{\partial v}{\partial x} + \frac{\partial^2 Z}{\partial v^2}$$

$$= 0 + \frac{\partial}{\partial v} \left[x \frac{\partial Z}{\partial v} \right] + \frac{\partial^2 Z}{\partial v^2} = x \left[\frac{\partial^2 Z}{\partial v^2} \right] + \frac{\partial^2 Z}{\partial v^2}$$

$$Z_{xx} = x^2 \frac{\partial^2 Z}{\partial v^2} + \frac{\partial^2 Z}{\partial v^2} = \left[2v \frac{\partial^2 Z}{\partial v^2} + \frac{\partial Z}{\partial v} \right] \frac{\partial v}{\partial x} = \frac{\partial v}{\partial x} x$$

$$1 = z_y = \left(\frac{\partial Z}{\partial y} \right) = \frac{\partial^2 Z}{\partial u \partial y} \frac{\partial u}{\partial y} + \frac{\partial^2 Z}{\partial v \partial y} \frac{\partial v}{\partial y} = \frac{\partial^2 Z}{\partial u \partial y} \cdot 1 + \frac{\partial^2 Z}{\partial v \partial y} \cdot 0$$

$$z_{yy} = \frac{\partial}{\partial y} \left(\frac{\partial Z}{\partial y} \right) \cdot 1 = \frac{\partial}{\partial u} \left(\frac{\partial Z}{\partial y} \right) \frac{\partial u}{\partial y} + \frac{\partial}{\partial v} \left(\frac{\partial Z}{\partial y} \right) \frac{\partial v}{\partial y}$$

$$z_{yy} = \frac{\partial^2 Z}{\partial u^2} = \frac{\partial^2 Z}{\partial u^2}$$

$$\text{by } \textcircled{1} \quad 2v \frac{\partial^2 Z}{\partial v^2} + x^2 \frac{\partial^2 Z}{\partial u^2} + \frac{\partial Z}{\partial v} = 0$$

$$\Rightarrow 2v Z_{vv} + 2x^2 Z_{uu} + Z_v = 0$$

$$\Rightarrow 2v (Z_{vv} + Z_{uu}) + Z_v = 0$$

$$\Rightarrow \boxed{\frac{\partial^2 Z}{\partial u^2} + \frac{\partial^2 Z}{\partial v^2} = -\frac{1}{2v} \frac{\partial Z}{\partial v}}$$

Classification of Second order PDE with three independent variable $\rightarrow$

Let pde of Second order with three inde. variable is

$$x_1, x_2, x_3$$

$$\sum_{i=1}^3 \sum_{j=1}^3 a_{ij} \frac{\partial^2 z}{\partial x_i \partial x_j} + \sum_{i=1}^3 b_i \frac{\partial z}{\partial x_i} + cz + f(x_1, x_2, x_3) = c$$

where $a_{ij} = a_{ji}$

a_{ij}, b_i, c are funcⁿ of x_1, x_2, x_3

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Eigen values by

$$|A - \lambda I| = 0$$

Result → ① Equ. (1) is Hyperbolic if all the Eigen values are Non zero & have same sign except precisely one of them.

② Equ. (1) is parabola if one of the eigen value is zero i.e. Determinant of $A = 0$

③ Equ. (1) is elliptical if all the eigen values are Non zero and has same sign.

Q.

$$u_{xx} + u_{yy} - u_{zz} = 0$$

$$R = -1, \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & -(1+\lambda) \end{vmatrix}$$

$$\Rightarrow -(1+\lambda)(1-\lambda)^2 = 0, \quad \lambda = 1, 1, -1$$

it is hyperbolic (2 have same sign)

$u_{xx} + u_{yy} + u_{zz} + u_{yz} + u_{zy} = 0$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} (1-\lambda) & 0 & 0 \\ 0 & (1-\lambda) & 1 \\ 0 & 1 & (1-\lambda) \end{vmatrix}$$

$$\Rightarrow -(1-\lambda)[(1-\lambda)^2 - 1] = 0 \Rightarrow 0, 1, 2$$

$\Rightarrow$ Parabolic.

#

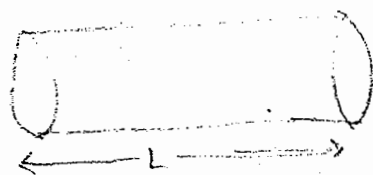
HEAT EQUATION : $\rightarrow$

(1) Heat conduction in infinite Rod : $\rightarrow$ Consider the problem of determining temp. distⁿ in cylindrical rod of length L which has insulated cylindrical surface

$u(x, t) \rightarrow$ Temp

$x \rightarrow$ displacement

$t \rightarrow$ time



[Boundary condition $\rightarrow u(0, t), u(L, t)$
 Initial " $\rightarrow u(x, 0) = f(x)$]

Insulated at the end $u_x(0, t) = 0$
 $u_x(L, t) = 0$

#

$$u_{xx} = \frac{1}{\alpha} u_t \quad 0 < x < L, \quad t > 0 \quad \text{--- (1)}$$

$$u(x, 0) = \phi(x), \quad 0 \leq x \leq L \quad \text{--- (2)}$$

$$u(0, t) = f_1(t), \quad u(L, t) = f_2(t), \quad t \geq 0 \quad \text{--- (3)}$$

Equ. ϕ, f_1, f_2 are assume to be continuous and satisfy the relation —

$$\phi(0) = f_1(0) \quad + \quad \phi(L) = f_2(0)$$

The problem is to find $u(x, t)$ defined and continuous in closed strip $0 \leq x \leq L$ + $t \geq 0$

and satisfy Equ. (1) in the interior strip

Satisfies equ. (2) & (3).

Result $\rightarrow$ $u(x, t)$ is solution if ϕ is twice continuously diff. and ϕ'' is piece wise continuous.

Maxima-Minima THM $\rightarrow$ let T be any no. > 0

Suppose $u(x, t)$ is continuous in closed rectangle

$$R: 0 \leq x \leq L, \quad 0 \leq t \leq T$$

$u(x, t)$ satisfies equ. (1) in $0 < x < L, 0 < t \leq L$
and u attain its maximum & Minimum value on
the lower base i.e. $t=0$ and on the vertical
sides $x=0$, or $x=L$ of R .

Result $\rightarrow$ ① There is atmost one solⁿ of this Equⁿ.
i.e. Unique Solution exist.

05/oct/2014

①
$$\left. \begin{array}{l} u(x, 0) = \phi(x), \quad 0 \leq x \leq L \\ u(0, t) = 0, \quad u(L, t) = 0 \end{array} \right\} \begin{array}{l} \text{Initial} \\ \text{Boundary} \end{array}$$

Heat Eqⁿ
$$u_{xx} = \frac{1}{\alpha} u_t, \quad 0 < x < L, t > 0$$

METHOD OF SEPARATION OF VARIABLE

$$u(x, t) = H(x) h(t)$$

$$u_{xx} = H''(x) h(t)$$

$$u_t = H(x) h'(t)$$

$$H''(x) h(t) = \frac{1}{\alpha} h'(t) H(x)$$

$$\boxed{\frac{H''(x)}{H(x)} = \frac{1}{\alpha} \frac{h'(t)}{h(t)} = K}$$

Case-I $\rightarrow$

$$K = 0$$

$$\frac{H''(x)}{H(x)} = 0$$

$$\Rightarrow H(x) = -C_1(x) + C_2$$

$$u(x, t) = H(x) \cdot G(t)$$

$$u(0, t) = H(0) \cdot G(t) = 0$$

$$\boxed{H(0) = 0}$$

$$u(L, T) = H(L) \cdot G(T) = 0$$

$$\boxed{H(L) = 0}$$

$$H(x) = C_1 x + C_2$$

$$H(0) = C_2 \Rightarrow \boxed{C_2 = 0}$$

$$H(L) = C_1 L + C_2$$

$$0 = C_1 L + 0 \Rightarrow \boxed{C_1 = 0} \quad \text{Trivial sol}^n \quad \times$$

if $G(t) = 0$ Then $u(x, t)$
Then Trivial solⁿ
But we find Non-trivial
solⁿ only.

Case-2

$$K > 0, \quad K = \mu^2, \quad (\mu \neq 0)$$

$$\frac{H''(x)}{H(x)} = \mu^2 \Rightarrow H''(x) - \mu^2 H(x) = 0$$

$$\lambda^2 - \mu^2 = 0$$

$$\lambda = \pm \mu$$

$$H(x) = C_1 e^{\mu x} + C_2 e^{-\mu x}$$

$$H(0) = C_1 + C_2 \Rightarrow C_1 + C_2 = 0$$

$$H(L) = C_1 e^{\mu L} + C_2 e^{-\mu L}$$

$$0 = C_1 e^{\mu L} + C_2 e^{-\mu L} \Rightarrow \boxed{C_1 = 0}, \boxed{C_2 = 0}$$

$$H(x) = 0 \quad \text{only Trivial sol}^n \quad \times$$

Case-3

$$K < 0, \quad K = -\mu^2 \quad (\mu \neq 0)$$

$$H''(x) + \mu^2 H(x) = 0$$

$$\lambda^2 + \mu^2 = 0 \Rightarrow \lambda = \pm \mu i$$

$$H(x) = C_1 \cos \mu x + C_2 \sin \mu x$$

$$H(0) = C_1 + 0 \Rightarrow \boxed{C_1 = 0}$$

$$H(x) = C_2 \sin \mu x$$

$$H(L) = C_2$$

$$H(L) = C_2 \sin \mu L \Rightarrow \boxed{C_2 = 0} \quad \text{Then Trivial}$$

$$\therefore \sin \mu L = 0 \Rightarrow \boxed{\mu L = n\pi}, \quad n = 1, 2, \dots$$

$$u = \frac{n\pi}{L}$$

$$H(x) = \sum \sin \frac{n\pi}{L} x$$

$$H_n(x) = c_n \sin \frac{n\pi}{L} x ; n = 1, 2, \dots$$

$$\text{Now } \frac{u'(t)}{\alpha u(t)} = -\frac{n^2 \pi^2}{L^2}$$

$$u'(t) + \alpha \frac{n^2 \pi^2}{L^2} u(t) = 0$$

$$\Rightarrow \lambda = -\alpha \frac{n^2 \pi^2}{L^2}$$

$$u(t) = c \cdot e^{-\alpha \frac{n^2 \pi^2}{L^2} t}, \quad n = 0, 1, 2, \dots$$

$$u_n(t) = E_n e^{-\alpha \frac{n^2 \pi^2}{L^2} t}, \quad n = 0, 1, 2, \dots$$

$$u_n(x, t) = f_n(t) \cdot u_n(t)$$

$$= c_n \sin \frac{n\pi x}{L} \cdot E_n e^{-\alpha \frac{n^2 \pi^2}{L^2} t}, \quad n = 1, 2, \dots$$

$$u_n(x, t) = D_n \sin \frac{n\pi x}{L} \cdot e^{-\alpha \frac{n^2 \pi^2}{L^2} t}$$

Initial condition-

$$u(x, 0) = \phi(x)$$

$$\phi(x) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{L}$$

By Fourier Series

$$D_n = \frac{2}{L} \int_0^L \phi(x) \cdot \sin \left(\frac{n\pi x}{L} \right) dx$$

Required Solⁿ is -

$$u(x, t) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{L} \cdot e^{-\alpha \frac{n^2 \pi^2}{L^2} t}$$

$$D_n = \frac{2}{L} \int_0^L \phi(x) \sin \frac{n\pi x}{L} dx$$

This is Heat Eqⁿ.

$$a_n = \frac{1}{L} \int_0^L \phi(x) \sin \frac{n\pi x}{L} dx$$

$$\phi(x) = \sum a_n \sin \frac{n\pi x}{L}$$

$$a_n = \frac{1}{L} \int_0^L \phi(x) \sin \frac{n\pi x}{L} dx$$

Q.

$$u_{xx} = u_t, \quad u(0, t) = u(\pi, t) = 0$$

$$u(x, 0) = 3 \sin 2x$$

Sol

$$\alpha = 1, \quad L = \pi$$

$$\Rightarrow \sum_{n=1}^{\infty} D_n \frac{\sin \frac{n\pi x}{\pi}}{\pi} \cdot e^{-n^2 t}$$

$$\Rightarrow \sum_{n=1}^{\infty} D_n \sin nx \cdot e^{-n^2 t}$$

$$u(x, 0) = 3 \sin 2x$$

$$3 \sin 2x = \sum_{n=1}^{\infty} D_n \sin 2x \cdot 1$$

$$D_2 = 3, \quad D_i = 0, \quad i \neq 2$$

$$u(x, t) = 3 \sin 2x e^{-4t}$$

OR

$$D_n = \frac{2}{\pi} \int_0^{\pi} 3 \sin 2x \cdot \sin nx \, dx$$

$$= \frac{2}{\pi} \times 3 \int_0^{\pi} \sin 2x \cdot \sin nx \, dx$$

$$D_2 = \frac{2}{\pi} \times 3 \times \frac{\pi}{2}$$

$$\int_0^{\pi} \sin mx \sin nx \, dx = \begin{cases} 0 & m \neq n \\ \pi/2 & m = n \end{cases}$$

$$D_2 = 3$$

#

$$u_t = u_{xx} \quad 0 < x < \pi, \quad t > 0$$

$$u(0, t) = u(\pi, t) = 0$$

$$u(x, 0) = \sin x + \sin 2x$$

$$(i) \quad u(x, t) \rightarrow 0 \quad \text{as } t \rightarrow \infty \quad \forall x \in (0, \pi)$$

$$(ii) \quad t^2 u(x, t) \rightarrow 0 \quad \text{as } t \rightarrow \infty \quad \forall x \in (0, \pi)$$

$$(iii) \quad e^t u(x, t) \text{ is bdd } \forall x \in (0, \pi)$$

$$(iv) \quad e^{2t} u(x, t) \rightarrow 0 \quad \text{as } t \rightarrow \infty \quad \forall x \in (0, \pi)$$

Solⁿ $u(x, t) = \sum D_n \sin \frac{n\pi x}{L} \cdot e^{-\frac{n^2 \pi^2}{L^2} t}$

$$\alpha = 1, L = \pi$$

$$= \sum D_n \sin n x \cdot e^{-n^2 t}$$

$$D_1 = 1, \phi(x),$$

$$u(x, 0) = \sin x + \sin 2x$$

$$D_1 = 1, D_2 = 1$$

$$u(x, t) = \sin x e^{-t} + \sin 2x e^{-4t}$$

$$t \rightarrow \infty, u(x, t) \rightarrow 0$$

$$t \rightarrow \infty, u(x, t) \rightarrow 0 \quad (\infty \cdot 0 = 0)$$

$$e^t u(x, t) = \sin x + \sin 2x e^{-3t}, \quad t > 0$$

It is bounded.

Heat Equation when both ends are insulated

$$u_{xx} = \frac{1}{\alpha} u_t \quad 0 \leq x \leq l, t > 0$$

$$u_x(0, t) = 0$$

$$u_x(l, t) = 0$$

$$u(x, 0) = \phi(x)$$

$$u(x, t) = \frac{D_0}{2} + \sum_{n=1}^{\infty} D_n \cos\left(\frac{n\pi x}{l}\right) \cdot e^{-\alpha \left(\frac{n^2 \pi^2}{l^2}\right) t}$$

$$D_0 = \frac{2}{l} \int_0^l \phi(x) dx$$

$$D_n = \frac{2}{l} \int_0^l \phi(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad n = 1, 2$$

GATE

Q.

$$u_{xx} = u_t$$

$$u_x(0, t) = 0$$

$$u_x(l, t) = 0$$

$$u(x, 0) = 2 \cos^2 \pi x$$

$$(b) \quad 1 - e^{-4\pi^2 t} \cos 2\pi x$$

$$(c) \quad 1 + e^{-4\pi^2 t} \cos 2\pi x$$

$$(d) \quad 1 - e^{-4\pi^2 t} \sin 2\pi x$$

$$(e) \quad 1 + e^{-4\pi^2 t} \sin 2\pi x$$

Solⁿ $\rightarrow \alpha = 1, \quad l = 1$

$$u(x, 0) = \frac{D_0}{2} + \sum D_n \cos(n\pi x) e^{-n^2 \pi^2 t}$$

$$u(x, 0) = 1 + \cos 2\pi x$$

$$1 + \cos 2\pi x = \frac{D_0}{2} + \sum D_n \cos n\pi x$$

$$D_0 = 2, \quad D_2 = 1$$

$$u(x, t) = 1 + \cos 2\pi x e^{-4\pi^2 t}$$

IInd $\rightarrow$

Put $t = 0$

$$u(x, 0) = 1 + \cos 2\pi x \quad (\text{By option})$$

NON-HOMOGENEOUS HEAT EQUATION $\rightarrow$

$$u_t - l u_{xx} = f(x, t)$$

$$u(0, t) = u(l, t) = 0, \quad u(x, 0) = \phi(x)$$

$$u(x, t) = \sum D_n \sin\left(\frac{n\pi x}{l}\right) e^{-\alpha\left(\frac{n^2 \pi^2}{l^2}\right)t} + \sum_{n=1}^{\infty} \sin \frac{n\pi x}{l} \int_0^t e^{-\alpha\left(\frac{n^2 \pi^2}{l^2}\right)(t-s)} f_n(s) ds.$$

where $D_n = \frac{2}{l} \int_0^l \phi(x) \sin\left(\frac{n\pi x}{l}\right) dx$

$$f_n(t) = \frac{2}{l} \int_0^l f(x, t) \sin\left(\frac{n\pi x}{l}\right) dx$$

Exp.

$$u_t - u_{xx} = e^t \sin 3x$$

$$u(0, t) = u(\pi, t) = 0, \quad t > 0$$

$$u(x, 0) = \sin x, \quad \text{exist } 0 < x < \pi$$

Solⁿ

$$u(x, t) = \sum_{n=1}^{\infty} D_n \sin\left(\frac{n\pi x}{l}\right) \cdot e^{-\frac{n^2 \pi^2}{l^2} t}$$

$$u(x, t) = D_1 \sin x e^{-t}$$

$$f_n(t) = \frac{2}{\pi} \int_0^{\pi} e^{-t} \sin 3x \cdot \sin nx \, dx$$

$$f_3(t) = \frac{2}{\pi} \times e^{-t} \times \frac{\pi}{2}$$

$$f_3(t) = e^{-t}$$

$$u(x, t) = D_1 \sin x e^{-t} + \sin 3x \int_0^t e^{g(s-t)} e^{-s} \, ds$$

$$= \sin x e^{-t} + \frac{\sin 3x}{\theta} e^{-9t} (e^{\theta t} - 1)$$

INITIAL Value problem for one-dimensional Heat Equation $\rightarrow$

$$u_t - u_{xx} = 0 \quad -\infty < x < \infty$$

$$u(x, 0) = \phi(x) \quad -\infty < x < \infty$$

Here ϕ is continuous over $[-\infty, \infty]$ while the desired solution $u(x, t)$ is to be continuous in closed upper half space. i.e. $t \geq 0, -\infty < x < \infty$

This problem has infinite solution if we impose one more condition that $u(x, t)$ is bounded. it will have atmost only one solⁿ.

RESULT / THM → Suppose $u(x, t)$ is continuous and bounded in closed half upper space. i.e. $-\infty < x < \infty$, $t \geq 0$ and also satisfied Heat Equⁿ

$$M = \sup_{-\infty < x < \infty} u(x, 0), \quad m = \inf_{-\infty < x < \infty} u(x, 0)$$

Then $\boxed{m \leq u(x, t) \leq M} \quad \forall (x, t)$
in $-\infty < x < \infty, t \geq 0$

(2) Let $\phi(x)$ be conti. & bdd in $-\infty < x < \infty$

$$u(x, t) = \begin{cases} \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-s)^2}{4t}} \phi(s) ds, & t > 0 \\ \phi(x), & t = 0 \end{cases}$$

12/Oct/2014.

I method $u_x = 4u_y$

$$u_x - 4u_y = 0$$

$$D_x - 4D_y = 0$$

$$u = \phi_1(y + 4x)$$

$$8e^{-3y} = \phi_1(y)$$

$$u(x, y) = 8e^{-3(y+4x)}$$

$$H'(x) - K H(x) = 0$$

$$H(x) = C e^{Kx}$$

$$V(x, y) = A e^{Kx + \frac{K}{4}y}$$

$$u(0, y) = 8e^{-3y}$$

$$8e^{-3y} = A e^{\frac{K}{4}y}$$

$$8e^{-3y} \cdot \frac{K}{4} = -3 \Rightarrow \boxed{K = -12}, \quad \boxed{A = 8}$$

$$8e^{-3y} = 8e^{-3y}$$

$$u(x, y) = 8e^{-3(y+4x)}$$

$$u(0, y) = 8e^{-3y}$$

IInd → $u(x, y) = H(x) G(y)$

$$H'(x) G(y) = 4 H(x) G'(y)$$

$$\frac{H'(x)}{H(x)} = 4 \frac{G'(y)}{G(y)} = K$$

$$\frac{H'(x)}{H(x)} = K$$

$$4 \frac{G'(y)}{G(y)} = K$$

$$4G'(y) - K G(y) = 0$$

$$G(y) = b e^{\frac{K}{4}y}$$

① 0+

$$\# \quad y^2 u_{xy} + 3x^2 u = 0$$

$$u(x, y) = H(x) \cdot G(y)$$

$$u_x = H'(x) G(y)$$

$$u_{xy} = H'(x) G'(y)$$

$$\Rightarrow y^2 H'(x) G'(y) + 3x^2 H(x) G(y) = 0$$

$$\Rightarrow y^2 H'(x) G'(y) = -3x^2 H(x) G(y)$$

$$\Rightarrow \frac{-1}{3x^2} \frac{H'(x)}{H(x)} = \frac{y^2 G'(y)}{G(y)}$$

$$\Rightarrow \frac{-1}{3x^2} \frac{H'(x)}{H(x)} = K$$

$$\Rightarrow H'(x) + 3x^2 K H(x) = 0$$

$$\Rightarrow H(x) = e^{\int 3x^2 K} = e^{x^3 K}$$

#

Wave Equation →

$$u_{xx} = \frac{1}{c^2} u_{tt}, \quad 0 \leq x \leq l$$

 $u(x, t) \rightarrow$ Disturbance

 $c \rightarrow$ Speed of wave

 $x \rightarrow$ Displacement

 $t \rightarrow$ Time

$$\left. \begin{aligned} u(0, t) &= 0 \\ u(l, t) &= 0 \end{aligned} \right\} \rightarrow \text{B.C.}$$

$$u(x, 0) = f(x)$$

$$u_t(x, 0) = g(x)$$

$$u(x, 0) = 0, \quad u_t(x, 0) = g(x)$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = 0$$

Solution is -

$$u(x, t) = \sum_{n=1}^{\infty} D_n \sin\left(\frac{n\pi x}{l}\right) \left[\cos\left(\frac{n\pi ct}{l}\right) + E_n \sin\left(\frac{n\pi ct}{l}\right) \right]$$

Case I

gen $\neq 0$

$$D_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$E_n = \frac{2}{n\pi c} \int_0^l g(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$u(0, t) = 0$$

$$u(l, t) = 0$$

$$u(x, 0) = f(x)$$

$$u_t(x, 0) = g(x)$$

$$u(x, 0) = f(x)$$

$$u_t(x, 0) = g(x)$$

Q.

$$u_{xx} = u_{tt} \quad 0 \leq x \leq \pi$$

$$u(0, t) = u(\pi, t) = 0$$

$$u(x, 0) = (\sin x - \sin 2x)$$

$$u_t(x, 0) = 0$$

$$(i) \quad \frac{1}{2} (\sin x \cos t - \sin 2x \cos 2t)$$

$$(ii) \quad \frac{1}{3} (\sin x \cos t - \sin 2x \cos 2t)$$

$$(iii) \quad \frac{1}{4} (\sin x \cos t - \sin 2x \cos 2t)$$

$$(iv) \quad \text{None of these}$$

Case II. $f(x) = 0, u_{tt} = c^2 u_{xx}$

$$u(0, t) = u(l, t) = 0$$

$$u(x, 0) = 0$$

$$u_t(x, 0) = g(x)$$

Soln

$$u(x, t) = \sum_{n=1}^{\infty} D_n \sin\left(\frac{n\pi x}{l}\right) \sin\left(\frac{n\pi ct}{l}\right)$$

$$D_n = \frac{2}{n\pi c} \int_0^l g(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

Case III. $f(x) \neq 0, g(x) \neq 0$

$$u_{tt} = c^2 u_{xx}$$

$$u(0, t) = u(l, t) = 0$$

$$u(x, 0) = f(x) \quad \& \quad u_t(x, 0) = g(x)$$

Soln

$$u(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \left[D_n \cos\left(\frac{n\pi ct}{l}\right) + E_n \sin\left(\frac{n\pi ct}{l}\right) \right]$$

$$\text{where } D_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$E_n = \frac{2}{n\pi c} \int_0^l g(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

Solⁿ

$$u(x, t) = \sum D_n \sin nx \cos nt$$

$$u(x, 0) = \sin x - \sin 2x = \sum D_n \sin nx$$

$$D_1 = 1$$

$$D_2 = -1$$

$$D_i = 0 \quad \forall i \geq 3$$

$$\Rightarrow \text{last } \sin x - \sin 2x \cos 2t$$

Q. (11)

Wave Equation of Infinite String : $\rightarrow$

$$u_{xx} = \frac{1}{c^2} u_{tt}$$

$$-\infty < x < \infty$$

$$t > 0$$

$$u(x, 0) = f(x)$$

$$u_t(x, 0) = g(x)$$

$$u(x, t) = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\xi) d\xi \quad (i)$$

#

D'Alembert Solution : $\rightarrow$

Exp.

$$u_{tt} - u_{xx} = 0$$

$$u(x, 0) = \sin x \rightarrow f(x)$$

$$u_t(x, 0) = 1 \rightarrow g(x)$$

$$u(\pi, \pi/2) = ?$$

$$u(x, t) = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\xi) d\xi$$

$$\frac{1}{2} \int_{x-ct}^{x+ct} 1 \cdot d\xi$$

$$= \frac{1}{2} [\sin(x+t) + \sin(x-t)] + \frac{1}{2} [(x+t) - (x-t)]$$

$$u(x, t) = \frac{1}{2} [\sin(x+t) + \sin(x-t)] + t$$

$$u(\pi, \pi/2) = \frac{1}{2} [\sin(\pi + \pi/2) + \sin(\pi - \pi/2)] + \pi/2$$

$$= \frac{1}{2} [-\sin \pi/2 + \sin \pi/2] + \pi/2$$

$$= \pi/2$$

RESULT → If $u(x, t)$ is a solution of equ. (1) and A, B, C, D are vertices of any $\square$ whose sides are characteristic curve.

i.e. (parallel to $x-ct$ & $x+ct$)

Then-

$$u(A) + u(C) = u(B) + u(D)$$

Q →

$$u_{tt} - u_{xx} = 0$$

$$u\left(\frac{1}{2}, 0\right) = \frac{1}{4}, \quad u\left(1, \frac{1}{2}\right) = 1$$

$$u\left(0, \frac{1}{2}\right) = \frac{1}{2}, \quad u\left(\frac{1}{2}, 1\right) = ?$$

Solⁿ →

$$D^2 - 1 = 0 \Rightarrow \left(\frac{1}{2}\right)^2 = 1 = \frac{1}{4} - 1 = +(\dots)$$

$$\frac{1}{4} + \frac{1}{2} = 1 + u\left(\frac{1}{2}, 1\right)$$

$$= \frac{3}{4} = 1 + u\left(\frac{1}{2}, 1\right)$$

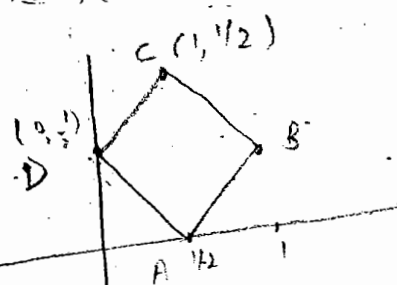
$$\Rightarrow \boxed{u\left(\frac{1}{2}, 1\right) = -\frac{1}{4}}$$

$$u(C) + u(A) = u(B) + u(D)$$

$$u\left(\frac{1}{2}, 0\right) + u\left(\frac{1}{2}, 1\right) = u\left(0, \frac{1}{2}\right) + u\left(1, \frac{1}{2}\right)$$

$$\Rightarrow \frac{1}{4} + u\left(\frac{1}{2}, 1\right) = \frac{1}{2} + 1$$

$$\Rightarrow u\left(\frac{1}{2}, 1\right) = \frac{3}{2} - \frac{1}{4} = \frac{6-1}{4} = \frac{5}{4} \text{ Ans.}$$



Method of characteristic for Non-linear Equⁿ:

$$\frac{dx}{f_p} = \frac{dy}{f_q} = \frac{dz}{p f_p + q f_q} = \frac{dp}{-f_x - p f_z} = \frac{dq}{-f_y - q f_z} = ds$$

$$\Rightarrow \text{Given, } x(\tau, 0) = \gamma_1(\tau)$$

$$y(\tau, 0) = \gamma_2(\tau)$$

$$z(\tau, 0) = \gamma_3(\phi_1(\tau))$$

$$\frac{dx}{ds} = f_p, \quad \frac{dy}{ds} = f_q, \quad \frac{dz}{ds} = p f_p + q f_q$$

$$p > \frac{dp}{ds} = \frac{dq}{ds}$$

$$p(\tau, 0) = \psi_1(\tau)$$

$$q(\tau, 0) = \psi_2(\tau)$$

$$I^{\Delta t} \quad F(x, y, z, p, q) = 0$$

$$f(\gamma_1(\tau), \gamma_2(\tau), \phi_1(\tau), \psi_1(\tau), \psi_2(\tau)) = 0$$

$$\phi'(\tau) = \psi_1(\tau) \gamma_1'(\tau) + \psi_2(\tau) \gamma_2'(\tau)$$

$$\psi_1 + \psi_2$$

Q.

$$4x^2 + uy^2 = 1$$

find the characteristics? $\begin{cases} x^2 + y^2 = 1 \\ z = 0 \end{cases}$

Solⁿ

$$F = p^2 + q^2 - 1$$

$$\frac{dx}{ds} = 2p, \quad \frac{dy}{ds} = 2q, \quad \frac{du}{ds} = 2[p^2 + q^2]$$

Initial condi. $(\cos \tau, \sin \tau, 0)$

$$x(\tau, 0) = \cos \tau = \gamma_1(\tau)$$

$$y(\tau, 0) = \sin \tau = \gamma_2(\tau)$$

$$u(\tau, 0) = 0 = \phi_1(\tau)$$

$$\psi_1(\tau, 0) =$$

$$\psi_2(\tau, 0) =$$

$$\frac{dp}{ds} = 0$$

$$p = c_1$$

$$\frac{dq}{ds} = 0$$

$$q = c_2$$

$$p(\tau, 0) = \psi_1$$

$$q_2$$

By (1)

$$\psi_1^2 + \psi_2^2 = 1 \quad \text{--- (1)}$$

$$0 = \psi_1(\epsilon)(-\sin \epsilon) + \psi_2(\epsilon) \cos \epsilon \quad \text{--- (2)}$$

$$\psi_1(\epsilon, 0) = \cos \epsilon$$

$$\psi_2(\epsilon, 0) = \sin \epsilon$$

$$-\cos \epsilon$$

$$-\sin \epsilon$$

Non-homogeneous
Wave Eqn

$$p = \cos \epsilon$$

$$a = \sin \epsilon$$

$$\frac{dx}{ds} = 2 \cos \epsilon$$

$$x(\epsilon, s) = 2 \cos \epsilon s + C_1$$

$$y(\epsilon, s) = 2 \sin \epsilon s + C_2$$

$$\Rightarrow x = 2 \cos \epsilon s + \cos \epsilon$$

$$y = 2 \sin \epsilon s + \sin \epsilon$$

$$u = 2(\cos^2 \epsilon + \sin^2 \epsilon) s$$

$$\left\{ \begin{array}{l} x = \cos \epsilon [2s+1] \\ y = \sin \epsilon [2s+1] \\ u = 2s \end{array} \right\}$$

$$\text{Let } -c^2 u_{xx} = f(x, t), \quad 0 < x < l, \quad t > 0$$

$$u(0, t) = u(l, t) = 0$$

$$u(x, 0) = f(x)$$

$$u_t(x, 0) = g(x)$$

$$u(x, t) = \sum_{n=1}^{\infty} \frac{\sin(n\pi x)}{n\pi} \left[D_n \cos \frac{n\pi c t}{l} + E_n \sin \frac{n\pi c t}{l} \right] + \sum_{n=1}^{\infty} \frac{\sin(n\pi x)}{n\pi} \int_0^t g(x) \sin \frac{n\pi c(t-\tau)}{l} d\tau$$

$$D_n(t) = \frac{2}{n\pi c} \int_0^t g(x) \sin \frac{n\pi c(t-\tau)}{l} d\tau$$

Ans.

19/10/2014

Laplace Equation $\rightarrow$ (Heat Eqn in two dimension nothing but Laplace)

$$u_{xx} + u_{yy} = \frac{1}{k} u_t \quad \text{if } u_t = 0$$

$$\text{then } u_{xx} + u_{yy} = 0$$

Dirichlet problem of rectangle $\rightarrow$ In this problem we have to find a harmonic function u in a given domain D . Such that u is prescribed on the boundary B of domain D . where D is region bounded by simple closed curve.

Consider a harmonic Eqn in u ,

$$u_{xx} + u_{yy} = 0$$

Boundary conditions are —

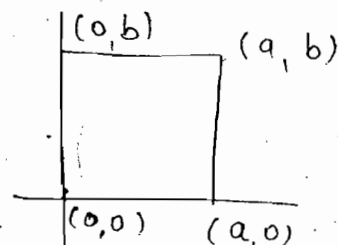
$$u(x, 0) = f(x)$$

$$u(x, b) = 0$$

$$u(0, y) = 0$$

$$u(a, y) = 0$$

$$\begin{array}{ccc} g(x) & 0 & 0 \\ 0 & -f(y) & 0 \\ 0 & 0 & f(y) \end{array}$$



Consider

$$u_{xx} + u_{yy} = 0$$

$$0 \leq x \leq a, \quad 0 \leq y \leq b$$

$$u(x, 0) = f(x)$$

$$u(x, b) = 0$$

$$u(0, y) = 0$$

$$u(a, y) = 0$$

$$u(x, y) = \psi(x) \cdot H(y)$$

$$\frac{\psi''(x)}{\psi(x)} = -\frac{H''(y)}{H(y)} = \lambda$$

$$\begin{array}{l} u(x, b) = 0 \\ \psi(x) \cdot H(b) = 0 \end{array}$$

$$\Rightarrow H(b) = 0$$

$$u(a, y) = 0$$

$$\psi(a) H(y) = 0$$

$$\Rightarrow \psi(a) = 0$$

$$u(a, y) = 0$$

$$\psi(a) \cdot H(y) = 0$$

$$\Rightarrow \psi(a) = 0$$

For $\lambda = 0$ + $\lambda > 0$

It gives trivial solution

for $\lambda = -\mu^2$ ($\mu \neq 0$)

$$\psi''(x) = -\mu^2 \psi(x)$$

$$\psi''(x) + \mu^2 \psi(x) = 0$$

$$\psi(x) = C_1 \cos \mu x + C_2 \sin \mu x$$

$$\psi(0) = 0 \Rightarrow C_1 = 0$$

$$\psi(a) = 0$$

$$C_2 \sin \mu a = 0$$

$$\sin \mu a = 0$$

$$\mu a = n\pi, \quad n = 1, 2, 3, \dots$$

$$\lambda = -\frac{n^2 \pi^2}{a^2}$$

$$\therefore \psi_n(x) = C_n \sin \frac{n\pi x}{a}$$

$$-\frac{H''(y)}{H(y)} = -\frac{n^2 \pi^2}{a^2}$$

$$\ddot{H}(y) - \frac{\pi^2 \pi^2}{a^2} H(y) = 0$$

$$H(y) = c_1 \cosh \frac{n\pi}{a} y + c_2 \sinh \frac{n\pi}{a} y \quad \text{--- (2)}$$

$$H(b) = 0$$

$$c_1 \cosh \frac{n\pi}{a} b + c_2 \sinh \frac{n\pi}{a} b = 0$$

$$c_2 = - \frac{c_1 \cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}}$$

Put in equ. (2)

$$H(y) = c_1 \cosh \frac{n\pi y}{a} - c_1 \cosh \frac{n\pi b}{a} \left(\frac{\sinh \frac{n\pi y}{a}}{\sinh \frac{n\pi b}{a}} \right)$$

$$H(y) = \frac{c_1 \cosh \frac{n\pi y}{a} \cdot \sinh \frac{n\pi b}{a} - c_1 \cosh \frac{n\pi b}{a} \cdot \sinh \frac{n\pi y}{a}}{\sinh \frac{n\pi b}{a}}$$

$$= \frac{c_1}{\sinh \frac{n\pi b}{a}} \left[\sinh \frac{n\pi}{a} (y-b) \right]$$

$$H_n(y) = B_n \sinh \frac{n\pi}{a} (b-y)$$

$$\therefore u_n(x, y) = c_n \sinh \frac{n\pi x}{a} \cdot B_n \sinh \frac{n\pi}{a} (b-y)$$

$$u_n(x, y) = D_n \sinh \frac{n\pi x}{a} \cdot \sinh \frac{n\pi}{a} (b-y)$$

$$u(x, y) = \sum_{n=1}^{\infty} D_n \sinh \frac{n\pi x}{a} \sinh \frac{n\pi}{a} (b-y)$$

NOTE → If $u(x, 0) = f(x)$

$$f(x) = \sum_{n=1}^{\infty} D_n \sinh \frac{n\pi x}{a} \cdot \sinh \frac{n\pi b}{a}$$

$$f(x) = \sum_{n=1}^{\infty} D_n \sinh \frac{n\pi b}{a} \cdot \sinh \frac{n\pi x}{a}$$

where $D_n \sinh \frac{n\pi b}{a} = \frac{z}{a} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$

$$D_n = \frac{z}{a \sinh \frac{n\pi b}{a}} \int_0^a f(x) \cdot \sin \frac{n\pi x}{a} dx$$

Eg $\rightarrow$ Ist

$$u_{xx} + u_{yy} = 0$$

$$u(x, 0) = f(x)$$

$$u(x, b) = 0, \quad u(0, y) = 0$$

$$u(a, y) = 0$$

$$u(x, y) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{a} \cdot \sinh \frac{n\pi}{a} (b-y)$$

$$D_n = \frac{z}{a \sinh \frac{n\pi b}{a}} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$$

IInd $\rightarrow$

$$u(x, 0) = 0$$

$$u(x, b) = f(x)$$

$$u(a, y) = 0, \quad u(0, y) = 0$$

$$u(x, y) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi y}{a}$$

where $D_n = \frac{z}{a \sinh \frac{n\pi b}{a}} \int_0^a f(x) \cdot \sin \frac{n\pi x}{a} dx$

IIIrd $\rightarrow$

$$u(x, 0) = 0$$

$$u(x, b) = 0$$

$$u(0, y) = 0$$

$$u(a, y) = f(y) \quad \text{Then}$$

$$u(x, y) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi y}{b} \sinh \frac{n\pi x}{b}$$

where $D_n = \frac{z}{b \sinh \frac{n\pi a}{b}} \int_0^b f(y) \cdot \frac{\sin n\pi y}{b} dy$

IVth →

$$u(x, 0) = 0$$

$$u(x, b) = 0$$

$$u(0, y) = f(y)$$

$$u(a, y) = 0$$

$$u(x, y) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi y}{b} \sinh \frac{n\pi}{b} (a-x)$$

$$\text{where } D_n = \frac{2}{b \sinh \frac{n\pi a}{b}} \int_0^b f(y) \sin \frac{n\pi y}{b} dy$$

Q →

$$u_{xx} + u_{yy} = 0$$

$$x \in (0, \pi), \quad y \in (0, \pi)$$

$$u(x, 0) = u(x, \pi) = u(0, y) = 0$$

Any solution of this boundary value problem is of type —

(a) $\sum_{n=1}^{\infty} a_n \sinh nx \cdot \sin ny$

(b) $\sum_{n=1}^{\infty} a_n \cosh nx \cdot \sin ny$

(c) $\sum_{n=1}^{\infty} a_n \sinh nx \cdot \cos ny$

(d) $\sum_{n=1}^{\infty} a_n \sin nx \cdot \sinh ny$

Same Que. →

$$u_x(\pi, y) = \sin y$$

$$u(x, \pi/2) = ?$$

(a) $\frac{\pi}{2} (e^{\pi} - e^{-\pi}) (e^x - e^{-x})$

(b) $\frac{\pi (e^x + e^{-x})}{(e^x - e^{-x})}$

(c) $\frac{e^x - e^{-x}}{e^{\pi} + e^{-\pi}}$

(d) $\frac{\pi}{2} (e^x - e^{-x}) (e^{\pi} + e^{-\pi})$

Solⁿ $\rightarrow u(x, y) = \sum_{n=1}^{\infty} a_n \sinh nx \cdot \sin ny$

where

$$u_x(x, y) = \sum_{n=1}^{\infty} a_n (n \cosh nx \cdot \sin ny)$$

$$u_x(\pi, y) = \sum_{n=1}^{\infty} a_n \cosh n\pi \cdot \sin ny$$

$$\sin y = \sum_{n=1}^{\infty} a_n \cosh n\pi \cdot \sin ny$$

exist only for $n=1$

$$a_n \cosh n\pi = \frac{2}{\pi} \int_0^{\pi} \sin y \sin ny \, dy$$

$$a_1 \cosh \pi = \frac{2}{\pi} \int_0^{\pi} \sin^2 y \, dy$$

$$a_1 \cosh \pi = 1$$

$$a_1 = \frac{2}{e^{\pi} + e^{-\pi}}$$

$$\therefore u(x, y) = a_1 \sinh x \cdot \sin y$$

$$u(x, \pi/2) = \frac{\sinh x}{e^{\pi} + e^{-\pi}}$$

$$u(x, \frac{\pi}{2}) = \frac{(e^x - e^{-x})/2}{2(e^{\pi} + e^{-\pi})} = \frac{e^x - e^{-x}}{e^{\pi} + e^{-\pi}}$$

Ex $\rightarrow$

$$u_{xx} + u_{yy} = 0$$

$$u(0, y) = u(a, y) = 0$$

$$u(x, 0) = f(x)$$

$$u(x, y) \rightarrow 0 \text{ as } y \rightarrow \infty$$

$$u(x, y) = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{a} \cdot e^{-\frac{n\pi y}{a}}$$

where $D_n = \frac{2}{a} \int_0^a f(x) \cdot \sin \frac{n\pi x}{a} \, dx$

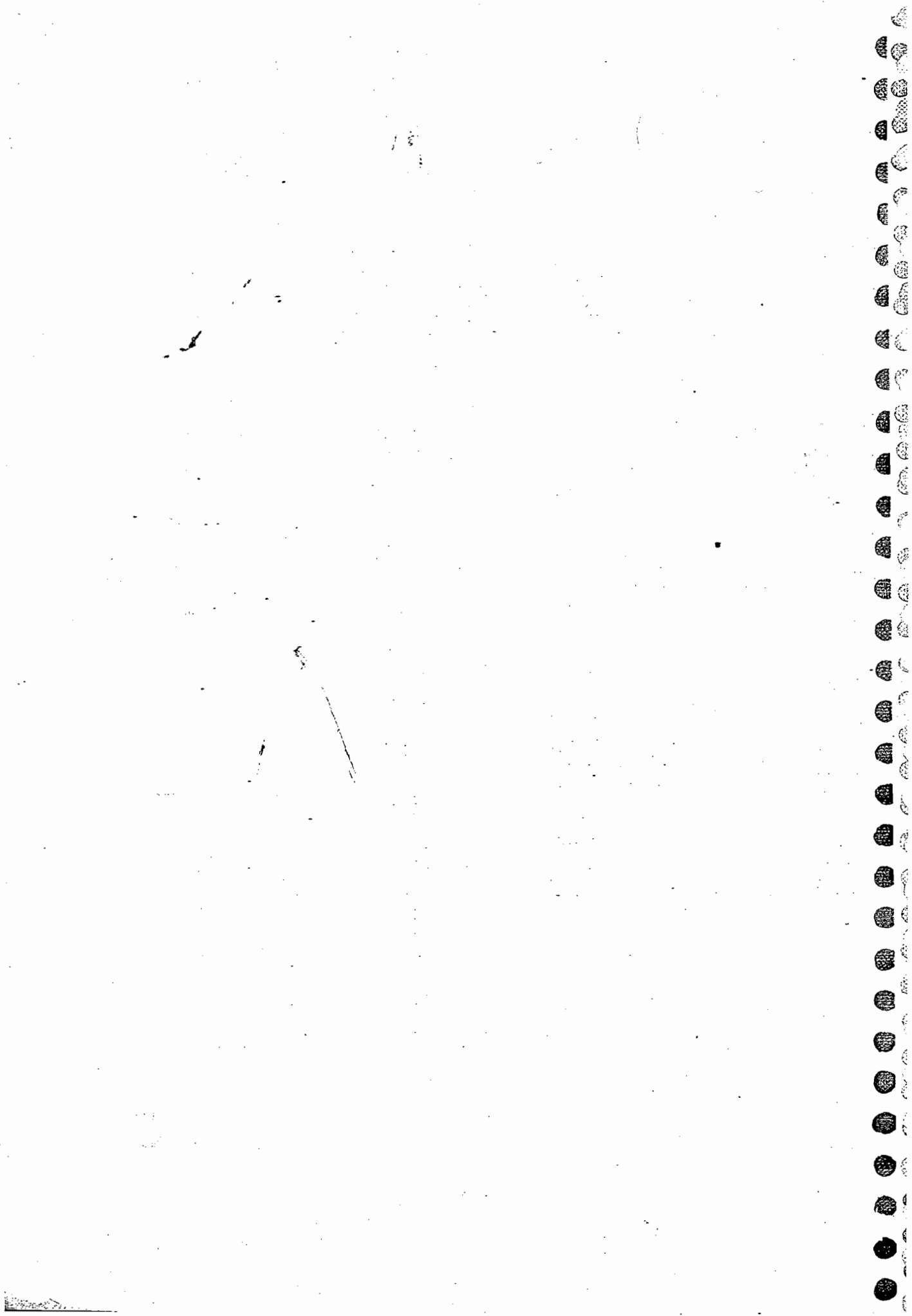
$$\begin{cases} u(0,t) = 0 = u(l,t) \\ u(x,0) = \sin\left(\frac{\pi x}{2}\right) \\ u_x(x,0) = 0 \end{cases}$$

Sol $u(x,t) = \sum D_m \sin\left(\frac{m\pi x}{l}\right) \cos\left(\frac{m\pi}{2}\right) t$

$$u(0,t) = 0 = u(l,t)$$

$$u(x,0) = 0$$

$$u(x,0) = f(x)$$





CALCULUS OF VARIATION

- 1) Definition (closeness of curves, etc. ...)
- 2) Functional involving $F(x, y, y')$
- 3) Functional involving $F(x, y, y' \dots y^n)$
- 4) Functional, involving $F(x, y, z, y', z')$
- 5) Functional involving $F(x, y, z, z')$
- 6) Moving boundary condition / Natural boundary condition
- 7) Isoparametric problem.
- 8) Variational method to solve ODE & PDE.
(Rayleigh Ritz Method)
- 9) Sufficient condition for maxima & minima

$$I[y(x)] = \int_{a_1}^{a_2} \sqrt{1+y'^2} dx$$

Functional $\rightarrow$ These are those variable quantities whose values are determined by one or several functions.

closeness of curves $\rightarrow$ The curve y_1 & y_2 are said to be closed in the sense of n^{th} order approxmately if,

$$|y_1(x) - y_2(x)| \\ \Rightarrow |y_1'(x) - y_2'(x)|$$

$|y_1^{(n)}(x) - y_2^{(n)}(x)|$ are closed $\forall x$ for which y_1 and y_2 are defined.

Continuity $\rightarrow$ A functional $I[y(x)]$ is said to be continuous at $y = y_0(x)$ in the sense of n^{th} order approxmately if for a given $\epsilon > 0$

$\exists \delta > 0$ Such that

$$|I(y(x)) - I(y_0(x))| < \epsilon \text{ whenever}$$

$$|y(x) - y_0(x)| < \delta, \quad |y'(x) - y_0'(x)| < \delta \dots$$

$$|y^{(n)}(x) - y_0^{(n)}(x)| < \delta$$

A functional $I[y(x)]$ attains a maximum on a curve $y = y_0(x)$ if the value of I on any curve close to y_0 do not exceed $I[y_0(x)]$.

$$\text{i.e. } \Delta I = I[y(x)] - I[y_0(x)] \leq 0$$

where $|y(x) - y_0(x)|$ is small.

further, $\Delta I < 0$ and $\Delta I = 0$ only on $y = y_0(x)$

maxima

In case of minima at $y = y_0$, $\Delta I > 0$ and if $\Delta I = 0$ only on $y = y_0(x)$ then strict minima.

Problem Statement in COV $\rightarrow$

Let us consider the problem of determining the function for which the integral takes maximum or minimum value which satisfies the boundary conditions.

$$y(a) = y_a \quad \& \quad y(b) = y_b$$

$$I[y(x)] = \int_a^b F(x, y, y') dx \quad \text{--- (1)}$$

The necessary condition to have Eqn (1) an extremal is given by an Euler Eqnⁿ.

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\text{i.e.} \quad F_y - \frac{d}{dx} (F_{y'}) = 0 \quad \text{--- (2)}$$

Using eqn. (2) we will form an ODE, the solution of that diff. Eqnⁿ will be an extremal & extremal can be unique, infinite or have no extremal depending on the boundary conditions.

Q $\rightarrow$ Find the Solⁿ of extremal of —
$$I = \int_0^{\pi/2} (y'^2 - y^2) dx, \quad y(0) = 0, \quad y\left(\frac{\pi}{2}\right) = 1$$

Solⁿ $\rightarrow$ Here $F = y'^2 - y^2$

N^y condition $\rightarrow$

$$F_y = \frac{d}{dx} (F_{y'}) = 0$$

$$-2y - \frac{d}{dx} (2y') = 0$$

$$-2y - 2y'' = 0$$

$$y'' + y = 0 \Rightarrow y^2 + 1 = 0 \Rightarrow y = \pm i$$

$$y(x) = C_1 \cos x + C_2 \sin x$$

$$y(0) = C_1 = 0$$

$$y(\pi/2) = C_2 = 1$$

$$y(x) = \sin x$$

unique Solⁿ

$$y(0) = 0, \quad y(\pi) = 1$$

$$0 = y(0) \Rightarrow C_1 = 0$$

$$y(\pi) = -C_1 = 1$$

No Solⁿ

$$\text{If } y(0) = 0, \quad y(\pi) = 0$$

$$y(0) = 0 \Rightarrow c_1 = 0$$

Infinite Solution.

Properties of Euler Equation $\rightarrow$

I) If, f is independent of x .

$$f - y' f y' = c$$

II) If f is independent of y .

$$f y' = c$$

$$f y - \frac{d}{dx} f y' = 0$$

$$\downarrow \quad f y' = \text{constant}$$

Q.7

$$I = \int_a^b \frac{\sqrt{1+y'^2}}{x} dx$$

$$f = \frac{\sqrt{1+y'^2}}{x}, \quad f \text{ is independent of } y$$

$$f y' = c$$

$$\frac{1}{x} \cdot \frac{1}{2\sqrt{1+y'^2}} (2y') = \frac{y'}{x\sqrt{1+y'^2}} = c$$

$$y'^2 = c^2 x^2 (1+y'^2)$$

$$y'^2 = c^2 x^2 + c^2 x^2 y'^2$$

$$y'^2 (1 - c^2 x^2) = c^2 x^2$$

$$y'^2 = \frac{c^2 x^2}{1 - c^2 x^2}$$

$$\frac{dy}{dx} = \frac{cx}{\sqrt{1 - c^2 x^2}}$$

$$\int dy = \int \frac{cx}{\sqrt{1 - c^2 x^2}} dx$$

$$y = -\frac{\sqrt{1 - c^2 x^2}}{c} + c_1$$

$$cy = -\sqrt{1 - c^2 x^2} + c_1$$

$$c^2 (y - c_1)^2 = 1 - c^2 x^2$$

$$\boxed{c^2 (y - c_1)^2 + c^2 x^2 = 1}$$

* Find the extremal of $\rightarrow$

$$I = \int_a^b \frac{1+y^2}{y'^2} dx$$

$$F = \frac{1+y^2}{y'^2} \text{ independent of } x.$$

$$F - y' \cdot f_{y'} = c$$

$$\Rightarrow \frac{1+y^2}{y'^2} - y' (1+y^2) \left(-\frac{2}{y'^3} \right) = c$$

$$\Rightarrow \frac{1+y^2}{y'^2} + \frac{2(1+y^2)}{y'^2} = c$$

$$\Rightarrow (1+y^2) + 2(1+y^2) = cy'^2$$

$$\Rightarrow 3(1+y^2) = cy'^2$$

$$\frac{dy}{dx} = \frac{\sqrt{3} \sqrt{1+y^2}}{c}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{3}}{c} \sqrt{1+y^2}$$

$$\Rightarrow \int \frac{dy}{\sqrt{1+y^2}} = \frac{\sqrt{3}}{c} \int dx$$

$$\log(y + \sqrt{1+y^2}) = Kx + c$$

$$\sinh^{-1}(y) = Kx + c$$

$$\boxed{\begin{aligned} & \int \frac{1}{\sqrt{a^2+x^2}} dx \\ &= \log(x + \sqrt{a^2+x^2}) \\ &= \sinh^{-1} \frac{x}{a} \end{aligned}}$$

* $I[y(x)] = \int_0^1 (2x - xy - y') y' dx$

✓ a) $2y'' + y = 2$

b) $2y'' - y = 2$

c) $y'' + 2y = 2$

d) $y'' - 2y = 2$

Solⁿ →

$$F = (2x - xy - y') y'$$

$$F = 2xy' - xy y' - y'^2$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$-xy' - \frac{d}{dx} (2x - xy - 2y') = 0$$

$$-xy' - (2 - y - y' - xy' - 2y'') = 0$$

$$-xy' - 2 + y + xy' + 2y'' = 0$$

$$\boxed{2y'' + y = 2}$$

Eq →

$$I = \int_0^1 (y'^2 + 12xy) dx$$

$$y(0) = 0$$

$$y(1) = 1$$

a) $y = \sin^2 \pi x / 2$

b) $y = \sin \pi x / 2$

c) $y = x^3$

d) $y = \frac{1}{2} (x^3 + \sin \frac{\pi x}{2})$

Solⁿ →

$$F = y'^2 + 12xy$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$12x - \frac{d}{dx} (2y') = 0$$

$$12x - 2y'' = 0$$

$$6x - y'' = 0$$

$$y'' = 6x$$

$$\frac{dy}{dx} = \frac{6x^2}{2} + c = 3x^2 + c$$

$$y(x) = \frac{3x^3}{3} + cx + K$$

$$y(x) = x^3 + cx + K$$

$$y(0) = 0, \quad K = 0$$

$$y(1) = 1$$

$$1 + c + k = 1 \Rightarrow c + k = 0$$

$$\boxed{c = 0}$$

$$y(x) = x^3$$

Q →

$$\int_0^2 y'^2 + 6xy + x^3, \quad y(0) = 0, \quad y(2) = 2$$

Solⁿ →

$$F = y'^2 + 6xy + x^3$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$6x - \frac{d}{dx} (2y') = 0$$

$$6x - 2y'' = 0 \Rightarrow 3x - y'' = 0$$

$$y'' = 3x$$

$$y' = \frac{3x^2}{2} + c_1$$

$$y(x) = \frac{3x^3}{6} + c_1 x + c_2$$

$$y(x) = \frac{x^3}{2} + c_1 x + c_2$$

$$y(0) = 0, \quad y(2) = 2$$

$$y(0) = 0 = c_2$$

$$y(2) = 2 = \frac{8}{2} + 2c_1$$

$$c_1 = -1$$

$$\therefore \boxed{y(x) = \frac{x^3}{2} - x}$$

Q →

$$\int_0^{\pi/2} (y'^2 + 2yy' - 16y^2) dx, \quad y(0) = 0, \quad y\left(\frac{\pi}{2}\right) = 1$$

Solⁿ →

$$F = y'^2 + 2yy' - 16y^2$$

independent of x

$$F - y'fy' = c$$

$$\begin{aligned}
 y'^2 + 2yy' - 16y^2 - y'(2y' + 2y) &= c \\
 \Rightarrow y'^2 + 2yy' - 16y^2 - 2y'^2 - 2yy' &= c \\
 \Rightarrow y'^2 - 16y^2 &= c^2 \\
 \Rightarrow y'^2 &= c^2 + 16y^2 \\
 y' &= \sqrt{c^2 + (4y)^2}
 \end{aligned}$$

$$\begin{aligned}
 \frac{dy}{dx} &= \sqrt{c^2 + (4y)^2} \\
 \frac{dy}{\sqrt{c^2 + (4y)^2}} &= dx
 \end{aligned}$$

$$\Rightarrow \int \frac{dy}{\sqrt{c^2 + (4y)^2}} = \int dx = x + c$$

$$\Rightarrow \boxed{\sinh^{-1}\left(\frac{4y}{c}\right) = x + c}$$

Brachistochrone problem : $\rightarrow$

Statement:- To find the equation of curve joining the points $(0,0)$ & (a,b) such that particle sliding along the curve takes minimum time

$$t = \frac{1}{\sqrt{2g}} \int_0^a \frac{\sqrt{1+y'^2}}{y} dx$$

Solve by $\rightarrow F - y'Fy' = c$

$$\begin{aligned}
 \text{ans} \rightarrow \quad \left. \begin{aligned} x &= l(\phi - \sin\phi) \\ y &= l(1 - \cos\phi) \end{aligned} \right\} \text{ parametric eqn of cycloid.}
 \end{aligned}$$

Minimum Surface of Revolution : $\rightarrow$

To find a curve passing through two points such that whenever it is revolved around the x-axis it generates minimum surface area

$$I = 2\pi \int_a^b y \sqrt{1+y'^2} dx$$

$$y(x) = c \cosh\left(\frac{x}{c} + c_1\right) \rightarrow \text{catenary}$$

Solⁿ: $F = (y \sqrt{1+y'^2})$ independent of x

$$F - y' F y' = c$$

$$(y \sqrt{1+y'^2}) - y y' \frac{1}{2 \sqrt{1+y'^2}} (2y') = c$$

$$y \sqrt{1+y'^2} - \frac{y y'^2}{\sqrt{1+y'^2}} = c$$

$$\Rightarrow y(1+y'^2) - y y'^2 = c \sqrt{1+y'^2}$$

$$y = c \sqrt{1+y'^2}$$

$$y^2 = c^2 (1+y'^2)$$

$$y'^2 = \frac{y^2}{c^2} - 1 \Rightarrow \frac{y^2 - c^2}{c^2}$$

$$y' = \frac{1}{c} \sqrt{y^2 - c^2}$$

$$\frac{dy}{dx} = \frac{1}{c} \sqrt{y^2 - c^2}$$

$$dy = \frac{1}{c} \sqrt{y^2 - c^2} dx$$

$$c \int \frac{dy}{\sqrt{y^2 - c^2}} = \int dx$$

$$\Rightarrow c \cosh\left(\frac{y}{c}\right) = x + c$$

$$\cosh\left(\frac{y}{c}\right) = \frac{x+c}{c} = \frac{x}{c} + 1$$

01 Nov 2014

Functional involving Higher order derivative : $\rightarrow$

$$I[y(x)] = \int_a^b F(x, y, y', y'' \dots y^{(n)}) dx$$

$$F(x, y, y', y'') = \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \dots + (-1)^n \frac{d^n}{dx^n} \left(\frac{\partial F}{\partial y^{(n)}} \right) = 0$$

#

Find the extremal -

$$I[y(x)] = \int_0^1 (1 + y'^2) dx$$

$$y(0) = 0, y'(0) = 1 \\ y(1) = 1, y'(1) = 1$$

$$F = 1 + y'^2$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right)$$

$$0 - 0 + \frac{d^2}{dx^2} (2y'')$$

$$\Rightarrow y''' = 0$$

$$y = \frac{C_1 x^3}{6} + \frac{C_2 x^2}{2} + C_3 x + C_4$$

$$y(0) = 0$$

$$\boxed{0 = C_4}$$

$$y' = \frac{C_1 x^2}{2} + C_2 x + C_3$$

$$\boxed{1 = C_3}$$

$$1 = \frac{C_1}{6} + \frac{C_2}{2} + C_3 + C_4$$

$$C_1 + 3C_2 = 0$$

$$1 = \frac{C_1}{2} + C_2 + C_3 \Rightarrow$$

$$\begin{array}{r} C_1 + 2C_2 = 0 \\ C_1 + 3C_2 = 0 \\ \hline C_2 = 0 \end{array}$$

$$\boxed{y = x}$$

Ans.

$$C_2 = 0, C_1 = 0$$

$$Q \rightarrow I[y(x)] = \int_0^1 (y + y'') dx$$

$$y(0) = 1, \quad y(1) = -1$$

$$y'(0) = 0, \quad y'(1) = -1$$

- a) Only one extremal
 b) Exactly two
 c) Infinite
 d) No extremal ✓

Use sufficient condition

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$1 - 0 + \frac{d^2}{dx^2} (1) = 0$$

$$1 = 0 \quad (\text{No extremal})$$

$$Q \rightarrow I = \int_0^{\pi/2} (y''^2 - y^2 + x^2) dx \quad y''$$

$$y(0) = 1, \quad y'(0) = 0$$

$$y(\pi/2) = 0, \quad y'(\pi/2) = -1$$

$$Sol^n \rightarrow \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$-2y - \frac{d}{dx} (0) + \frac{d^2}{dx^2} (2y'') = 0$$

$$y''' - 1 = 0$$

$$\Rightarrow \pm 1, \pm i$$

$$y = C_1 e^x + C_2 e^{-x} + C_3 \sin x + C_4 \cos x$$

$$y(0) = 1$$

$$1 = C_1 + C_2 + C_4$$

$$y(\pi/2) = 0$$

$$0 = C_1 e^{\pi/2} + C_2 e^{-\pi/2} + C_3$$

$$y' = c_1 e^x - c_2 e^{-x} + c_3 \cos x - c_4 \sin x$$

$$0 = c_1 - c_2 + c_3$$

$$-1 = c_1 e^{\pi/2} - c_2 e^{-\pi/2} - c_4$$

$$0 = c_1 e^{\pi/2} + c_2 e^{\pi/2} + c_3$$

$$c_1 + c_2 = c_4 - 1$$

$$c_1 + c_2 + c_4 = 1$$

$$c_1 - c_2 + c_3 = 0$$

$$2c_1 + c_3 + c_4 = 1$$

$$2c_1 e^{\pi/2} + c_3 - c_4 = -1$$

$$2c_1 + c_3 + c_4 = 1$$

$$2c_1 + 2c_3 + 2c_1 e^{\pi/2} = 0$$

$$2c_1 (1 + e^{\pi/2}) + 2c_3 = 0$$

$$c_3 = -c_1 (1 + e^{\pi/2})$$

$$c_3 = 1$$

$$\text{Ans} \rightarrow \boxed{y = \cos x}$$

note
Q#

An extremal of the function —

$$I[y(x)] = \int_a^b F(x, y, \frac{dy}{dx})$$

$$y(a) = y_1$$

$$y(b) = y_2$$

Satisfies Euler's Equation which is in general.

- Is a second order L.O.D.E.
- Is a Non linear of order greater than two.
- Admits unique solⁿ satisfy $y(a) = y_1$, $y(b) = y_2$
- May not admit a solⁿ satisfying $y(a) = y_1$, $y(b) = y_2$

3. Functional Involving more than one dependent Variable :-

$$I = \int_a^b F(x, y, y', \dots, y^{(n)}, z, z', \dots, z^{(n)}) dx \rightarrow \text{Independent}$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) - \dots - (-1)^n \frac{d^n}{dx^n} \left(\frac{\partial F}{\partial y^{(n)}} \right) = 0$$

$$+ \frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) + \dots - (-1)^n \frac{d^n}{dx^n} \left(\frac{\partial F}{\partial z^{(n)}} \right) = 0$$

It will form a curve (parametric) in space.

Put $y =$ in terms of x

$z =$ " " " "

Find the extremal —

$$I = \int_0^{\pi/2} (y'^2 + z'^2 + 2yz) dx$$

$$y(0) = 0, y(\pi/2) = 1$$

$$z(0) = 0, z(\pi/2) = -1$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$2z - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow z - y'' = 0 \quad \text{--- (1)} \Rightarrow z = y''$$

$$z'' = y''''$$

$$+ \frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0$$

$$z'' - y = 0 \quad \text{--- (2)}$$

$$y'''' - y = 0$$

$$y = c_1 e^x + c_2 e^{-x} + c_3 \sin x + c_4 \cos x$$

$$y(0) = 0$$

$$0 = c_1 + c_2 + c_4$$

$$1 = c_1 e^{\pi/2} + c_2 e^{-\pi/2} + c_3$$

$$y' = c_1 e^x - c_2 e^{-x} + c_3 \cos x - c_4 \sin x$$

$$z = y' = c_1 e^x + c_2 e^{-x} - c_3 \sin x - c_4 \cos x$$

$$0 = c_1 + c_2 - c_4 \quad \text{--- (3)}$$

$$-1 = c_1 e^{\pi/2} + c_2 e^{-\pi/2} - c_3 \quad \text{--- (4)}$$

$$c_1 + c_2 - c_4 = 0$$

$$c_1 + c_2 + c_4 = 0$$

$$c_4 = 0$$

$$c_1 e^{\pi/2} + c_2 e^{-\pi/2} + c_3 = 1$$

$$c_1 e^{\pi/2} + c_2 e^{-\pi/2} - c_3 = -1$$

$$c_3 = 1$$

$$c_1 + c_2 = 0, \quad c_1 + c_2 = 0 \Rightarrow c_1 = 0, c_2 = 0$$

$$y = \sin x$$

$$z = -\sin x$$

Q →

$$I = \int_1^2 (y'^2 + z'^2 + z^2) dx$$

$$F_y - \frac{d}{dx} (F_{y'}) = 0$$

$$0 - \frac{d}{dx} (2y') = 0$$

$$y' = c_1$$

$$y = c_1 x + c_2$$

in O.D.E., y is independent of z .

$$F_z - \frac{d}{dx} (F_{z'}) = 0$$

$$z'' - z = 0$$

$$z = c_3 e^x + c_4 e^{-x}$$

$$y(1) = 1$$

$$y(2) = 2$$

$$z(1) = 0$$

$$z(2) = 1$$

$$y(1) = 1 \quad + \quad y(2) = 2$$

$$1 = c_1 + c_2$$

$$2 = 2c_1 + c_2$$

$$\begin{array}{r} c_1 + c_2 = 1 \\ 2c_1 + c_2 = 2 \\ \hline -c_1 = -1 \\ c_1 = 1 \end{array} \quad + \quad c_2 = 0$$

$$\boxed{y = x}$$

Now $z(1) = 0$

$$z(2) = 1$$

$$0 = c_3 e + c_4 e^{-1} \times e$$

$$1 = c_3 e^2 + c_4 e^{-2}$$

$$\Rightarrow \quad c_3 e^2 + c_4 = 0$$

$$\begin{array}{r} c_3 e^2 + c_4 = 0 \\ c_3 e^2 + c_4 e^{-2} = 1 \\ \hline -c_4 = 1 \\ c_4 = -1 \end{array}$$

$$c_4 (1 - e^{-2}) = -1$$

$$c_4 \left(\frac{e^2 - 1}{e^2} \right) = -1$$

$$c_4 = \frac{e^2}{1 - e^2} \quad + \quad c_3 = \frac{1}{e^2 - 1}$$

$$\boxed{Z = \frac{1}{e^2 - 1} \cdot e^x + \frac{e^2}{1 - e^2} e^{-x}}$$

$$Z = \frac{e^x - e^2 e^{-x}}{e^2 - 1} \times \frac{e^{-1}}{e^{-1}}$$

$$= \frac{e^x e^{-1} - e^2 e^{-x}}{e - e^{-1}}$$

$$= \frac{e^{(x-1)} - e^{(x-1)} \times 2}{2(e - e^{-1})}$$

$$\Rightarrow \frac{\sinh(x-1)}{\sinh 1}$$

Q2

$$\int_0^1 (y^2 + 2y'^2 + y''^2) dx$$

$$y(0) = 0, \quad y'(0) = 1$$

$$y(1) = 0, \quad y'(1) = -\sinh$$

$$Q \rightarrow \int_0^{\pi/4} (2z - 4y^2 + y'^2 - z'^2) dx$$

$$y(0) = 0$$

$$z(0) = 0$$

$$y(\pi/4) = 1$$

$$z(\pi/4) = 1$$

$$Q \rightarrow \int_0^1 (y'^2 + z'^2 + 2y) dx$$

$$y(0) = 1$$

$$z(0) = 0$$

$$y(1) = 3/2$$

$$z(1) = 1$$

4. Function Involving two independent variables $\rightarrow$

$$I = \iint F(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}) dx dy$$

$$\frac{\partial F}{\partial z} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial z_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial z_y} \right) = 0$$

It is called Euler - Ostrogradsky Equation.

Q $\rightarrow$ Find the extremal -

$$\iiint \left[\left(\frac{\partial z}{\partial y} \right)^2 + \left(\frac{\partial z}{\partial x} \right)^2 \right] dx dy$$

$$\Rightarrow \iint [z_x^2 + z_y^2] dx dy$$

$$= \frac{\partial F}{\partial z} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial z_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial z_y} \right) = 0$$

$$= 0 - \frac{\partial}{\partial x} (2z_x) - \frac{\partial}{\partial y} (2z_y) = 0$$

$$\Rightarrow \boxed{z_{xx} + z_{yy} = 0} \rightarrow \text{Laplace Eqn}$$

Its Solⁿ is Harmonic fun^c
 $\downarrow$
 extremal.

is R.H.S. $\neq 0$ then it is Poisson Eqn.

$$\underline{Q \rightarrow} \iint [z_x^2 + z_y^2 + 2zf(x, y)] dx dy$$

$$\frac{\partial F}{\partial z} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial z_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial z_y} \right) = 0$$

$$\Rightarrow 2f(x, y) - \frac{\partial}{\partial x} (2zx) - \frac{\partial}{\partial y} (2zy) = 0$$

$$\Rightarrow \boxed{z_{xx} + z_{yy} = f(x, y)}$$

$$\underline{Q \rightarrow} \iiint \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial z} \right)^2 + 2uf(x, y, z) \right] dy dx dz$$

$$\Rightarrow \frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial F}{\partial u_z} \right) = 0$$

$$\Rightarrow 2f(x, y, z) - 2u_{xx} - 2u_{yy} - 2u_{zz} = 0$$

$$\Rightarrow \boxed{u_{xx} + u_{yy} + u_{zz} = f(x, y, z)} \rightarrow \text{Poisson}$$

Isoparametric problem : $\rightarrow$

Let there are two given funcⁿ

$$F(x, y, y') \text{ \& \& } G(x, y, y')$$

we seek to determinat $y(x)$ for the functional

$$I = \int_a^b F(x, y, y') dx$$

Subject to constraint —

$$\int_a^b G(x, y, y') dx = K$$

$$\text{where } y(a) = y_a$$

$$y(b) = y_b \text{ are fixed constant.}$$

Method to solve isoparametric Eqnⁿ:-

Create a new functional -

$$I[y(x)] = \int_a^b \underbrace{[F(x, y, y') + \lambda G(x, y, y')]}_{H(x, y, y')} dx$$

$$\frac{\partial H}{\partial y} - \frac{d}{dx} \left(\frac{\partial H}{\partial y'} \right) = 0$$

Q→ Find the extremal of Isoparametric Eqnⁿ→

$$I[y(x)] = \int_1^4 y'^2 dx$$

$$y(1) = 3$$

$$y(4) = 24$$

Sub to constraint -

$$\int_1^4 y dx = 36$$

Solⁿ:-

$$I[y(x)] = \int_1^4 [y'^2 + \lambda y] dx$$

$$F = y'^2 + \lambda y$$

$$F_y - \frac{d}{dx} (F_{y'}) = 0$$

$$\lambda - \frac{d}{dx} (2y') = 0$$

$$\lambda - 2y'' = 0$$

$$y'' = \lambda/2$$

$$y(x) = \frac{\lambda x^2}{4} + C_1 x + C_2$$

$$3 = \frac{\lambda}{4} + C_1 + C_2$$

$$\Rightarrow 12 = \lambda + 4C_1 + 4C_2$$

$$24 = 4\lambda + 4C_1 + C_2$$

$$12 = \lambda + 4C_1 + 4C_2$$

$$12 = 3\lambda - 3C_2$$

$$4 - \lambda = -C_2$$

$$4C_1 + 4C_2 = 12 - \lambda$$

$$C_1 + 4C_2 = 24 - 4\lambda$$

$$3C_1 = -12 + 3\lambda$$

$$C_1 = \lambda - 4$$

$$C_2 = 3 - \frac{\lambda}{4} - \lambda + 4$$

$$= 7 - \frac{\lambda}{4}$$

$$y = (d-4) + \left(7 - \frac{5d}{4}\right)x + \frac{d}{4}x^2$$

$$\Rightarrow \int_1^4 y \, dx = 36$$

$$\Rightarrow \int_1^4 \left((d-4) + \left(7 - \frac{5d}{4}\right)x + \frac{d}{4}x^2 \right) dx = 36$$

$$\Rightarrow (d-4)(4-1) + \left(7 - \frac{5d}{4}\right) \frac{1}{2} (16-1) + \frac{d}{4} \cdot \frac{1}{3} (64-1) = 36$$

$$\Rightarrow 3d - 12 + \frac{105}{2} - \frac{75d}{8} + \frac{31d}{4} = 36$$

$$\begin{array}{r} 420 \\ 60 \\ \hline 356 \end{array}$$

$$= 24d - 96 + 420 - 75d + 31d = 36 \times 8$$

$$\Rightarrow 8(24 - 75)d + 324 = 288$$

$$\Rightarrow 10d = 324 - 288$$

$$20d = 68 \quad 10d = 36$$

$$d = \frac{34}{10} = \frac{17}{5}$$

$$\boxed{d = 4}$$

Q →

$$\int_0^\pi y'^2 \, dx$$

$$y(0) = 0$$

$$y(\pi) = 0$$

$$\text{st. constraint } \int_0^\pi y^2 \, dx = 1$$

$$F = y'^2 + \lambda y^2$$

$$2\lambda y - \frac{d}{dx}(2y') = 0$$

$$\Rightarrow y'' - \lambda y = 0$$

It is Eigen value problem.

λ = -ve → Non Trivial solⁿ

λ = +ve, 0 → Trivial solⁿ.

$$\lambda = -\mu^2 \quad (\mu \neq 0)$$

$$y'' + \mu^2 y = 0$$

$$y = c_1 \cos \mu x + c_2 \sin \mu x$$

$$0 = C_1 \cdot 1 + 0$$

$$C_1 = 0$$

$$y = C_2 \sin \mu x$$

$$0 = C_2 \sin \mu \pi$$

$$\Rightarrow \sin \mu \pi = 0$$

$$\Rightarrow \mu \pi = n\pi \Rightarrow \mu = n, \quad n \in \mathbb{Z} - \{0\}$$

$$y = C_2 \sin nx, \quad \lambda = -n^2$$

$$\text{Now, } \int_0^\pi y^2 dx = 1$$

$$(\cos x = 1 - 2 \sin^2 x)$$

$$= \int_0^\pi C_2^2 \sin^2 nx = 1$$

$$= C_2^2 \int_0^\pi \left(\frac{1 - \cos 2x}{2} \right) dx = 1$$

$$C_2^2 \left[\frac{1}{2}x - \frac{\sin 2x}{4} \right]_0^\pi = 1$$

$$C_2^2 \left[\frac{\pi}{2} - \frac{1}{4} \sin \frac{2\pi}{2} \right] = 1$$

$$C_2^2 \left[\frac{\pi}{2} - 0 \right] = 1$$

$$C_2 = \sqrt{\frac{2}{\pi}}$$

$$\text{Hence, } y = \sqrt{\frac{2}{\pi}} \sin n\pi$$

$$\underline{Q \rightarrow} \quad I = \int_b^a y'^2 dx$$

$$y(a) = 0$$

$$y(b) = 0$$

$$\int_b^a y dx = 2$$

Q $\rightarrow$

$$\int (x^2 + y'^2) dx$$

$$y(0) = 0$$

$$\int_0^1 y^2 dx = 2$$

$$y(1) = 0$$

02/Nov/2014

Moving Boundary Condition : $\rightarrow$

$$(i) \quad I = \int_{x_1}^{x_2} F(x, y, y') dx \quad y(x_1) = y_1$$

+ (x_2, y_2) is moving along some curve $y_2 = \phi(x_2)$

$$* \quad [F - y' F y']_{x=x_2} \delta x_2 + F y' \Big|_{x=x_2} \delta y_2 = 0 \rightarrow \text{Transversality}$$

(i) If (x_2, y_2) moves along $x = c$
 $\delta x_2 = 0$

$$\boxed{[F y']_{x=x_2} = 0}$$

(ii) If (x_2, y_2) moves along $y = c$

$$\boxed{[F - y' F y']_{x=x_2} = 0}$$

(iii) If (x_2, y_2) moves along $y = \phi(x)$
 $y_2 = \phi(x_2)$

$$\delta y_2 = \phi'(x_2) \delta x_2$$

$$[F - y' F y'] \delta x_2 + [F y'] [\phi'(x_2)] \delta x_2 = 0$$

$$\Rightarrow \boxed{[F + (\phi' - y') F y']_{x=x_2} = 0} \rightarrow$$

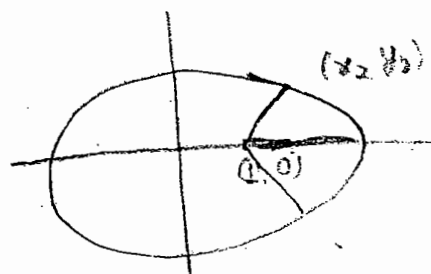
Find the shortest distance b/w $(1, 0)$ & ellipse $4x^2 + 9y^2 = 36$

Solⁿ $\rightarrow$ Shortage distance is straight line.

$$I[y(x)] = \int_1^{x_2} \sqrt{1 + y'^2} dx$$

$$F = 1 + y'^2$$

$$y = c_1 x + c_2$$



$$0 = c_1 + c_2 \Rightarrow c_1 = -c_2$$

$$y = c_1 x - c_1$$

$$y = \phi(x)$$

$$y = \frac{2}{3} \sqrt{9-x^2}$$

$$\phi(x) = \frac{2}{3} \sqrt{9-x^2}$$

$$[F + (\phi' - y') f y']_{x=x_2} = 0$$

$$\Rightarrow \sqrt{1+y'^2} + \left(\frac{2}{3} \cdot \frac{1}{2} \cdot \frac{-2x}{\sqrt{9-x^2}} - c_1 \right) \frac{y'}{\sqrt{1+y'^2}} = 0$$

$$\Rightarrow (1+y'^2) + \frac{2x}{3} \frac{y'}{\sqrt{9-x^2}} c_1 = 0$$

$$(\because y' = c_1)$$

$$= (1+c_1^2) + \frac{2x_2 c_1^2}{3 \sqrt{9-x_2^2}} - c_1^2 = 0$$

$$\Rightarrow c_1^2 \left(1 + \frac{2x_2}{3 \sqrt{9-x_2^2}} \right) = -1$$

$$\Rightarrow \boxed{3 \sqrt{9-x_2^2} = 2x_2 c_1} \quad \text{--- (2)}$$

Now (x_2, y_2) lies on

$$y_2 = c_1 x_2 - c_1 \quad \text{--- (3)}$$

$$4x_2^2 + 9y_2^2 = 36 \quad \text{--- (4)}$$

Put (3) in (4)

$$4x_2^2 + 9(c_1 x_2 - c_1)^2 = 36$$

$$4x_2^2 + 9c_1^2 (x_2 - 1)^2 = 36 \quad \text{--- (5)}$$

$$x_2 = \frac{3 \sqrt{9-x_2^2}}{2c_1} \Rightarrow 4c_1^2 x_2^2 = 9(9-x_2^2)$$

$$\Rightarrow 4(9-x_2^2) = 9c_1^2 (x_2 - 1)^2$$

$$\Rightarrow \frac{4}{9} x_2^2 = 9c_1^2 (x_2 - 1)^2$$

$$\Rightarrow 4x_2^2 = 81(x_2 - 1)^2$$

$$\Rightarrow 77x_2^2 = 81$$

$$x_2^2 = \frac{81}{77} = \frac{9}{5}$$

$$4c_1^2 \left(\frac{81}{77} \right) = 9 \left(9 - \frac{81}{77} \right)$$

$$c_1 = 2$$

$$c_1^2 = \frac{77}{36} \left(\frac{396-81}{77} \right) \Rightarrow \frac{315}{36}$$

$$y_2 = \frac{10}{5} - 2 \Rightarrow y_2 = \frac{0}{5}$$

Other point $(9/5, 0/5)$ & $q=2$

$(0, 1)$
 $(9/5, 0/5)$
 $\frac{0}{5} = 0$

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{\frac{81}{25} + \frac{9}{25}} = \sqrt{\frac{90}{25}} = \frac{3\sqrt{10}}{5}$$

Dec NET-11

Q.

$$I[y] = \int_{y_2}^{x_2} \sqrt{1+y'^2} dx \quad \left(\frac{1}{2}, \frac{1}{4}\right)$$

$$(x_1, y_1) = y = x = 5$$

(i) $2c_1 + 4c_2 = 1$ The general solⁿ of Euler Eqnⁿ

(ii) The transversality condition that hold at right end yields.

$$\sqrt{1+c_1^2} + (1-c_1) \times \frac{c_1}{\sqrt{1+c_1^2}} = 0$$

(iii) The relation $c_1 x_1 + c_2 = x_1 - 5$ is true.

(iv) The extremal is $y = -x + 3/4$

Solⁿ:-

$$F = \sqrt{1+y'^2}$$

$$y = c_1 x + c_2$$

$$\frac{1}{4} = \frac{1}{2} c_1 + c_2 \Rightarrow 2c_1 + 4c_2 = \frac{1}{2}$$

$$\Rightarrow c_1 + 4c_2 = 1$$

$$(ii) \left[F + [\phi' - y'] F y' \right]_{x=x_1} = 0$$

$$\phi(x) = x - 5$$

$$\left[\sqrt{1+y'^2} + (1-y') \times \frac{y'}{\sqrt{1+y'^2}} \right]_{x=x_1} = 0$$

$$i) \sqrt{1+c_1^2} + (1-c_1) \times \frac{c_1}{\sqrt{1+c_1^2}} = 0$$

$$y_1 = c_1 x + c_2 \quad \text{--- (1)}$$

$$y_1 = x_1 - 5 \quad \text{--- (2)}$$

$$c_1 x + c_2 = x - 5 \quad y_1$$

$$\frac{1}{2} c_1 + c_2 = \frac{1}{4} \quad y \quad 2c_1 + 4c_2 = 1$$

Steps →

- ① $y = c_1 x + c_2$
 $f(c_1, c_2) = (a, b)$
- ② $[F - (\phi' - y') f_{y'}]_{x=x_1} = 0$
- ③ (x_2, y_2)
 $y_2 = c_1 x_2 + c_2$ — (i)
- ④ $y_2 = \phi(x_2)$ — (ii)

Q. Find the shortest distance b/w $\phi(x) = x^2$
 $y = x^2$ + $y = x - 5$ ① $[F + \phi' - y'] f_{y'}]_{x=x_1}$
 (x_1, y_1) (x_2, y_2)

Ans: - $y = -x + \frac{3}{4}$

* Variational problem with moving Boundaries
for the functional involving more than one
dependent Variable :-

$$I = \int_{x_1}^{x_2} F(x, y, z, y', z') dx$$

General condition →

$$[F - y' f_{y'} - z' f_{z'}]_{x=x_2} \delta x_2 + [f_{y'}]_{x=x_2} \delta y_2 + [f_{z'}]_{x=x_2} \delta z_2 = 0$$

Case-I:- If variation in x-axis, y-axis, z-axis
 i.e. $\delta x_2, \delta y_2, \delta z_2$ are independent of each other is

$$\Rightarrow [F - y' f_{y'} - z' f_{z'}]_{x=x_2} \delta x_2 = 0$$

$$[f_{y'}]_{x=x_2} \delta y_2 = 0$$

$$\Rightarrow [f_{z'}]_{x=x_2} \delta z_2 = 0$$

(If $y = c$ then the second term $\delta y = 0$)

Case-2: $\rightarrow$ when a point $P(x_2, y_2, z_2)$ moves along some curve $y_2 = \phi(x_2)$, & $z_2 = \psi(x_2)$

$$\left[F + (\phi' - y') F_y + (\psi' - z') F_z \right]_{x=x_2} = 0$$

Case-3: If a point $P(x_2, y_2, z_2)$ moves along surface $z_2 = \phi_2(x_2, y_2)$

Then $\rightarrow$

$$\left[F - y' F_y + (\phi_x - z') F_z \right]_{x=x_2} = 0$$

$$+ \left[F_y + \phi_y F_z \right]_{x=x_2} = 0$$

* Extremal will always be perpendicular to the curve on which Boundary point is moving.

Q:- Find the extremal of the functional —

$$I = \int_{x_1}^{x_2} (y'^2 + z'^2 + 2yz) dx, \quad y(0) = 0, \quad z(0) = 0$$

& (x_2, y_2, z_2) moves over a fixed plane $x = x_2$.

Solⁿ: $\rightarrow$ Case-1 $\rightarrow$ Type problem

$$\delta y_2 \neq 0$$

$$\text{So } F_y' |_{x=x_2} = 0$$

$$\Rightarrow 2z + 2y' = 0 \Rightarrow y' + z = 0 \quad \text{--- (1)}$$

$$\text{f } 2y + 2z' = 0 \Rightarrow z' + y = 0 \quad \text{--- (2)}$$

Now Euler's eqn.

$$F_y - \frac{d}{dx} (F_y') = 0 \Rightarrow 2z - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow y'' = z$$

$$+ F_z - \frac{d}{dx} (F_z') = 0$$

$$2y - \frac{d}{dx} (2z') = 0 \Rightarrow z'' = y$$

$$y'''' - y = 0$$

$$y = c_1 e^x + c_2 e^{-x} + c_3 \sin x + c_4 \cos x$$

$$z = y'' = c_1 e^x + c_2 e^{-x} - c_3 \sin x - c_4 \cos x$$

$$y = c_1 \cosh x + c_2 \sinh x + c_3 \cos x + c_4 \sin x \quad (1)$$

$$z = c_1 \cosh x + c_2 \sinh x - c_3 \cos x - c_4 \sin x \quad (2)$$

$$y(0) = 0$$

$$\left[\begin{array}{l} 0 = c_1 + c_2 + c_3 \\ 0 = c_1 + c_2 - c_3 \end{array} \right] \therefore c_3 = 0$$

$$y(x) = c_1 e^x + c_2 e^{-x} + c_4 \sin x$$

$$z(x) = c_1 e^x + c_2 e^{-x} - c_4 \sin x$$

$$c_1 + c_3 = 0 \quad (3)$$

$$c_1 - c_3 = 0 \quad (4)$$

$$\Rightarrow c_1 = c_3 = 0$$

$$y = c_2 \sinh x + c_4 \sin x \quad (5)$$

$$z = c_2 \sinh x - c_4 \sin x \quad (6)$$

$$\text{Now } \delta x_2 = 0$$

$$[f_y']_{x=x_2} = 0 \quad (7)$$

$$[f_z']_{x=x_2} = 0 \quad (8)$$

$$[2y']_{x=x_2} = 0$$

$$\Rightarrow [2z']_{x=x_2} = 0$$

$$\Rightarrow y'(x_2) = 0 \text{ \& } z'(x_2) = 0 \quad (9)$$

$$c_2 \cosh x_2 + c_4 \cos x_2 = 0 \quad (10)$$

$$c_2 \cosh x_2 - c_4 \cos x_2 = 0 \quad (11)$$

on solving.

Case-I: $\cos x_2 \neq 0, \quad c_2 = c_4 = 0$

$$\Rightarrow \cos x_2 = 0 \Rightarrow x_2 = (2n+1) \frac{\pi}{2}$$

$$\cos x_2 = 0$$

$$\Rightarrow \quad \square \cos h x_2 = 0$$

$$\Rightarrow \quad \boxed{S_2 = 0}$$

$$\left. \begin{array}{l} y = c_4 \sin x \\ z = -c_4 \sin x \end{array} \right\} \underline{\text{Ans}} \rightarrow$$

08/10/2014

Natural Boundary Condition $\rightarrow$

$$I(y) = \int_{x_1}^{x_2} F(x, y, y') dx$$

If the boundary condition is not given at $x = x_2$

The natural Boundary condition at the point will be -

$$\boxed{\left[\frac{\partial F}{\partial y'} \right]_{x=x_2} = 0}$$

Find the extremal of -

$$\int_0^1 (y'^2 - y^2) dx$$

(i) $y(0) = 0, \quad y(1) = 1$

(ii) $y(0) = 0$

(iii) $y(1) = 1$

(iv) No end condition are satisfied.

Solⁿ (i) $F(x, y, y') = (y'^2 - y^2)$

Euler's Equⁿ

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \Rightarrow (D^2 + 1)y = 0$$

$$\Rightarrow y = \pm i$$

$$y = c_1 \cos x + c_2 \sin x$$

$$y(0) = 0, \quad y(1) = 1$$

$$0 = c_1, \quad 1 = c_2 \sin 1 + c_1 \cos 1$$

$$c_2 = \frac{1}{\sin 1}$$

$$y = \frac{\sin x}{\sin 1} \quad \underline{\underline{Ans.}}$$

(ii)

$$y(0) = 0$$

$$0 = c_1 + 0$$

$$\Rightarrow c_1 = 0$$

$$y = c_2 \sin x$$

$$\left[\frac{\partial f}{\partial y'} \right]_{x=1} = 0$$

[condition given only
 $y(0) = 0$]

$$2y'(1) = 0 \Rightarrow \boxed{y'(1) = 0}$$

$$c_2 \cos x + c_1 \sin x = y'$$

$$c_2 \cos 1 + 0 = 0$$

$$\Rightarrow c_2 = 0$$

$$\boxed{y = 0}$$

(iii)

$$y = c_1 \cos x + c_2 \sin x$$

$$y(1) = 1$$

$$1 = c_1 \cos 1 + c_2 \sin 1 \Rightarrow c_1 = \frac{1}{\cos 1}$$

$$\left[\frac{\partial f}{\partial y'} \right]_{x=0} = 0$$

$$2y'(0) = 0 \Rightarrow y'(0) = 0$$

$$y' = -c_1 \sin x + c_2 \cos x$$

$$0 = c_2$$

$$\therefore y = \frac{\cos x}{\cos 1} \quad \underline{\underline{Ans.}}$$

(p.v) No end condition are satisfied

$$\left[\frac{\partial F}{\partial y'} \right]_{x=0} = 0$$

$$y'(0) = 0$$

$$\# \left[\frac{\partial F}{\partial y'} \right]_{x=1} = 0$$

$$y'(1) = 0$$

$$y = C_1 \cos x + C_2 \sin x$$

$$y' = -C_1 \sin x + C_2 \cos x$$

$$\boxed{0 = C_2}$$

$$0 = -C_1 \cos x \Rightarrow \boxed{C_1 = 0}$$

$\boxed{y = 0}$ only trivial solⁿ exist

#

$$I = \int_0^1 (y'^2 - 2\alpha y y' - 2\beta y') dx$$

$$(i) \quad y(0) = 0, \quad y(1) = 1$$

$$(ii) \quad y(0) = 0$$

$$(iii) \quad y(1) = 1$$

(iv) No end condition

Ans:- (i) $y = x$

$$(ii) \quad y = \frac{\beta x}{1-\alpha}$$

$$(iii) \quad y = \frac{(\alpha + \beta)x + (1 - \beta)}{\alpha + 1}$$

$$(iv) \quad y = -\beta / \alpha$$

Imp

One-Sided Variation $\rightarrow$



Equation $\rightarrow$

$$y' = \left\{ \begin{array}{l} y = c_1 x + c_2 x^2 \\ x^2 + y^2 \\ \text{straightline} \end{array} \right\} =$$

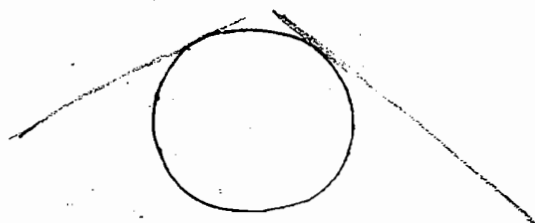
Let the functional $I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$

The problem is to find extremal of this function with the restriction that the extremal can not pass through certain point of certain region R bounded by the curve $\boxed{\psi(x, y) = 0}$

Condition $\rightarrow$ If $F_{y'y'}(x, y, y') \neq 0$

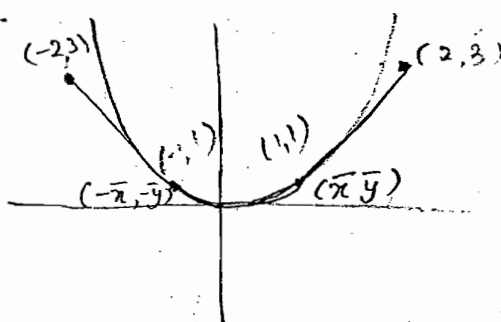
Tangent $\leftarrow y'(\bar{x}) = \phi'(\bar{x})$

where $\bar{x}$ is point of contact b/w curve and region R .



Find the shortest path from the point $P(-2, 3)$ to the point $(2, 3)$ located at the region $y \leq x^2$.

Solⁿ:-



Let the point of contacts are $(\bar{x}, \bar{y}) + (-\bar{x}, \bar{y})$

$$I = \int_{-2}^2 \sqrt{1+y'^2} dx$$

$$y = 4x + c_2 \longrightarrow \text{Extremal.}$$

$x^2 = y$ is Equation of parabola

$$\boxed{x\bar{x} = \frac{(y+\bar{y})}{2}} \longrightarrow \text{Equation of tangent at point } (\bar{x}, \bar{y})$$

$$2x\bar{x} = (y+\bar{y}) \text{ --- (1)}$$

Now check $F_{y'} y' \neq 0$

$$F = \sqrt{1+y'^2}$$

$$F_{y'} = \frac{1 \cdot 2y'}{2\sqrt{1+y'^2}} \Rightarrow \frac{y'}{\sqrt{1+y'^2}}$$

$$F_{y'} y' = \frac{y' \cdot \frac{1 \cdot 2y'}{2\sqrt{1+y'^2}}}{(1+y'^2)}$$

$$= \frac{y'^2 - 1 - y'^2}{(1+y'^2)^{3/2}} \Rightarrow \frac{-1}{\sqrt{(1+y'^2)^3}} \neq 0$$

By given condition Equ. (1) satisfied (2, 3)

(2, 3) is straightline So satisfied the equ. (1)

$$4\bar{x} = (3+\bar{y}) \text{ --- (2)}$$

Also $(\bar{x}, \bar{y})$ lies on $x^2 = y$

$$\bar{x}^2 = \bar{y} \text{ --- (3)}$$

$$\text{by (2)} \quad 4\bar{x} = 3 + \bar{x}^2$$

$$4\bar{x} - (4\bar{x}) = 3 - 3$$

$$\Rightarrow \bar{x}^2 - 4\bar{x} + 3 = 0$$

$$\bar{x}^2 - 3\bar{x} - \bar{x} + 3 = 0$$

$$\bar{x}(\bar{x}-3) - 1(\bar{x}-3) = 0$$

$$\bar{x} = 3, 1$$

$\bar{x}$ can not be 3

$$\bar{x} = 1$$

$$\bar{y} = 1$$

(1, 1) & other will be (-1, 1)

Due to symmetric.

$$(-2, 3) \text{ \& } (-1, 1)$$

Eqn of straight line

$$\sqrt{(2+1)^2 + (3-1)^2} \Rightarrow \sqrt{9+4} = \sqrt{5}$$

Eqn of straight line joining two point

$$(2, 3) \text{ \& } (-1, 1)$$

$$(y - y_1) = \frac{y_2 - y_1}{(x_2 - x_1)} (x - x_1)$$

$$\Rightarrow (y - 3) = \frac{1-3}{-1-2} (x-2)$$

$$\Rightarrow y - 3 = \frac{2}{3} (x-2)$$

$$\Rightarrow 3y - 9 = 2x - 4$$

$$\Rightarrow 2x - 3y + 5 = 0$$

$$y = 2x - 1$$

Ans $\rightarrow$

$$y = \begin{cases} -2x - 1 & -2 \leq x \leq -1 \\ x^2 & -1 \leq x \leq 1 \\ 2x - 1 & 1 \leq x \leq 2 \end{cases}$$

#

Find the extremal

$$I = \int_{-3}^3 y'^2 dx$$

$$y(-3) = 1$$

$$y(3) = 1$$

$$x^2 + y^2 = 5$$

Points are $(-3, 1) + (3, 1)$

$$\Rightarrow x\bar{x} + y\bar{y} = 5 \text{ --- (1)}$$

Satisfied $(3, 1)$

$$3\bar{x} + \bar{y} = 5 \text{ --- (2)}$$

$$\bar{x}^2 + \bar{y}^2 = 5 \text{ --- (3)}$$

$$\bar{x}^2 + (5 - 3\bar{x})^2 = 5$$

$$\bar{x}^2 + 25 + 9\bar{x}^2 - 30\bar{x} = 5$$

$$\Rightarrow 10\bar{x}^2 - 30\bar{x} + 20 = 0$$

$$\Rightarrow \bar{x}^2 - 3\bar{x} + 2 = 0$$

$$\bar{x}^2 - 2\bar{x} - \bar{x} + 2 = 0$$

$$\bar{x}(\bar{x} - 2) - 1(\bar{x} - 2) = 0$$

$$\bar{x} = 2, 1$$

$$\bar{y} = 3 \times 2 + \bar{y} = 5$$

$$\bar{y} = -1$$

$\therefore$ reject $2, 1$

$$\bar{y} = -1$$

$$(2, -1) + (-2, 1)$$

at $x = 1$

$$\text{For } 3 \times 1 + \bar{y} = 5$$

$$\bar{y} = 2$$

$$(1, 2) + (-1, -2)$$

$$\sqrt{-4 + 9} = \sqrt{5}$$

or

$$(2, -1) + (-2, 1)$$

$$\sqrt{2^2 + 1^2} = \sqrt{5}$$

Eqn of straight line joining two points

$$(-3, 1) (-1, 1)$$

Ans ->

$$(y - 1) = \frac{1 - 1}{-1 + 3} (x + 3)$$

$$2y - 2 = 0 \Rightarrow y = 1$$

$$(-3, 1) (-1, -2)$$

$$(y - 1) = \frac{-2 - 1}{-1 + 3} (x + 3)$$

$$2y - 2 = -3x - 9$$

$$2y = -3x - 7$$

$$y = -\frac{3x}{2} - \frac{7}{2}$$

$$y = \begin{cases} x/2 + 5/2 & -3 \leq x \leq -1 \\ \sqrt{5 - x^2} & -1 \leq x \leq 1 \\ -\frac{x}{2} + \frac{5}{2} & 1 \leq x \leq 3 \end{cases}$$

Rayleigh Ritz Method $\rightarrow$

Let a functional $I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$ — (1)

(It is direct method to find solve functional)

$$\text{Let } y = \sum_{i=1}^n \alpha_i w_i(x) \text{ — (2)}$$

Where α_i is constant and (i) w_i should be L.I.

(ii) continuous and smooth

(iii) y should satisfied boundary condition.

$$w_1(x) = x(1-x)$$

$$w_2(x) = x^2(1-x)$$

upto given n

$n \rightarrow \infty$ solution will be exist exact

by (1) & (2), $I[y(x)] = \phi(\alpha_i)$ — (3)

$$I = \phi(\alpha_1, \dots, \alpha_n)$$

These coefficients are selected in such a way that

ϕ is extremal.

$$\frac{\partial \phi}{\partial \alpha_1} = 0, \quad \frac{\partial \phi}{\partial \alpha_2} = 0 \quad \dots \quad \frac{\partial \phi}{\partial \alpha_n} = 0$$

Galerkin Method :-

$$a(x)y'' + b(x)y' + c(x)y = f(x) \quad \text{--- (1)}$$

$$\& y(x_1) = y_1 + y(x_2) = y_2$$

Then the corresponding function will be

$$I[y(x)] = \int_{x_1}^{x_2} \left\{ (P(x)) y'^2 + Q(x) y'^2 + R(x) y \right\} dx \quad \text{--- (2)}$$

$$\text{where } P(x) = e^{\int \frac{b}{a} dx}$$

$$Q = -\frac{cP}{a}$$

$$R = \frac{2fP}{a}$$

After that apply rayleigh method.

Q -> Find the approximate solⁿ upto 1st approximation to given ODE.

$$y'' = 1, \quad y(0) = 0, \quad y(1) = 0$$

Solⁿ

$$y'' = 1$$

$$y' = x + c_1$$

$$y = \frac{x^2}{2} + c_1 x + c_2$$

Compare with eqn (1)

$$a = 1, \quad b = 0, \quad c = 0, \quad f = 1$$

$$P = e^{\int \frac{b}{a} dx} = e^{\int \frac{0}{1} dx} = 1$$

$$Q = \frac{-0 \times 1}{1} = 0$$

$$R = \frac{2fP}{a} = \frac{2 \times 1 \times 1}{1} = 2$$

$$I = \int_0^1 (y'^2 + 2y) dx \quad \text{by eqn. (2)}$$

Replace $y'' \rightarrow -y'^2$

$y \rightarrow y^2$

$y' \rightarrow 0$

x or constant $\rightarrow$ Multiply by $2y$

(If P, Q, R Not Known)

Same Exp.

$$y'' - 1 = 0$$

$$-y'^2 - 2y = 0$$

$$F = y'^2 + 2y = 0$$

If xy'' Then add
apply only rule at
 y''
i.e. $xy'' = -xy'^2$

$$y(0) = 0, y(1) = 0$$

$$y = C_1 x(1-x)$$

If two approximate then -

$$y' = C_1 x(1-x) + C_1 x^2(1-x)$$

$$y' = C_1(1-2x)$$

$$I = \int_0^1 [C_1^2(1-2x)^2 + 2C_1 x(1-x)] dx$$

$$= \left[C_1^2 \frac{(1-2x)^3}{3(-2)} + \frac{2C_1 x^2}{2} - \frac{2C_1 x^3}{3} \right]_0^1$$

$$= C_1^2 \frac{(1-2)^3}{-6} - C_1 \frac{1}{-6} + \frac{2C_1}{2} - \frac{2C_1}{3}$$

$$= \frac{C_1^2}{6} + \frac{C_1^2}{6} + \frac{6C_1 - 4C_1}{6}$$

$$\frac{2C_1^2}{6} + \frac{2C_1}{6} = 0$$

$$C_1^2 + C_1 = 0$$

$$C_1(C_1 + 1) = 0 \Rightarrow C_1 = 0, C_1 = -1$$

$$\frac{\partial I}{\partial c_1} = 2c_1 + 1 = 0 \Rightarrow c_1 = -\frac{1}{2}$$

Ans $\rightarrow$ $\boxed{y = -\frac{1}{2}x(1-x)}$ A

Q $\rightarrow$ Find the approximate solⁿ of this Eqⁿ upto one approximate

$$y'' + y = -x$$

$$y(0) = y(1) = 0$$

$$-y'^2 + y^2 = -2yx$$

$$y'^2 - y^2 = 2yx$$

$$F = y'^2 - y^2 - 2xy$$

$$y(0) = 0, y(1) = 0$$

$$y = c_1 x(1-x)$$

$$I = \int_0^1 c_1^2 (1-2x)^2 - c_1^2 x^2 (1-x)^2 - 2x^2 c_1 (1-x) dx$$

$$= \left[c_1^2 \cdot \frac{2(1-2x)^3}{-6} - c_1^2 \left(\frac{x^3}{3} \right) - \left(c_1^2 \frac{x^5}{5} \right) + 2c_1^2 \frac{x^4}{4} \right]_0^1$$

$$- 2 \frac{x^3}{3} c_1 + 2c_1 \frac{x^4}{4}$$

$$\Rightarrow \frac{c_1^2}{6} - \frac{c_1^2}{6} - \frac{c_1^2}{3} - \frac{c_1^2}{5} + \frac{2c_1^2}{4} - \frac{2c_1}{3} + \frac{2c_1}{4}$$

$$\Rightarrow \frac{2c_1^2}{6} - \left[\frac{5c_1^2 + 3c_1^2}{15} \right] + \frac{2c_1^2}{4} - \left[\frac{4c_1 - 3c_1}{12} \right]$$

$$\Rightarrow \left[\frac{c_1^2}{3} - \frac{8c_1^2}{15} + \frac{c_1^2}{4} - \frac{c_1}{6} \right]$$

70-64

$$\frac{2c_1}{3} - \frac{16c_1}{15} + \frac{2c_1}{4} - \frac{1}{6} = 0 \Rightarrow c_1 \left[\frac{40 - 64 + 30}{60} \right] = \frac{1}{6}$$

$$c_1 = \frac{10}{8} = \frac{5}{4}$$

Ans $\rightarrow$ $\boxed{c = \frac{5}{10}}$ A

If condition are given

$$y(0) = 0, y(1) = 1$$

$$\boxed{y = x + cx(1-x)}$$

If $y(\pm 1) = 0$

$$\boxed{y = c(1-x^2)}$$

Q $\rightarrow$ $xy'' + y' + y = x$

$$y(0) = 0$$

$$y(1) = 1$$

Ans $\rightarrow 5/4$

find c upto first approximation.

Q $\rightarrow$ $y'' + (1+x^2)y = 0$

$$y(\pm 1) = 0$$

Ans $\rightarrow 3.5138$

9/11/2014

* Rayleigh method can be also used to find smallest Eigen Value.

Q $\rightarrow$ find the smallest Eigen value —

$$y'' + \lambda y = 0, \quad y(0) = y(1) = 0$$

Solⁿ

$$I = \int_0^1 (y'^2 - \lambda y^2) dy$$

$$y = cx(1-x)$$

$$y' = c[1-2x]$$

$$= \int_0^1 c^2 (1+4x^2-4x) - \lambda c^2 x^2 (1+x^2-2x) dx$$

$$I = c^2 \left[x + \frac{4x^3}{3} - \frac{4x^2}{2} - \lambda \left(\frac{x^3}{3} + \frac{x^5}{5} - \frac{2x^4}{4} \right) \right]_0^1$$

$$= c^2 \left[1 + \frac{4}{3} - \frac{4}{2} - \lambda \left(\frac{1}{3} + \frac{1}{5} - \frac{1}{2} \right) \right]$$

$$= c^2 \left[\frac{1}{3} - \frac{4}{2} - \lambda \left(\frac{10+6-15}{30} \right) \right]$$

$$2I = c^2 \left[\frac{14-12}{6} - \lambda \left(\frac{1}{30} \right) \right]$$

$$= c^2 \left[\frac{2}{3} - \lambda \left(\frac{1}{30} \right) \right]$$

$$\frac{\partial I}{\partial c} = 2c \left[\frac{1}{3} - \lambda \left(\frac{1}{30} \right) \right] = 0$$

$$\Rightarrow \frac{1}{3} - \lambda \left(\frac{1}{30} \right) = 0$$

$$\boxed{\lambda = 10}$$

Q:- Find the smallest value of Eigen value?
 $y(0) = y(1) = 0$, $(1+x)y'' + y' + \lambda y = 0$

$$I = \int_0^1 \left(-y \left[-(1+x)y'^2 + 0 + \lambda y^2 \right] \right) dx = 0$$

$$I = \int_0^1 - (1+x)y'^2 + \lambda y^2 dx$$

$$y = x(1-x)$$

$$= \int_0^1 \left(- \left(c^2(1+4x^2-4x) - 2cx(1+4x^2-4x) + \lambda c^2 x^2(1+x^2-2x) \right) \right) dx$$

$$\Rightarrow \left[c^2 \left(-x + \frac{4x^3}{3} + \frac{4x^2}{2} - \frac{x^2}{2} - \frac{4x^4}{4} + \frac{4x^3}{3} + \lambda \frac{x^3}{3} + \lambda \frac{x^5}{5} - \frac{2x^4}{4} \right) \right]_0^1$$

$$c^2 \left[-1 - \frac{4}{3} + \frac{4}{2} - \frac{1}{2} - 1 + \frac{4}{3} + \lambda \left(\frac{1}{3} + \frac{1}{5} - \frac{2}{4} \right) \right]$$

$$c^2 \left[-\frac{4}{3} - \frac{1}{2} + \frac{4}{2} + \lambda \left(\frac{20+12-30}{60} \right) \right]$$

$$2c \left[-\frac{1}{2} + \lambda \left(\frac{1}{30} \right) \right] = 0 \therefore$$

$$\frac{\lambda}{30} - \frac{1}{2} = 0 \quad \therefore \quad \boxed{\lambda = 15}$$

Variation Method to Solve PDE $\rightarrow$

Consider the Boundary value problem

$$-P(u_{xx} + u_{yy}) + qu = r$$

Corresponding variation problem will be

$$\iint [P(u_x^2 + u_y^2) + qu^2 - 2ru] \, dx \, dy \quad \text{--- (1)}$$

Put $u_{xx} = -u_x^2$, $u_{yy} = -u_y^2$

$u_x \text{ or } u_y = 0$, $u \rightarrow u^2$

r or constant $\rightarrow$ Multiply by $2u$.

The method to solve variation problem (1)

Assume $u = \sum c_i \phi_i(x, y)$ --- (2)

where ϕ_i are v.i. , c_i 's are constant

ϕ_i should be smooth and continuous.

and their partial derivative should be conti.

It is called Kantorovich Method. $\rightarrow$

Using (2) in (1)

$$I = \Psi(c_1, c_2, \dots, c_n)$$

$$\frac{\partial I}{\partial c_i} = 0 \quad \forall i = 1, 2, \dots, n$$

ϕ_i should satisfied the boundary condition.

#

$$u_{xx} + u_{yy} = -1$$

$$|x| \leq 1$$

$$|y| \leq 1$$

$$u = 0 \text{ at } x = \pm 1, y = \pm 1$$

$$u_{xx} + u_{yy} + 1 = 0$$

$$-u_x^2 - u_y^2 + 2u = 0$$

$$I = \iint_{-1}^1 (u_x^2 + u_y^2 - 2u) \, dx \, dy$$

Let $u = c(1-x^2)(1-y^2)$ Q.15

$u_x = c(-2x)(1-y^2)$

$u_y = c(1-x^2)(-2y)$

$$I = \iint_{-1}^1 4c^2 x^2 (1-y^2)^2 + 4c^2 y^2 (1-x^2)^2 - 2c(1-x^2)(1-y^2) dx dy$$

$$\Rightarrow \int_{-1}^1 \int_{-1}^1 4c^2 x^2 (1+y^4-2y^2) + 4c^2 y^2 (1+x^4-2x^2) - 2c(1-x^2)(1-y^2) dx dy$$

$$= \int_{-1}^1 \left[4c^2 x^2 y + \frac{4c^2 x^2 y^5}{5} - 4c^2 x^2 \cdot \frac{2y^3}{3} + \frac{4c^2 y^3 (1-x^2)^2}{3} - 2c(1-x^2) \left(y - \frac{y^3}{3} \right) \right] dx$$

$$\Rightarrow \int_{-1}^1 \left[4c^2 x^2 (1+1) + \frac{4c^2 x^2}{5} (1+1) - \frac{8c^2 x^2}{3} (1+1) + \frac{4c^2 y^3}{3} (1-x^2)^2 (1+1) - 2c(1-x^2) \left(2 - \left(\frac{1}{3} + \frac{1}{3} \right) \right) \right] dy$$

$$\Rightarrow \int_{-1}^1 \left[8c^2 x^2 + \frac{8c^2 x^2}{5} - \frac{16c^2 x^2}{3} + \frac{8c^2}{3} (1-x^2)^2 - \frac{8c}{3} (1-x^2) \right] dy$$

$$= \left[\frac{8c^2 x^3}{3} + \frac{8c^2 x^3}{15} - \frac{16c^2 x^3}{9} + \frac{8c^2}{3} \left(x + \frac{x^5}{5} - \frac{2x^3}{3} \right) - \frac{8c}{3} \left(x - \frac{x^3}{3} \right) \right]_{-1}^1$$

$$= \left[\frac{8c^2}{3} (2) + \frac{8c^2}{15} (2) - \frac{16c^2}{9} (2) + \frac{8c^2}{3} \left(2 + \frac{2}{5} - \frac{4}{3} \right) - \frac{8c}{3} \left(2 - \frac{2}{3} \right) \right]$$

$$= \left[\frac{16c^2}{3} + \frac{16c^2}{15} - \frac{32c^2}{9} + \frac{16c^2}{3} + \frac{16c^2}{15} - \frac{32c^2}{9} - \frac{16c}{3} + \frac{16c}{9} \right]$$

$$= \left[\frac{16c^2 \times 6}{15} - \frac{64c^2}{9} + \left\{ \frac{96c^2}{15} \right\} - \left\{ \frac{16c(3-1)}{9} \right\} \right] = 0$$

$$= \frac{96 \times 2c}{15} - \frac{64 \times 2c}{9} + \frac{96 \times 2c}{15} - \frac{32}{9} = 0$$

$$\Rightarrow \frac{2c}{90} \left[\frac{96 \times 6 - 64 \times 10 + 96 \times 6}{90} \right] = \frac{32 \times 16}{9}$$

$$\Rightarrow 16c \left[\frac{72 - 40}{10} \right] = 16 \Rightarrow \boxed{c = \frac{10}{32} = \frac{5}{16}}$$

Path independence : $\rightarrow$

For a functional $I = \int_{x_1}^{x_2} F(x, y, y') dx$

If the functional is path independent then I will be constant for every Extremal.

In this case,

Euler's Equⁿ became Indentically Zero..
and $F(x, y, y')$ can be written in the form of -

$$F(x, y, y') = \frac{d}{dx} [\phi(x, y)]$$

In other word, it is Exact.

Sufficient condition for Extremal : $\rightarrow$

Weierstrass function : $\rightarrow$

$$E(x, y, p, y') = F(x, y, y') - F(x, y, p) - (y' - p) F_p(x, y, p)$$

$$I = \int F(x, y, y') dx$$

p = Differential of functional.

$$p = \frac{dy}{dx} \quad \text{where } (y \text{ is an Extremal})$$

- (i) If $E > 0$, Minima
- (ii) If $E \leq 0 \rightarrow$ Maxima

- 1) If E does not change its sign for (x, y) close to the curve c and for the value of y' close to p on the extremal. It is weak.
- 2) If E does not change its sign for (x, y) close to curve c and for arbitrary value of y' is Strong.

Every Strong maxima/minima is weak maxima/minima

Q →

$$I[y(x)] = \int_0^1 \left[x + 2y + \frac{y'^2}{2} \right] dx, \quad y(0) = 0, \quad y(1) = 0$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$2 - y'' = 0$$

$$y'' = 2 \Rightarrow y' = 2x + C_1$$

$$y = x^2 + C_1 x + C_2$$

$$0 = C_2$$

$$0 = 1 + C_1 \Rightarrow C_1 = -1$$

$$\Rightarrow y = x^2 - x$$

$$y' = p = 2x - 1$$

$$E = \left(x + 2y + \frac{y'^2}{2} \right) - \left(x + 2y + \frac{p^2}{2} \right) - (y' - p) \cdot (p)$$

$$E = \frac{y'^2}{2} - \frac{(2x-1)^2}{2} - (y' - 2x + 1)(2x - 1)$$

$$E = \left(\frac{y'^2}{2} - \frac{p^2}{2} \right) - (y' - p) p$$

$$= \frac{(y' - p)^2}{2}$$

$$= \frac{(y' - p)^2}{2} \geq 0 \quad \text{for every } y$$

So it is Strong Minima.

For weak → find Extremal

For Strong → Donot find Extremal.

Q.

$$I = \int_0^a y^3 dx$$

$$y(0) = 0$$

$$y(a) = b$$

$$a, b > 0$$

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right)$$

$$= -\frac{d}{dx} (3y^2) = -6y'y'' = 0$$

$$y'' = 0$$

$$y' = c_1 \Rightarrow y = c_1 x + c'$$

$$0 = 0 + c', \quad \boxed{b = c_1 a} \Rightarrow c_1 = \frac{b}{a}$$

$$c' = 0$$

$$y = \frac{bx}{a}$$

$$p = \frac{b}{a}$$

$$E = \frac{b}{a} x (y'^3 - p^3) - (y' - p) \times 3p^2$$

$$\Rightarrow (y' - p) (y'^2 + p^2 + y'p - 3p^2)$$

$$= (y' - p) (y'^2 + y'p - 2p^2)$$

$$= (y' - p) (y'^2 + p^2 - 3p^2 + 2y'p + 3y'p)$$

$$(y' - p) [(y' - p)^2 + 3p(y' - p)]$$

$$= (y' - p)(y' - p)(y' - p + 3p)$$

$$= (y' - p)^2 (y' + 2p)$$

y' It will change its sign.

$$p > 0$$

$$\therefore E \geq 0$$

But y' close to $p \therefore E \geq 0$

$\therefore$ It is weak minima.

(Not true for arbitrary y')

Legendre Condition: $\rightarrow$

Maxima $f_{y'y'} < 0$ for every (x, y) close to
Curve C .

and y' close to P then weak maxima.

$\forall$ f_0

Minima, $f_{y'y'} > 0$ $\forall$ (x, y) close to C &
 y' close to P then
weak minima.

$f_{y'y'} > 0$ $\forall$ y' Strong minima

$f_{y'y'} < 0$ $\forall$ y' Strong maxima.

Q.

$$f = y'^3$$

$f_{y'y'} = 6y' > 0$ for y' close to P ($P > 0$),
Weak minima.

Q.

$$I = \int_0^1 y^2 e^x + \frac{e^x}{2} y'^2$$

$$f_{y'y'} = 2e^x > 0 \quad \forall y'$$

Strong minima on e^x .

Sufficient Condition for an Extremal

① Legendre's Condition:

Extremal

$$f_{y'y'} > 0$$

$\downarrow$
minima

$\downarrow$
strong
 $\forall y'$

$\downarrow$
weak
for y'
close to P

$$f_{y'y'} = 0$$

test-fail

$$f_{y'y'} < 0$$

maximisation

$\downarrow$
strong
 $\forall y'$

$\downarrow$
weak
for y' close to
 P

Green function

Q.

Construct Green's function for the BVP

$$u'' - u = x, \quad u(0) = u(1) = 0$$

Solⁿ:-

Comparing with -

$$L[u(x)] + \phi(x) = 0$$

$$u'' - u - x = 0$$

$$\text{Here } L[u(x)] = u'' - u$$

$$\phi(x) = -x$$

The associated BVP is given by -

$$L[u(x)] = 0$$

$$u'' - u = 0$$

$$m^2 - 1 = 0$$

$$m = \pm 1$$

$$u = c_1 e^x + c_2 e^{-x} = c_1 \cosh x + c_2 \sinh x$$

$$u(0) = 0, \quad u(1) = 0$$

$$0 = c_1 + c_2$$

$$0 = c_1 e + c_2 e^{-1}$$

$$0 = c_1 e + c_2 e^{-1}$$

$$c_2 \left(\frac{1}{e} - e \right) = 0 \Rightarrow c_2 \left(\frac{e^2 - 1}{e} \right) = 0$$

$$c_2 = 0, \quad c_1 = 0$$

$u(x) = 0$ only trivial solⁿ exist

$\therefore$ Green's function exist.

When $u(0) = 0$ then, $a = 0$

Hence, $u(x) = b \sinh x$

for $b = 1 \Rightarrow u_1(x) = \sinh x$

and when $u(1) = 0$

$$\Rightarrow a \cosh 1 + b \sinh 1 = 0$$

$$\Rightarrow b = -a \frac{\cosh 1}{\sinh 1}$$

$$\text{Hence, } u(x) = a \cosh x - a \frac{\cosh 1}{\sinh 1} \sinh x$$

$$= a \left[\frac{\cosh x \sinh 1 - \cosh 1 \sinh x}{\sinh 1} \right]$$

$$= -a \frac{\sinh(x-1)}{\sinh 1}$$

$$\text{For } a = -\sinh 1$$

$$u_2(x) = \sinh(x-1)$$

$$W(u_1(t), u_2(t)) = \begin{vmatrix} u_1(t) & u_2(t) \\ u_1'(t) & u_2'(t) \end{vmatrix}$$

$$= \begin{vmatrix} \sinh t & \sinh(t-1) \\ \cosh t & \cosh(t-1) \end{vmatrix}$$

$$= \sinh t \cosh(t-1) - \cosh t \sinh(t-1)$$

$$= \sinh [ht - (t-1)h] = \sinh 1$$

$$C = P(t) \cdot W[u_1(t), u_2(t)] = \sinh 1$$

$$G(x, t) = \begin{cases} -\frac{u_1(x) u_2(t)}{C} & ; 0 \leq x < t \\ -\frac{u_2(x) u_1(t)}{C} & ; t < x \leq 1 \end{cases}$$

$$= \begin{cases} +\frac{\sinh x \sinh(1-t)}{\sinh 1} & ; 0 \leq x < t \\ -\frac{\sinh(x-1) \sinh t}{\sinh 1} & ; t < x \leq 1 \end{cases}$$

Ans.

Q. Consider the Green's functions for BVP

$$u'' + \lambda u = x$$

$$u'' + \lambda u - x = 0$$

On comparing,

$$L[u(x)] = u'', \quad \phi(x) = -x$$

$$L[u(x)] + \phi(x) = 0$$

$$u'' = 0$$

$$u = ax + b$$

$$\text{When } u(0) = 0, \quad u(\pi/2) = 0$$

$$0 = a + b \Rightarrow b = 0$$

$$0 = a\frac{\pi}{2} + b \Rightarrow a = 0$$

$$a = 0, \quad b = 0 \therefore u(x) = 0$$

So Green func exist.

$$\text{When } u(0) = 0 \quad b = 0$$

$$\therefore u_1(x) = ax$$

$$\text{for } a = 1, \quad u_1(x) = x$$

$$\text{When } u(\frac{\pi}{2}) = 0, \quad a = a$$

$$\therefore u(x) = b + a\frac{\pi}{2} = 0$$

$$a = -\frac{2b}{\pi}$$

$$u(x) = -\frac{2b}{\pi}x + b = \left(1 - \frac{2}{\pi}x\right)b$$

$$\text{for } b = 1, \quad u_2(x) = \left(1 - \frac{2}{\pi}x\right)$$

$$\begin{aligned} W[u_1(t), u_2(t)] &= \begin{vmatrix} t & 1 - \frac{2}{\pi}t \\ 1 & -\frac{2}{\pi} \end{vmatrix} \\ &= -\frac{2t}{\pi} - 1 + \frac{2t}{\pi} = -1 \end{aligned}$$

$$G(x,t) = \begin{cases} x \left(1 - \frac{2t}{\pi}\right) & 0 \leq x < t \\ t \left(1 - \frac{2x}{\pi}\right) & t < x \leq \frac{\pi}{2} \end{cases} \quad \underline{\text{Ans.}}$$

